



中华人民共和国国家标准

GB/T 4087—2009
代替 GB/T 4087.3—1985



数据的统计处理 and 解释 二项分布可靠度单侧置信下限

Statistical interpretation of data—
One-sided confidence lower limit of reliability
for binomial distribution

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前 言

“数据的统计处理和解释”包括以下国家标准：

- GB/T 3359 数据的统计处理和解释 统计容忍区间的确定
- GB/T 3361 数据的统计处理和解释 在成对观测值情形下两个均值的比较
- GB/T 4087 数据的统计处理和解释 二项分布可靠度单侧置信下限
- GB/T 4088 数据的统计处理和解释 二项分布参数的估计与检验
- GB/T 4089 数据的统计处理和解释 泊松分布参数的估计和检验
- GB/T 4882 数据的统计处理和解释 正态性检验
- GB/T 4883 数据的统计处理和解释 正态样本离群值的判断和处理
- GB/T 4885 正态分布完全样本可靠度置信下限
- GB/T 4889 数据的统计处理和解释 正态分布均值和方差的估计与检验
- GB/T 4890 数据的统计处理和解释 正态分布均值和方差检验的功效
- GB/T 8055 数据的统计处理和解释 Γ 分布(皮尔逊Ⅲ型分布)的参数估计
- GB/T 8056 数据的统计处理和解释 指数分布样本离群值的判断和处理
- GB/T 6380 数据的统计处理和解释 I 型极值分布样本离群值的判断和处理
- GB/T 10092 数据的统计处理和解释 测试结果的多重比较
- GB/T 10094 正态分布分位数与变异系数的置信限

本标准代替 GB/T 4087.3—1985《数据的统计处理和解释 二项分布可靠度单侧置信下限》。

本标准与 GB/T 4087.3—1985 相比主要变化如下：

- 按 GB/T 1.1—2000《标准化工作导则 第1部分：标准的结构和编写规则》的要求对标准格式进行了修订；
- 按 GB/T 3358.1—2009《统计学词汇及符号 第1部分：一般统计术语与用于概率的术语》修改了术语，并增加了若干术语、符号和定义；
- 附录 A 中的有效数字由原来的 5 位提高到 7 位；
- 附录 B 中的计算程序由 BASIC 语言更改为 S-PLUS 语言。

本标准的附录 A 为规范性附录，附录 B 为资料性附录。

本标准由全国统计方法应用标准化技术委员会(SAC/TC 21)提出并归口。

本标准起草单位：东北师范大学、北京大学、中国标准化研究院。

本标准主要起草人：张宝学、房祥忠、李丹、丁文兴、于振凡等。

本标准所代替标准的历次版本发布情况为：

- GB/T 4087.3—1985。

引 言

本标准适用于产品试验结果为成功、失败两种状态的可靠度单侧置信下限估计,也可用于可化为此种情形的其他场合。从产品总体中随机地、独立地抽取若干个个体作为样本,本标准规定了基于这类样本确定产品可靠度单侧置信下限的方法。

本标准的主要内容以数表形式列出,但有些场合的参数可能不在表中,这时可以通过附录 B 中的近似公式计算,还可以用附录 B 中的 S-PLUS 程序直接计算出来。



数据的统计处理和解释

二项分布可靠度单侧置信下限

1 范围

从产品总体中随机地、独立地抽取若干个个体作为样本,本标准规定了基于这类样本确定产品可靠度单侧置信下限的方法。对有限总体,设其大小为 N ,样本大小为 n 。当抽取是有放回抽样时,或当抽取是无放回抽样,但 $n/N < 0.1$ 时, n 次抽取可以认为是独立的。

本标准适用于产品试验结果为成功、失败两种状态的可靠度单侧置信下限估计,也可用于可化为此种情形的其他场合。

2 规范性引用文件

下列文件中的条款通过本标准的引用而成为本标准的条款。凡是注日期的引用文件,其随后所有的修改单(不包括勘误的内容)或修订版均不适用于本标准,然而,鼓励根据本标准达成协议的各方研究是否可使用这些文件的最新版本。凡是不注日期的引用文件,其最新版本适用于本标准。

GB/T 2900.13 电工术语 可信性与服务质量(GB/T 2900.13—2008,IEC 60050(191):1990,Amd.1:1999 及 Amd.2:2002,IDT)

GB/T 3358.1 统计学词汇及符号 第1部分:一般统计术语与用于概率的术语(GB/T 3358.1—2009,ISO 3534-1:2006,IDT)

GB/T 3358.2 统计学词汇及符号 第2部分:应用统计(GB/T 3358.2—2009,ISO 3534-2:2006,IDT)

3 术语、定义和符号

3.1 术语和定义

GB/T 2900.13、GB/T 3358.1 和 GB/T 3358.2 确立的术语和定义适用于本标准。为便于参考,某些术语直接引自上述标准。

3.2 符号

GB/T 3358.1 和 GB/T 3358.2 确立的以及下列符号适用于本标准。

n	样本量
γ	置信水平
R_L	可靠度单侧置信下限
F	试验结果为失败的个数

4 可靠度单侧置信下限的确定

4.1 可靠度单侧置信下限的计算公式

对样本大小 n ,失败数 F ,在置信水平 γ 下,可靠度 R 的单侧置信下限 R_L 按以下公式计算:

$$F = 0, R_L = \sqrt[n]{1 - \gamma} \quad \dots\dots\dots (1)$$

$$F > 0, \sum_{x=0}^F \binom{n}{x} R_L^{n-x} (1 - R_L)^x = 1 - \gamma \quad \dots\dots\dots (2)$$

$$F = n - 1, R_L = 1 - \sqrt[n]{\gamma} \quad \dots\dots\dots (3)$$

$$F = n, R_L = 0 \quad \dots\dots\dots (4)$$

式(1)、式(3)、式(4)中直接给出了 R_L 的计算公式。

式(2)的计算方法参见附录 B 中推荐的方法。

4.2 查表法确定可靠度单侧置信下限(见附录 A 中表 A.1)

4.2.1 数表的参数范围与表距

$$\begin{aligned} \gamma &= 0.50(0.10)0.90, 0.95, 0.99 \\ n &= 1(1)70(5)100(10)140(20)200(50)1\ 000 \\ F &= 0(1)20 \end{aligned}$$

4.2.2 数表的应用

由给定的置信水平 γ 、样本量 n 及失败数 F , 查本标准附录 A 所列数表(表 A.1), 即得产品可靠度单侧置信下限 R_L 。表 A.1 未给出的 R_L 值可利用本标准提供的数据内插计算, 也可参照本标准附录 B 给出的近似公式计算, 附录 B 中还提供了近似计算的 S-PLUS 程序。

4.2.3 应用示例

示例 1:

某爆炸螺栓试验了 100 件, 失败一件, 规定置信水平为 0.90, 求可靠度单侧置信下限。

相应于 $n=100, F=1, \gamma=0.90$, 查表 A.1 得:

$$R_L = 0.961\ 660\ 4$$

示例 2:

某电子设备使用寿命服从指数分布, 其任务时间 $t_0=250\text{ h}$, 今试验数 $n=80$, 失败数 $F=2$, 未记录失效时间, 置信水平为 0.90, 求该设备失效率上限 λ_u 。

根据指数分布可靠性估计理论有:

$$R_L = e^{-\lambda_u t_0}$$

因此

$$\lambda_u = \frac{1}{t_0} \ln \frac{1}{R_L}$$

相应于 $n=80, F=2, \gamma=0.90$, 查表 A.1 得

$$R_L = 0.934\ 840\ 0$$

故

$$\lambda_u = \frac{1}{250} \ln \frac{1}{0.934\ 840\ 0} = 0.000\ 269\ 52$$



附 录 A
(规范性附录)

二项分布可靠度单侧置信下限 R_L 数表

表 A.1 二项分布可靠度单侧置信下限 R_L 数表

N/F	$\gamma=0.50$						
	0	1	2	3	4	5	6
1	0.5000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.7071068	0.2928932	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.7937005	0.5000000	0.2062995	0.0000000	0.0000000	0.0000000	0.0000000
4	0.8408964	0.6138010	0.3840238	0.1591036	0.0000000	0.0000000	0.0000000
5	0.8705506	0.6859450	0.5000000	0.3117948	0.1294494	0.0000000	0.0000000
6	0.8908987	0.7354098	0.5780276	0.4201154	0.2642594	0.1091013	0.0000000
7	0.9057237	0.7714233	0.6355293	0.5000000	0.3640622	0.2283141	0.0942763
8	0.9170040	0.7988119	0.6792462	0.5592889	0.4401409	0.3204631	0.2009700
9	0.9258747	0.8203413	0.7136007	0.6065285	0.5000000	0.3930646	0.2861825
10	0.9330330	0.8377093	0.7413084	0.6446216	0.5483119	0.4516879	0.3550785
11	0.9389309	0.8520160	0.7641279	0.6759891	0.5881187	0.5000000	0.4118814
12	0.9438743	0.8640049	0.7832478	0.7022673	0.6214819	0.5404961	0.4595043
13	0.9480775	0.8741972	0.7995003	0.7246021	0.6498474	0.5749278	0.5000000
14	0.9516952	0.8829683	0.8134854	0.7438192	0.6742599	0.6045625	0.5348551
15	0.9548416	0.8905963	0.8256468	0.7605289	0.6954911	0.6303351	0.5651699
16	0.9576033	0.8972908	0.8363192	0.7751919	0.7141246	0.6529560	0.5917766
17	0.9600467	0.9032133	0.8457605	0.7881626	0.7306097	0.6729679	0.6153163
18	0.9622238	0.9084901	0.8541719	0.7997180	0.7452975	0.6907986	0.6362898
19	0.9641760	0.9132213	0.8617134	0.8100779	0.7584674	0.7067857	0.6550948
20	0.9659363	0.9174873	0.8685132	0.8194186	0.7703418	0.7212009	0.6720516
21	0.9675318	0.9213536	0.8746756	0.8278835	0.7811039	0.7342661	0.6874191
22	0.9689845	0.9248738	0.8802862	0.8355902	0.7909025	0.7461613	0.7014117
23	0.9703128	0.9280923	0.8854160	0.8426364	0.7998612	0.7570368	0.7142048
24	0.9715319	0.9310465	0.8901242	0.8491033	0.8080842	0.7670195	0.7259471
25	0.9726549	0.9337675	0.8944608	0.8550597	0.8156580	0.7762142	0.7367627
26	0.9736927	0.9362819	0.8984680	0.8605637	0.8226572	0.7847106	0.7467572
27	0.9746546	0.9386124	0.9021821	0.8656649	0.8291440	0.7925854	0.7560206
28	0.9755486	0.9407784	0.9056340	0.8704061	0.8351728	0.7999049	0.7646300
29	0.9763818	0.9427968	0.9088506	0.8748239	0.8407909	0.8067247	0.7726524
30	0.9771600	0.9446822	0.9118552	0.8789505	0.8460386	0.8130959	0.7801464
31	0.9778885	0.9464472	0.9146680	0.8828137	0.8509516	0.8190597	0.7871618
32	0.9785721	0.9481031	0.9173068	0.8864379	0.8555606	0.8246555	0.7937435
33	0.9792146	0.9496596	0.9197873	0.8898447	0.8598932	0.8299145	0.7999303
34	0.9798197	0.9511255	0.9221234	0.8930530	0.8639733	0.8348682	0.8057571
35	0.9803906	0.9525085	0.9243272	0.8960797	0.8678229	0.8395414	0.8112538
36	0.9809301	0.9538153	0.9264097	0.8989398	0.8714606	0.8439567	0.8164483
37	0.9814407	0.9550522	0.9283807	0.9016467	0.8749031	0.8481362	0.8213640
38	0.9819246	0.9562245	0.9302488	0.9042123	0.8781661	0.8520975	0.8260236
39	0.9823840	0.9573372	0.9320220	0.9066475	0.8812634	0.8558574	0.8304468
40	0.9828206	0.9583948	0.9337072	0.9089620	0.8842067	0.8594310	0.8346500
41	0.9832361	0.9594012	0.9353109	0.9111644	0.8870081	0.8628315	0.8386503
42	0.9836319	0.9603601	0.9368389	0.9132628	0.8896766	0.8660716	0.8424612
43	0.9840095	0.9612747	0.9382963	0.9152644	0.8922226	0.8691613	0.8460962
44	0.9843701	0.9621481	0.9396880	0.9171756	0.8946534	0.8721123	0.8495673
45	0.9847148	0.9629829	0.9410183	0.9190026	0.8969772	0.8749335	0.8528860
46	0.9850445	0.9637817	0.9422912	0.9207507	0.8992003	0.8776328	0.8560608
47	0.9853604	0.9645468	0.9435103	0.9224249	0.9013296	0.8802178	0.8591015
48	0.9856632	0.9652802	0.9446789	0.9240298	0.9033712	0.8826958	0.8620166
49	0.9859537	0.9659839	0.9458002	0.9255697	0.9053298	0.8850735	0.8648133
50	0.9862327	0.9666596	0.9468770	0.9270484	0.9072104	0.8873561	0.8674988

表 A.1 (续)

N/F	$\gamma=0.50$						
	0	1	2	3	4	5	6
51	0.9865008	0.9673090	0.9479118	0.9284695	0.9090178	0.8895504	0.8700799
52	0.9867587	0.9679336	0.9489070	0.9298363	0.9107565	0.8916608	0.8725620
53	0.9870069	0.9685347	0.9498649	0.9311518	0.9124292	0.8936925	0.8749512
54	0.9872460	0.9691138	0.9507876	0.9324189	0.9140412	0.8956485	0.8772529
55	0.9874764	0.9696719	0.9516769	0.9336402	0.9155947	0.8975342	0.8794714
56	0.9876987	0.9702102	0.9525347	0.9348182	0.9170926	0.8993529	0.8816102
57	0.9879132	0.9707297	0.9533625	0.9359550	0.9185387	0.9011085	0.8836753
58	0.9881203	0.9712315	0.9541619	0.9370529	0.9199348	0.9028041	0.8856695
59	0.9883205	0.9717163	0.9549344	0.9381138	0.9212845	0.9044422	0.8875960
60	0.9885140	0.9721850	0.9556813	0.9391395	0.9225892	0.9060255	0.8894596
61	0.9887012	0.9726385	0.9564039	0.9401317	0.9238507	0.9075580	0.8912614
62	0.9888825	0.9730774	0.9571032	0.9410922	0.9250724	0.9090410	0.8930058
63	0.9890580	0.9735024	0.9577805	0.9420223	0.9262559	0.9104767	0.8946952
64	0.9892280	0.9739142	0.9584367	0.9429234	0.9274021	0.9118686	0.8963321
65	0.9893929	0.9743135	0.9590729	0.9437970	0.9285131	0.9132170	0.8979186
66	0.9895527	0.9747007	0.9596898	0.9446443	0.9295905	0.9145252	0.8994576
67	0.9897079	0.9750764	0.9602885	0.9454664	0.9306362	0.9157948	0.9009503
68	0.9898584	0.9754411	0.9608696	0.9462644	0.9316515	0.9170273	0.9024000
69	0.9900047	0.9757953	0.9614339	0.9470394	0.9326368	0.9182234	0.9038077
70	0.9901468	0.9761394	0.9619822	0.9477924	0.9335946	0.9193867	0.9051750
75	0.9908006	0.9777229	0.9645054	0.9512574	0.9380023	0.9247369	0.9114691
80	0.9913731	0.9791093	0.9667145	0.9542911	0.9418605	0.9294208	0.9169788
85	0.9918785	0.9803333	0.9686647	0.9569693	0.9452671	0.9335566	0.9218430
90	0.9923279	0.9814218	0.9703991	0.9593510	0.9482965	0.9372342	0.9261689
95	0.9927303	0.9823961	0.9719515	0.9614829	0.9510080	0.9405259	0.9300416
100	0.9930925	0.9832733	0.9733492	0.9634023	0.9534497	0.9434894	0.9335275
110	0.9937185	0.9847892	0.9757646	0.9667192	0.9576682	0.9486109	0.9395520
120	0.9942404	0.9860532	0.9777786	0.9694849	0.9611860	0.9528817	0.9445751
130	0.9946823	0.9871233	0.9794835	0.9718262	0.9641644	0.9564965	0.9488279
140	0.9950612	0.9880408	0.9809454	0.9738338	0.9667177	0.9595970	0.9524740
160	0.9956772	0.9895325	0.9833223	0.9770977	0.9708698	0.9646366	0.9584026
180	0.9961566	0.9906934	0.9851719	0.9796377	0.9741000	0.9685589	0.9630156
200	0.9965403	0.9916225	0.9866522	0.9816705	0.9766861	0.9716976	0.9667082
250	0.9972313	0.9932957	0.9893182	0.9853315	0.9813422	0.9773506	0.9733575
300	0.9976922	0.9944118	0.9910965	0.9877735	0.9844484	0.9811211	0.9777931
350	0.9980215	0.9952094	0.9923672	0.9895185	0.9866682	0.9838157	0.9809623
400	0.9982686	0.9958077	0.9933205	0.9908276	0.9883333	0.9858371	0.9833402
450	0.9984609	0.9962731	0.9940621	0.9918460	0.9896283	0.9874091	0.9851899
500	0.9986147	0.9966456	0.9946555	0.9926608	0.9906652	0.9886675	0.9866698
550	0.9987405	0.9969503	0.9951410	0.9933276	0.9915130	0.9896971	0.9878812
600	0.9988454	0.9972043	0.9955457	0.9938833	0.9922198	0.9905554	0.9888902
650	0.9989342	0.9974193	0.9958882	0.9943536	0.9928180	0.9912811	0.9897442
700	0.9990103	0.9976035	0.9961818	0.9947567	0.9933308	0.9919038	0.9904767
750	0.9990762	0.9977632	0.9964362	0.9951061	0.9937751	0.9924433	0.9911115
800	0.9991339	0.9979030	0.9966588	0.9954119	0.9941641	0.9929157	0.9916664
850	0.9991849	0.9980263	0.9968553	0.9956816	0.9945071	0.9933321	0.9921564
900	0.9992301	0.9981359	0.9970299	0.9959215	0.9948120	0.9937024	0.9925921
950	0.9992706	0.9982339	0.9971862	0.9961360	0.9950856	0.9940337	0.9929819
1000	0.9993071	0.9983222	0.9973268	0.9963292	0.9953312	0.9943320	0.9933328

表 A.1 (续)

N/F	$\gamma=0.50$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0829960	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.1794712	0.0741253	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.2585231	0.1621257	0.0669670	0.0000000	0.0000000	0.0000000	0.0000000
11	0.3237823	0.2357365	0.1478364	0.0610691	0.0000000	0.0000000	0.0000000
12	0.3785185	0.2975541	0.2166400	0.1358610	0.0561257	0.0000000	0.0000000
13	0.4250718	0.3501530	0.2752547	0.2004046	0.1256803	0.0519225	0.0000000
14	0.4651447	0.3954381	0.3257406	0.2560637	0.1864326	0.1169183	0.0483049
15	0.5000000	0.4348303	0.3696643	0.3045090	0.2393736	0.1742809	0.1092980
16	0.5305927	0.4694072	0.4082233	0.3470447	0.2858753	0.2247257	0.1636165
17	0.5576593	0.5000000	0.4423407	0.3846837	0.3270321	0.2693903	0.2117669
18	0.5817757	0.5272585	0.4727413	0.4182241	0.3637099	0.3092019	0.2547022
19	0.6033993	0.5517001	0.5000000	0.4483000	0.3966007	0.3449053	0.2932144
20	0.6228969	0.5737392	0.5245799	0.4754199	0.4262607	0.3771030	0.3279483
21	0.6405684	0.5937138	0.5468569	0.5000000	0.4531432	0.4062863	0.3594317
22	0.6566575	0.6118995	0.5671408	0.5223805	0.4776195	0.4328592	0.3881005
23	0.6713681	0.6285285	0.5856865	0.5428438	0.5000000	0.4571568	0.4143133
24	0.6848702	0.6437903	0.6027088	0.5616257	0.5205420	0.4794582	0.4383744
25	0.6973067	0.6578484	0.6183878	0.5789257	0.5394628	0.5000000	0.4605370
26	0.7087992	0.6708389	0.6328763	0.5949122	0.5569474	0.5189825	0.4810176
27	0.7194512	0.6828788	0.6463049	0.6097294	0.5731532	0.5365770	0.5000000
28	0.7293506	0.6940689	0.6587857	0.6235009	0.5882154	0.5529299	0.5176429
29	0.7385763	0.7044971	0.6704165	0.6363342	0.6022513	0.5681675	0.5340838
30	0.7471930	0.7142373	0.6812801	0.6483214	0.6153619	0.5824016	0.5494413
31	0.7552600	0.7233559	0.6914504	0.6595432	0.6276354	0.5957267	0.5638181
32	0.7628284	0.7319110	0.7009921	0.6700717	0.6391506	0.6082286	0.5773067
33	0.7699432	0.7399529	0.7099612	0.6799679	0.6499746	0.6199798	0.5899858
34	0.7766428	0.7475263	0.7184075	0.6892880	0.6601684	0.6310473	0.6019263
35	0.7829631	0.7546709	0.7263765	0.6980812	0.6697852	0.6414884	0.6131909
36	0.7889360	0.7614222	0.7339069	0.7063900	0.6788724	0.6513540	0.6238357
37	0.7945895	0.7678120	0.7410337	0.7142538	0.6874732	0.6606919	0.6339105
38	0.7999474	0.7738681	0.7477880	0.7217072	0.6956249	0.6695426	0.6434595
39	0.8050331	0.7796171	0.7541997	0.7287814	0.7033632	0.6779434	0.6525236
40	0.8098659	0.7850803	0.7602939	0.7355052	0.7107166	0.6859279	0.6611377
41	0.8144661	0.7902795	0.7660922	0.7419042	0.7177153	0.6935257	0.6693354
42	0.8188485	0.7952335	0.7716170	0.7480005	0.7243825	0.7007645	0.6771457
43	0.8230281	0.7999585	0.7768873	0.7538153	0.7307426	0.7076691	0.6845957
44	0.8270201	0.8044706	0.7819196	0.7593678	0.7368160	0.7142626	0.6917093
45	0.8308355	0.8087834	0.7867298	0.7646754	0.7426210	0.7205651	0.6985100
46	0.8344864	0.8129098	0.7913324	0.7697543	0.7481753	0.7265957	0.7050160
47	0.8379828	0.8168626	0.7957409	0.7746184	0.7534952	0.7323719	0.7112479
48	0.8413344	0.8206513	0.7999668	0.7792814	0.7585953	0.7379085	0.7172217
49	0.8445509	0.8242861	0.8040206	0.7837552	0.7634881	0.7432211	0.7229533
50	0.8476384	0.8277773	0.8079146	0.7880512	0.7681870	0.7483220	0.7284571

表 A.1 (续)

N/F	$\gamma=0.50$						
	7	8	9	10	11	12	13
51	0.8506063	0.8311320	0.8116561	0.7921795	0.7727021	0.7532247	0.7337466
52	0.8534610	0.8343584	0.8152550	0.7961501	0.7770452	0.7579396	0.7388339
53	0.8562084	0.8374641	0.8187190	0.7999724	0.7812257	0.7624784	0.7437310
54	0.8588549	0.8404554	0.8220551	0.8036533	0.7852515	0.7668498	0.7484472
55	0.8614056	0.8433390	0.8252708	0.8072019	0.7891330	0.7710634	0.7529929
56	0.8638660	0.8461195	0.8283722	0.8106241	0.7928761	0.7751265	0.7573777
57	0.8662398	0.8488035	0.8313657	0.8139271	0.7964885	0.7790492	0.7616091
58	0.8685326	0.8513950	0.8342566	0.8171167	0.7999768	0.7828361	0.7656955
59	0.8707483	0.8538999	0.8370499	0.8201992	0.8033477	0.7864962	0.7696447
60	0.8728905	0.8563208	0.8397502	0.8231789	0.8066069	0.7900348	0.7734627
61	0.8749633	0.8586637	0.8423634	0.8260622	0.8097603	0.7934585	0.7771558
62	0.8769691	0.8609308	0.8448918	0.8288521	0.8128123	0.7967718	0.7807305
63	0.8789115	0.8631270	0.8473409	0.8315541	0.8157673	0.7999805	0.7841930
64	0.8807933	0.8652538	0.8497135	0.8341724	0.8186313	0.8030895	0.7875469
65	0.8826179	0.8673165	0.8520143	0.8367105	0.8214068	0.8061030	0.7907993
66	0.8843877	0.8693163	0.8542449	0.8391720	0.8240990	0.8090261	0.7939524
67	0.8861042	0.8712574	0.8564091	0.8415608	0.8267117	0.8118626	0.7970128
68	0.8877711	0.8731415	0.8585104	0.8438793	0.8292474	0.8146156	0.7999829
69	0.8893897	0.8749709	0.8605514	0.8461311	0.8317101	0.8172891	0.8028680
70	0.8909625	0.8767485	0.8625338	0.8483183	0.8341027	0.8198872	0.8056709
75	0.8981991	0.8849291	0.8716576	0.8583860	0.8451138	0.8318407	0.8185684
80	0.9045352	0.8920909	0.8796451	0.8671993	0.8547534	0.8423068	0.8298603
85	0.9101286	0.8984135	0.8866976	0.8749810	0.8632636	0.8515469	0.8398295
90	0.9151028	0.9040359	0.8929683	0.8819007	0.8708323	0.8597632	0.8486948
95	0.9195557	0.9090691	0.8985817	0.8880943	0.8776068	0.8671187	0.8566297
100	0.9235642	0.9136000	0.9036359	0.8936710	0.8837061	0.8737404	0.8637747
110	0.9304916	0.9214312	0.9123693	0.9033082	0.8942462	0.8851836	0.8761209
120	0.9362684	0.9279603	0.9196521	0.9113432	0.9030343	0.8947254	0.8864157
130	0.9411577	0.9334875	0.9258166	0.9181456	0.9104739	0.9028022	0.8951305
140	0.9453510	0.9382272	0.9311026	0.9239781	0.9168536	0.9097282	0.9026037
160	0.9521678	0.9459323	0.9396968	0.9334613	0.9272251	0.9209888	0.9147525
180	0.9574730	0.9519289	0.9463848	0.9408406	0.9352958	0.9297517	0.9242068
200	0.9617182	0.9567281	0.9517372	0.9467464	0.9417555	0.9367647	0.9317739
250	0.9693644	0.9653705	0.9613773	0.9573827	0.9533888	0.9493949	0.9454002
300	0.9744644	0.9711356	0.9678069	0.9644781	0.9611486	0.9578198	0.9544903
350	0.9781090	0.9752557	0.9724016	0.9695475	0.9666934	0.9638393	0.9609852
400	0.9808432	0.9783455	0.9758486	0.9733509	0.9708539	0.9683562	0.9658585
450	0.9829700	0.9807501	0.9785301	0.9763102	0.9740895	0.9718696	0.9696489
500	0.9846720	0.9826736	0.9806759	0.9786774	0.9766789	0.9746804	0.9726820
550	0.9860646	0.9842479	0.9824312	0.9806146	0.9787979	0.9769805	0.9751639
600	0.9872251	0.9855600	0.9838948	0.9822289	0.9805638	0.9788979	0.9772327
650	0.9882072	0.9866703	0.9851326	0.9835957	0.9820580	0.9805203	0.9789833
700	0.9890489	0.9876219	0.9861941	0.9847663	0.9833392	0.9819114	0.9804836
750	0.9897789	0.9884464	0.9871146	0.9857820	0.9844494	0.9831168	0.9817843
800	0.9904172	0.9891687	0.9879195	0.9866702	0.9854210	0.9841710	0.9829218
850	0.9909807	0.9898050	0.9886293	0.9874536	0.9862779	0.9851022	0.9839265
900	0.9914817	0.9903714	0.9892610	0.9881507	0.9870403	0.9859292	0.9848189
950	0.9919300	0.9908781	0.9898263	0.9887736	0.9877218	0.9866699	0.9856173
1000	0.9923335	0.9913343	0.9903343	0.9893350	0.9883358	0.9873358	0.9863366

表 A.1 (续)

N/F	$\gamma=0.50$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0451584	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.1026103	0.0423967	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.1541816	0.0966940	0.0399533	0.0000000	0.0000000	0.0000000	0.0000000
18	0.2002209	0.1457754	0.0914223	0.0377762	0.0000000	0.0000000	0.0000000
19	0.2415327	0.1898685	0.1382387	0.0866958	0.0358240	0.0000000	0.0000000
20	0.2787990	0.2296581	0.1805339	0.1314426	0.0824337	0.0340637	0.0000000
21	0.3125802	0.2657340	0.2188962	0.1720744	0.1252832	0.0785713	0.0324682
22	0.3433425	0.2985883	0.2538387	0.2090975	0.1643716	0.1196754	0.0750540
23	0.3714721	0.3286316	0.2857950	0.2429629	0.2001385	0.1573286	0.1145477
24	0.3972914	0.3562099	0.3151299	0.2740530	0.2329807	0.1919160	0.1508650
25	0.4210741	0.3816127	0.3421521	0.3026930	0.2632370	0.2237856	0.1843418
26	0.4430528	0.4050879	0.3671238	0.3291612	0.2912009	0.2532429	0.2152895
27	0.4634230	0.4268468	0.3902706	0.3536951	0.3171212	0.2805488	0.2439794
28	0.4823566	0.4470703	0.4117840	0.3764985	0.3412145	0.3059305	0.2706488
29	0.5000000	0.4659163	0.4318326	0.3977488	0.3636658	0.3295836	0.2955029
30	0.5164803	0.4835193	0.4505591	0.4175980	0.3846385	0.3516790	0.3187203
31	0.5319095	0.5000000	0.4680907	0.4361820	0.4042726	0.3723647	0.3404569
32	0.5463840	0.5154613	0.4845386	0.4536159	0.4226940	0.3917713	0.3608493
33	0.5599902	0.5299954	0.5000000	0.4700051	0.4400095	0.4100147	0.3800199
34	0.5728044	0.5436826	0.5145607	0.4854389	0.4563170	0.4271952	0.3980741
35	0.5848934	0.5565958	0.5282975	0.5000000	0.4717025	0.4434042	0.4151066
36	0.5963173	0.5687981	0.5412790	0.5137599	0.4862407	0.4587208	0.4312017
37	0.6071292	0.5803470	0.5535649	0.5267820	0.5000000	0.4732178	0.4464356
38	0.6173764	0.5912933	0.5652094	0.5391256	0.5130417	0.4869579	0.4608740
39	0.6271031	0.6016833	0.5762620	0.5508415	0.5254209	0.5000000	0.4745791
40	0.6363475	0.6115573	0.5867671	0.5619770	0.5371860	0.5123950	0.4876048
41	0.6451450	0.6209547	0.5967636	0.5725732	0.5483821	0.5241910	0.5000000
42	0.6535269	0.6299073	0.6062885	0.5826690	0.5590494	0.5354299	0.5118096
43	0.6615214	0.6384480	0.6153730	0.5922987	0.5692245	0.5461495	0.5230745
44	0.6691560	0.6466027	0.6240486	0.6014945	0.5789404	0.5563856	0.5338315
45	0.6764533	0.6543974	0.6323407	0.6102841	0.5882274	0.5661707	0.5441141
46	0.6834363	0.6618559	0.6402754	0.6186950	0.5971138	0.5755334	0.5539522
47	0.6901239	0.6689991	0.6478744	0.6267503	0.6056248	0.5845000	0.5633753
48	0.6965348	0.6758472	0.6551596	0.6344720	0.6137844	0.5930960	0.5724084
49	0.7026855	0.6824178	0.6621492	0.6418807	0.6216121	0.6013436	0.5810750
50	0.7085921	0.6887264	0.6688614	0.6489957	0.6291292	0.6092635	0.5893978

表 A.1 (续)

N/F	$\gamma=0.50$						
	14	15	16	17	18	19	20
51	0.7142684	0.6947903	0.6753114	0.6558325	0.6363536	0.6168747	0.5973958
52	0.7197283	0.7006219	0.6815155	0.6624091	0.6433019	0.6241955	0.6050883
53	0.7249828	0.7062347	0.6874866	0.6687384	0.6499895	0.6312414	0.6124925
54	0.7300447	0.7116413	0.6932380	0.6748347	0.6564314	0.6380281	0.6196240
55	0.7349225	0.7168521	0.6987816	0.6807112	0.6626400	0.6445696	0.6264984
56	0.7396282	0.7218786	0.7041290	0.6863787	0.6686284	0.6508788	0.6331285
57	0.7441690	0.7267289	0.7092888	0.6918487	0.6744078	0.6569677	0.6395269
58	0.7485548	0.7314134	0.7142727	0.6971313	0.6799891	0.6628476	0.6457062
59	0.7527925	0.7359402	0.7190879	0.7022357	0.6853827	0.6685296	0.6516774
60	0.7568899	0.7403171	0.7237435	0.7071707	0.6905971	0.6740235	0.6574499
61	0.7608532	0.7445505	0.7282478	0.7119444	0.6956418	0.6793384	0.6630349
62	0.7646899	0.7486486	0.7326073	0.7165661	0.7005240	0.6844827	0.6684406
63	0.7684054	0.7526171	0.7368287	0.7210412	0.7052528	0.6894637	0.6736754
64	0.7720050	0.7564624	0.7409198	0.7253765	0.7098339	0.6942913	0.6787479
65	0.7754948	0.7601895	0.7448850	0.7295797	0.7142752	0.6989699	0.6836647
66	0.7788787	0.7638050	0.7487306	0.7336569	0.7185824	0.7035080	0.6884335
67	0.7821630	0.7673124	0.7524625	0.7376119	0.7227621	0.7079115	0.6930609
68	0.7853503	0.7707176	0.7560850	0.7414516	0.7268189	0.7121855	0.6975521
69	0.7884462	0.7740244	0.7596026	0.7451808	0.7307590	0.7163365	0.7019147
70	0.7914539	0.7772376	0.7630205	0.7488035	0.7345864	0.7203694	0.7061523
75	0.8052953	0.7920223	0.7787492	0.7654754	0.7522023	0.7389285	0.7256554
80	0.8174137	0.8049663	0.7925197	0.7800723	0.7676250	0.7551776	0.7427303
85	0.8281114	0.8163940	0.8046766	0.7929584	0.7812402	0.7695221	0.7578039
90	0.8376256	0.8265565	0.8154873	0.8044182	0.7933483	0.7822791	0.7712100
95	0.8461416	0.8356534	0.8251645	0.8146755	0.8041866	0.7936977	0.7832088
100	0.8538091	0.8438426	0.8338770	0.8239105	0.8139449	0.8039784	0.7940120
110	0.8670590	0.8579963	0.8489336	0.8398702	0.8308075	0.8217448	0.8126814
120	0.8781060	0.8697971	0.8614874	0.8531777	0.8448680	0.8365576	0.8282479
130	0.8874588	0.8797863	0.8721146	0.8644429	0.8567704	0.8490979	0.8414262
140	0.8954784	0.8883531	0.8812278	0.8741025	0.8669764	0.8598511	0.8527258
160	0.9085163	0.9022800	0.8960430	0.8898067	0.8835697	0.8773334	0.8710964
180	0.9186619	0.9131170	0.9075722	0.9020273	0.8964824	0.8909375	0.8853927
200	0.9267822	0.9217914	0.9167998	0.9118082	0.9068173	0.9018257	0.8968341
250	0.9414063	0.9374117	0.9334178	0.9294231	0.9254285	0.9214346	0.9174399
300	0.9511608	0.9478321	0.9445025	0.9411730	0.9378435	0.9345140	0.9311845
350	0.9581311	0.9552770	0.9524229	0.9495689	0.9467148	0.9438599	0.9410058
400	0.9633608	0.9608631	0.9583653	0.9558676	0.9533699	0.9508722	0.9483737
450	0.9674289	0.9652083	0.9629876	0.9607676	0.9585469	0.9563262	0.9541063
500	0.9706835	0.9686850	0.9666865	0.9646881	0.9626896	0.9606911	0.9586926
550	0.9733472	0.9715305	0.9697131	0.9678965	0.9660791	0.9642624	0.9624450
600	0.9755668	0.9739017	0.9722358	0.9705699	0.9689047	0.9672388	0.9655729
650	0.9774456	0.9759079	0.9743702	0.9728325	0.9712948	0.9697579	0.9682202
700	0.9790558	0.9776280	0.9762002	0.9747724	0.9733446	0.9719168	0.9704890
750	0.9804517	0.9791184	0.9777858	0.9764532	0.9751207	0.9737881	0.9724555
800	0.9816725	0.9804233	0.9791741	0.9779248	0.9766748	0.9754256	0.9741764
850	0.9827500	0.9815743	0.9803986	0.9792229	0.9780464	0.9768707	0.9756950
900	0.9837085	0.9825974	0.9814871	0.9803767	0.9792656	0.9781553	0.9770442
950	0.9845654	0.9835135	0.9824609	0.9814090	0.9803564	0.9793045	0.9782519
1000	0.9853366	0.9843373	0.9833381	0.9823381	0.9813389	0.9803389	0.9793396

表 A.1 (续)

N/F	$\gamma=0.60$						
	0	1	2	3	4	5	6
1	0.4000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.6324555	0.2254033	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.7368063	0.4319305	0.1565673	0.0000000	0.0000000	0.0000000	0.0000000
4	0.7952707	0.5549569	0.3277439	0.1198883	0.0000000	0.0000000	0.0000000
5	0.8325532	0.6347052	0.4453037	0.2640632	0.0971196	0.0000000	0.0000000
6	0.8583742	0.6903662	0.5286019	0.3719138	0.2224740	0.0816141	0.0000000
7	0.8773067	0.7313640	0.5903758	0.4530713	0.3205716	0.1914740	0.0703760
8	0.8917795	0.7627997	0.6379250	0.5159453	0.3974636	0.2810466	0.1680614
9	0.9032011	0.7876616	0.6756248	0.5659742	0.4590212	0.3535428	0.2502137
10	0.9124435	0.8078131	0.7062357	0.6066882	0.5093074	0.4130794	0.3183910
11	0.9200759	0.8244754	0.7315794	0.6404496	0.5511146	0.4627293	0.3755358
12	0.9264849	0.8384815	0.7529048	0.6688908	0.5863997	0.5047189	0.4240055
13	0.9319427	0.8504192	0.7710956	0.6931731	0.6165672	0.5406710	0.4655859
14	0.9366465	0.8607148	0.7867946	0.7141441	0.6426498	0.5717874	0.5016222
15	0.9407423	0.8696851	0.8004805	0.7324364	0.6654206	0.5989752	0.5331409
16	0.9443408	0.8775705	0.8125168	0.7485316	0.6854717	0.6229305	0.5609325
17	0.9475274	0.8845564	0.8231845	0.7628025	0.7032598	0.6441944	0.5856173
18	0.9503689	0.8907884	0.8327043	0.7755422	0.7191477	0.6631954	0.6076857
19	0.9529186	0.8963822	0.8412520	0.7869844	0.7334239	0.6802743	0.6275306
20	0.9552191	0.9014311	0.8489691	0.7973175	0.7463207	0.6957088	0.6454700
21	0.9573053	0.9060109	0.8559710	0.8066951	0.7580295	0.7097238	0.6617653
22	0.9592058	0.9101842	0.8623526	0.8152439	0.7687055	0.7225071	0.6766322
23	0.9609444	0.9140027	0.8681930	0.8230691	0.7784806	0.7342142	0.6902492
24	0.9625408	0.9175098	0.8735581	0.8302586	0.7874642	0.7449743	0.7027683
25	0.9640119	0.9207421	0.8785036	0.8368869	0.7957475	0.7548976	0.7143163
26	0.9653718	0.9237307	0.8830769	0.8430172	0.8034105	0.7640792	0.7250020
27	0.9666327	0.9265022	0.8873185	0.8487036	0.8105192	0.7725979	0.7349185
28	0.9678050	0.9290793	0.8912632	0.8539926	0.8171323	0.7805239	0.7441451
29	0.9688977	0.9314818	0.8949411	0.8589244	0.8232998	0.7879160	0.7527519
30	0.9699187	0.9337269	0.8983784	0.8635340	0.8290648	0.7948273	0.7607990
31	0.9708748	0.9358296	0.9015980	0.8678520	0.8344652	0.8013023	0.7683393
32	0.9717720	0.9378030	0.9046198	0.8719052	0.8395358	0.8073818	0.7754192
33	0.9726156	0.9396586	0.9074616	0.8757172	0.8443049	0.8131000	0.7820791
34	0.9734102	0.9414067	0.9101390	0.8793090	0.8487992	0.8184894	0.7883559
35	0.9741600	0.9430564	0.9126659	0.8826990	0.8530410	0.8235761	0.7942822
36	0.9748687	0.9446158	0.9150545	0.8859038	0.8570510	0.8283859	0.7998856
37	0.9755395	0.9460920	0.9173160	0.8889382	0.8608490	0.8329408	0.8051912
38	0.9761755	0.9474916	0.9194602	0.8918154	0.8644492	0.8372599	0.8102239
39	0.9767792	0.9488203	0.9214960	0.8945473	0.8678693	0.8413615	0.8150026
40	0.9773531	0.9500835	0.9234314	0.8971447	0.8711200	0.8452617	0.8195468
41	0.9778993	0.9512858	0.9252738	0.8996172	0.8742152	0.8489742	0.8238737
42	0.9784198	0.9524316	0.9270295	0.9019736	0.8771652	0.8525134	0.8279974
43	0.9789163	0.9535247	0.9287047	0.9042220	0.8799797	0.8558908	0.8319330
44	0.9793905	0.9545687	0.9303046	0.9063695	0.8826684	0.8591163	0.8356925
45	0.9798439	0.9555669	0.9318344	0.9084229	0.8852394	0.8622008	0.8392874
46	0.9802777	0.9565221	0.9332984	0.9103881	0.8877005	0.8651534	0.8427284
47	0.9806933	0.9574371	0.9347009	0.9122708	0.8900572	0.8679826	0.8460254
48	0.9810917	0.9583144	0.9360456	0.9140760	0.8923178	0.8706948	0.8491870
49	0.9814739	0.9591563	0.9373360	0.9158085	0.8944877	0.8732983	0.8522218
50	0.9818411	0.9599648	0.9385755	0.9174724	0.8965712	0.8757984	0.8551361

表 A.1 (续)

N/F	$\gamma=0.60$						
	0	1	2	3	4	5	6
51	0.9821940	0.9607419	0.9397668	0.9190719	0.8985740	0.8782024	0.8579384
52	0.9825334	0.9614895	0.9409128	0.9206105	0.9005011	0.8805150	0.8606335
53	0.9828601	0.9622091	0.9420160	0.9220918	0.9023561	0.8827412	0.8632294
54	0.9831748	0.9629023	0.9430788	0.9235187	0.9041436	0.8848865	0.8657294
55	0.9834782	0.9635706	0.9441033	0.9248944	0.9058665	0.8869538	0.8681403
56	0.9837708	0.9642152	0.9450916	0.9262214	0.9075284	0.8889487	0.8704660
57	0.9840532	0.9648373	0.9460456	0.9275024	0.9091332	0.8908745	0.8727111
58	0.9843260	0.9654383	0.9469669	0.9287396	0.9106826	0.8927348	0.8748792
59	0.9845896	0.9660190	0.9478574	0.9299354	0.9121804	0.8945324	0.8769759
60	0.9848445	0.9665805	0.9487184	0.9310916	0.9136292	0.8962713	0.8790027
61	0.9850911	0.9671238	0.9495515	0.9322104	0.9150303	0.8979533	0.8809640
62	0.9853298	0.9676497	0.9503579	0.9332933	0.9163871	0.8995821	0.8828625
63	0.9855610	0.9681590	0.9511390	0.9343423	0.9177010	0.9011589	0.8847015
64	0.9857850	0.9686526	0.9518958	0.9353587	0.9189739	0.9026874	0.8864833
65	0.9860021	0.9691310	0.9526296	0.9363442	0.9202085	0.9041693	0.8882117
66	0.9862127	0.9695951	0.9533413	0.9373000	0.9214064	0.9056072	0.8898880
67	0.9864171	0.9700455	0.9540319	0.9382276	0.9225681	0.9070019	0.8915142
68	0.9866155	0.9704827	0.9547024	0.9391282	0.9236963	0.9083564	0.8930936
69	0.9868082	0.9709073	0.9553537	0.9400028	0.9247922	0.9096723	0.8946280
70	0.9869954	0.9713199	0.9559864	0.9408527	0.9258569	0.9109501	0.8961182
75	0.9878571	0.9732189	0.9588990	0.9447649	0.9307589	0.9168351	0.9029807
80	0.9886117	0.9748820	0.9614501	0.9481916	0.9350528	0.9219905	0.9089923
85	0.9892780	0.9763507	0.9637030	0.9512181	0.9388447	0.9265436	0.9143028
90	0.9898706	0.9776571	0.9657071	0.9539104	0.9422189	0.9305951	0.9190278
95	0.9904012	0.9788267	0.9675015	0.9563212	0.9452402	0.9342225	0.9232591
100	0.9908789	0.9798799	0.9691174	0.9584922	0.9479611	0.9374904	0.9270700
110	0.9917047	0.9817005	0.9719108	0.9622455	0.9526649	0.9431391	0.9336591
120	0.9923933	0.9832190	0.9742408	0.9653762	0.9565893	0.9478516	0.9391559
130	0.9929764	0.9845048	0.9762138	0.9680276	0.9599126	0.9518434	0.9438115
140	0.9934764	0.9856075	0.9779061	0.9703017	0.9627634	0.9552666	0.9478056
160	0.9942895	0.9874008	0.9806583	0.9740003	0.9673999	0.9608356	0.9543025
180	0.9949224	0.9887968	0.9828008	0.9768797	0.9710093	0.9651715	0.9593611
200	0.9954290	0.9899142	0.9845160	0.9791850	0.9738991	0.9686435	0.9634108
250	0.9963415	0.9919273	0.9876059	0.9833382	0.9791068	0.9748986	0.9707093
300	0.9969504	0.9932704	0.9896678	0.9861097	0.9825815	0.9790726	0.9755799
350	0.9973855	0.9942304	0.9911415	0.9880907	0.9850653	0.9820571	0.9790618
400	0.9977119	0.9949507	0.9922473	0.9895772	0.9869295	0.9842960	0.9816747
450	0.9979659	0.9955111	0.9931076	0.9907337	0.9883801	0.9860389	0.9837076
500	0.9981691	0.9959595	0.9937961	0.9916593	0.9895406	0.9874330	0.9853346
550	0.9983354	0.9963265	0.9943595	0.9924167	0.9904899	0.9885741	0.9866666
600	0.9984740	0.9966323	0.9948291	0.9930480	0.9912821	0.9895254	0.9877764
650	0.9985913	0.9968912	0.9952266	0.9935823	0.9919521	0.9903297	0.9887157
700	0.9986919	0.9971131	0.9955672	0.9940403	0.9925259	0.9910203	0.9895208
750	0.9987790	0.9973054	0.9958625	0.9944373	0.9930244	0.9916186	0.9902189
800	0.9988553	0.9974737	0.9961210	0.9947848	0.9934600	0.9921421	0.9908295
850	0.9989226	0.9976222	0.9963490	0.9950913	0.9938441	0.9926035	0.9913683
900	0.9989824	0.9977543	0.9965517	0.9953638	0.9941856	0.9930143	0.9918475
950	0.9990359	0.9978724	0.9967331	0.9956077	0.9944920	0.9933814	0.9922762
1000	0.9990841	0.9979787	0.9968963	0.9958272	0.9947667	0.9937125	0.9926621

表 A.1 (续)

N/F	$\gamma=0.60$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0618573	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.1497525	0.0551777	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.2254856	0.1350421	0.0497998	0.0000000	0.0000000	0.0000000	0.0000000
11	0.2896125	0.2052105	0.1229638	0.0453769	0.0000000	0.0000000	0.0000000
12	0.3442709	0.2656144	0.1882839	0.1128698	0.0416755	0.0000000	0.0000000
13	0.3912919	0.3178249	0.2452949	0.1739382	0.1043065	0.0385323	0.0000000
14	0.4321186	0.3632824	0.2951612	0.2278670	0.1616257	0.0969522	0.0358299
15	0.4678735	0.4031622	0.3390262	0.2755204	0.2127539	0.1509418	0.0905663
16	0.4994304	0.4384044	0.3778568	0.3178150	0.2583347	0.1995227	0.1415828
17	0.5274805	0.4697579	0.4124449	0.3555509	0.2991070	0.2431696	0.1878419
18	0.5525711	0.4978243	0.4434329	0.3893995	0.3357383	0.2824822	0.2296885
19	0.5751446	0.5230883	0.4713463	0.4199149	0.3688015	0.3180224	0.2676117
20	0.5955585	0.5459461	0.4966144	0.4475574	0.3987758	0.3502788	0.3020862
21	0.6141081	0.5667233	0.5195933	0.4727082	0.4260657	0.3796704	0.3335330
22	0.6310357	0.5856903	0.5405768	0.4956845	0.4510089	0.4065515	0.3623169
23	0.6465451	0.6030722	0.5598122	0.5167543	0.4738925	0.4312245	0.3887525
24	0.6608056	0.6190588	0.5775089	0.5361443	0.4949576	0.4539455	0.4131089
25	0.6739630	0.6338112	0.5938425	0.5540447	0.5144101	0.4749350	0.4356171
26	0.6861399	0.6474663	0.6089643	0.5706210	0.5324288	0.4943815	0.4564777
27	0.6974412	0.6601425	0.6230040	0.5860135	0.5491634	0.5124468	0.4758621
28	0.7079585	0.6719398	0.6360730	0.6003442	0.5647466	0.5292727	0.4939200
29	0.7177702	0.6829479	0.6482683	0.6137192	0.5792921	0.5449810	0.5107821
30	0.7269446	0.6932420	0.6596744	0.6262304	0.5929002	0.5596790	0.5265623
31	0.7355426	0.7028901	0.6703658	0.6379582	0.6056589	0.5734611	0.5413610
32	0.7436160	0.7119502	0.6804065	0.6489742	0.6176433	0.5864093	0.5552669
33	0.7512115	0.7204751	0.6898555	0.6593411	0.6289236	0.5985969	0.5683565
34	0.7583704	0.7285108	0.6987619	0.6691136	0.6395584	0.6100886	0.5807005
35	0.7651294	0.7360972	0.7071725	0.6783432	0.6496016	0.6209425	0.5923604
36	0.7715204	0.7432719	0.7151264	0.6870716	0.6591016	0.6312094	0.6033912
37	0.7775737	0.7500675	0.7226598	0.6953406	0.6681014	0.6409371	0.6138422
38	0.7833145	0.7565127	0.7298056	0.7031838	0.6766384	0.6501647	0.6237581
39	0.7887665	0.7626341	0.7365933	0.7106341	0.6847482	0.6589317	0.6331785
40	0.7939510	0.7684558	0.7430484	0.7177196	0.6924617	0.6672702	0.6421390
41	0.7988876	0.7739992	0.7491954	0.7244672	0.6998076	0.6752114	0.6506739
42	0.8035928	0.7792827	0.7550551	0.7309000	0.7068112	0.6827827	0.6588115
43	0.8080837	0.7843258	0.7606473	0.7370398	0.7134956	0.6900101	0.6665796
44	0.8123732	0.7891431	0.7659902	0.7429051	0.7198825	0.6969157	0.6740015
45	0.8164762	0.7937504	0.7711002	0.7485156	0.7259912	0.7035210	0.6811020
46	0.8204026	0.7981606	0.7759912	0.7538866	0.7318393	0.7098446	0.6878994
47	0.8241652	0.8023866	0.7806782	0.7590323	0.7374429	0.7159046	0.6944136
48	0.8277731	0.8064393	0.7851734	0.7639685	0.7428178	0.7217166	0.7006613
49	0.8312368	0.8103289	0.7894882	0.7687062	0.7479768	0.7272963	0.7066600
50	0.8345632	0.8140657	0.7936331	0.7732578	0.7529335	0.7326566	0.7124224

表 A.1 (续)

N/F	$\gamma=0.60$						
	7	8	9	10	11	12	13
51	0.8377614	0.8176577	0.7976173	0.7776333	0.7576989	0.7378103	0.7179629
52	0.8408381	0.8211137	0.8014511	0.7818435	0.7622846	0.7427693	0.7232952
53	0.8437999	0.8244415	0.8051432	0.7858976	0.7666994	0.7475446	0.7284287
54	0.8466547	0.8276479	0.8086999	0.7898037	0.7709533	0.7521456	0.7333761
55	0.8494061	0.8307390	0.8121292	0.7935697	0.7750553	0.7565820	0.7381461
56	0.8520610	0.8337209	0.8154373	0.7972033	0.7790135	0.7608626	0.7427491
57	0.8546233	0.8366003	0.8186315	0.8007115	0.7828342	0.7649959	0.7471926
58	0.8570992	0.8393810	0.8217170	0.8041002	0.7865254	0.7689880	0.7514857
59	0.8594919	0.8420690	0.8246995	0.8073757	0.7900932	0.7728472	0.7556356
60	0.8618058	0.8446692	0.8275836	0.8105439	0.7935438	0.7765796	0.7596490
61	0.8640449	0.8471845	0.8303745	0.8136088	0.7968827	0.7801917	0.7635335
62	0.8662124	0.8496195	0.8330762	0.8165764	0.8001155	0.7836889	0.7672936
63	0.8683112	0.8519781	0.8356939	0.8194516	0.8032467	0.7870762	0.7709369
64	0.8703464	0.8542644	0.8382305	0.8222377	0.8062816	0.7903591	0.7744670
65	0.8723190	0.8564812	0.8406899	0.8249391	0.8092249	0.7935427	0.7778903
66	0.8742330	0.8586313	0.8430755	0.8275600	0.8120797	0.7966314	0.7812121
67	0.8760899	0.8607174	0.8453907	0.8301030	0.8148503	0.7996281	0.7844350
68	0.8778926	0.8627434	0.8476385	0.8325725	0.8175401	0.8025390	0.7875653
69	0.8796446	0.8647117	0.8498222	0.8349709	0.8201531	0.8053658	0.7906053
70	0.8813465	0.8666243	0.8519442	0.8373022	0.8226930	0.8081128	0.7935600
75	0.8891812	0.8754282	0.8617142	0.8480345	0.8343853	0.8207628	0.8071647
80	0.8960460	0.8831424	0.8702747	0.8574390	0.8446315	0.8318485	0.8190883
85	0.9021101	0.8899571	0.8778376	0.8657479	0.8536841	0.8416432	0.8296229
90	0.9075054	0.8960205	0.8845668	0.8731414	0.8617396	0.8503599	0.8389993
95	0.9123375	0.9014511	0.8905944	0.8797630	0.8689551	0.8581671	0.8473974
100	0.9166893	0.9063422	0.8960233	0.8857280	0.8754549	0.8652001	0.8549628
110	0.9242142	0.9147998	0.9054105	0.8960434	0.8866947	0.8773634	0.8680467
120	0.9304930	0.9218568	0.9132434	0.9046499	0.8960740	0.8875133	0.8789663
130	0.9358102	0.9278341	0.9198786	0.9119414	0.9040194	0.8961120	0.8882167
140	0.9403720	0.9329621	0.9255706	0.9181965	0.9108362	0.9034889	0.8961530
160	0.9477931	0.9413036	0.9348308	0.9283725	0.9219264	0.9154918	0.9090663
180	0.9535713	0.9477991	0.9420421	0.9362973	0.9305633	0.9248391	0.9191233
200	0.9581980	0.9530004	0.9478157	0.9426425	0.9374793	0.9323244	0.9271779
250	0.9665346	0.9623729	0.9582218	0.9540792	0.9499441	0.9458159	0.9416939
300	0.9720993	0.9686294	0.9651671	0.9617132	0.9582647	0.9548223	0.9513852
350	0.9760773	0.9731011	0.9701326	0.9671709	0.9642138	0.9612613	0.9583134
400	0.9790625	0.9764580	0.9738596	0.9712665	0.9686780	0.9660941	0.9635140
450	0.9813854	0.9790693	0.9767594	0.9744540	0.9721524	0.9698547	0.9675607
500	0.9832438	0.9811591	0.9790798	0.9770043	0.9749325	0.9728646	0.9707990
550	0.9847653	0.9828700	0.9809794	0.9790918	0.9772087	0.9753280	0.9734495
600	0.9860334	0.9842958	0.9825620	0.9808320	0.9791050	0.9773812	0.9756596
650	0.9871063	0.9855022	0.9839020	0.9823048	0.9807106	0.9791187	0.9775292
700	0.9880266	0.9865370	0.9850504	0.9835677	0.9820872	0.9806083	0.9791325
750	0.9888245	0.9874332	0.9860464	0.9846620	0.9832798	0.9819000	0.9805224
800	0.9895223	0.9882181	0.9869177	0.9856197	0.9843239	0.9830304	0.9817385
850	0.9901377	0.9889109	0.9876863	0.9864648	0.9852449	0.9840272	0.9828111
900	0.9906852	0.9895260	0.9883699	0.9872161	0.9860638	0.9849138	0.9837653
950	0.9911755	0.9900770	0.9889817	0.9878886	0.9867971	0.9857071	0.9846193
1000	0.9916164	0.9905729	0.9895324	0.9884935	0.9874569	0.9864210	0.9853875

表 A.1 (续)

N/F	$\gamma=0.60$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0334817	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0849697	0.0314223	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.1333175	0.0800246	0.0296016	0.0000000	0.0000000	0.0000000	0.0000000
18	0.1774540	0.1259642	0.0756233	0.0279803	0.0000000	0.0000000	0.0000000
19	0.2176245	0.1681560	0.1193796	0.0716812	0.0265274	0.0000000	0.0000000
20	0.2542308	0.2067661	0.1597835	0.1134502	0.0681301	0.0252179	0.0000000
21	0.2876740	0.2421256	0.1969411	0.1522067	0.1080812	0.0649139	0.0240316
22	0.3183181	0.2745763	0.2311222	0.1880076	0.1453157	0.1031982	0.0619872
23	0.3464843	0.3044335	0.2626207	0.2210765	0.1798497	0.1390211	0.0987373
24	0.3724508	0.3319812	0.2917123	0.2516638	0.2118678	0.1723710	0.1332502
25	0.3964594	0.3574657	0.3186460	0.2800124	0.2415863	0.2033968	0.1654895
26	0.4187173	0.3811041	0.3436420	0.3063417	0.2692161	0.2322850	0.1955774
27	0.4394072	0.4030843	0.3668972	0.3308504	0.2949548	0.2592231	0.2236746
28	0.4586864	0.4235725	0.3885808	0.3537133	0.3189787	0.2843851	0.2499457
29	0.4766930	0.4427130	0.4088437	0.3750872	0.3414467	0.3079306	0.2745480
30	0.4935479	0.4606334	0.4278196	0.3951081	0.3625011	0.3300047	0.2976243
31	0.5093562	0.4774438	0.4456244	0.4138981	0.3822672	0.3507346	0.3193051
32	0.5242122	0.4932453	0.4623630	0.4315669	0.4008579	0.3702373	0.3397098
33	0.5381999	0.5081235	0.4781272	0.4482094	0.4183725	0.3886158	0.3589437
34	0.5513910	0.5221570	0.4929978	0.4639118	0.4348998	0.4059626	0.3771016
35	0.5638523	0.5354159	0.5070489	0.4787514	0.4505217	0.4223608	0.3942708
36	0.5756424	0.5479615	0.5203462	0.4927950	0.4653087	0.4378857	0.4105268
37	0.5868144	0.5598499	0.5329480	0.5061064	0.4793250	0.4526031	0.4259408
38	0.5974141	0.5711311	0.5449069	0.5187398	0.4926293	0.4665744	0.4405753
39	0.6074856	0.5818507	0.5562707	0.5307457	0.5052740	0.4798542	0.4544871
40	0.6170665	0.5920482	0.5670833	0.5421695	0.5173068	0.4924930	0.4677280
41	0.6261913	0.6017622	0.5773826	0.5530526	0.5287700	0.5045346	0.4803450
42	0.6348929	0.6110246	0.5872051	0.5634322	0.5397043	0.5160207	0.4923813
43	0.6431994	0.6198681	0.5965825	0.5733412	0.5501441	0.5269883	0.5038744
44	0.6511369	0.6283181	0.6055443	0.5828124	0.5601218	0.5374723	0.5148618
45	0.6587295	0.6364019	0.6141172	0.5918736	0.5696689	0.5475024	0.5253740
46	0.6659993	0.6441427	0.6223265	0.6005500	0.5788109	0.5571091	0.5354425
47	0.6729662	0.6515606	0.6301948	0.6088663	0.5875737	0.5663169	0.5450945
48	0.6796487	0.6586765	0.6377417	0.6168435	0.5959804	0.5751509	0.5543542
49	0.6860641	0.6655080	0.6449884	0.6245032	0.6040523	0.5836335	0.5632459
50	0.6922286	0.6720714	0.6519501	0.6318631	0.6118082	0.5917845	0.5717914

表 A.1 (续)

N/F	$\gamma=0.60$						
	14	15	16	17	18	19	20
51	0.6981552	0.6783833	0.6586457	0.6389402	0.6192668	0.5996231	0.5800091
52	0.7038584	0.6844567	0.6650879	0.6457518	0.6264447	0.6071674	0.5879184
53	0.7093502	0.6903053	0.6712924	0.6523108	0.6333582	0.6144330	0.5955354
54	0.7146417	0.6959408	0.6772712	0.6586314	0.6400198	0.6214357	0.6028775
55	0.7197446	0.7013758	0.6830369	0.6647276	0.6464451	0.6281885	0.6099579
56	0.7246684	0.7066198	0.6886009	0.6706095	0.6526448	0.6347052	0.6167909
57	0.7294229	0.7116830	0.6939728	0.6762893	0.6586310	0.6409979	0.6233892
58	0.7340155	0.7165749	0.6991627	0.6817763	0.6644152	0.6470777	0.6297638
59	0.7384553	0.7213040	0.7041793	0.6870806	0.6700064	0.6529557	0.6359272
60	0.7427488	0.7258769	0.7090317	0.6922117	0.6754154	0.6586411	0.6418890
61	0.7469043	0.7303034	0.7137283	0.6971770	0.6806493	0.6641437	0.6476594
62	0.7509280	0.7345884	0.7182748	0.7019848	0.6857176	0.6694719	0.6532467
63	0.7548251	0.7387400	0.7226800	0.7066430	0.6906281	0.6746337	0.6586592
64	0.7586025	0.7427638	0.7269496	0.7111576	0.6953868	0.6796367	0.6639064
65	0.7622654	0.7466656	0.7310895	0.7155355	0.7000021	0.6844894	0.6689949
66	0.7658195	0.7504513	0.7351060	0.7197828	0.7044802	0.6891967	0.6739315
67	0.7692678	0.7541249	0.7390043	0.7239057	0.7088262	0.6937658	0.6787237
68	0.7726168	0.7576927	0.7427900	0.7279087	0.7130464	0.6982024	0.6833767
69	0.7758699	0.7611582	0.7464671	0.7317973	0.7171459	0.7025128	0.6878972
70	0.7790317	0.7645255	0.7500414	0.7355764	0.7211305	0.7067021	0.6922905
75	0.7935895	0.7800342	0.7664987	0.7529807	0.7394803	0.7259951	0.7125252
80	0.8063480	0.7936275	0.7809238	0.7682376	0.7555660	0.7429088	0.7302654
85	0.8176225	0.8056388	0.7936719	0.7817195	0.7697816	0.7578567	0.7459448
90	0.8276563	0.8163293	0.8050183	0.7937202	0.7824352	0.7711631	0.7599024
95	0.8366437	0.8259061	0.8151814	0.8044705	0.7937717	0.7830837	0.7724071
100	0.8447408	0.8345333	0.8243387	0.8141564	0.8039855	0.7938245	0.7836742
110	0.8587444	0.8494544	0.8401766	0.8309094	0.8216514	0.8124034	0.8031644
120	0.8704316	0.8619083	0.8533949	0.8448922	0.8363979	0.8279120	0.8194337
130	0.8803322	0.8724590	0.8645951	0.8567395	0.8488923	0.8410527	0.8332201
140	0.8888278	0.8815118	0.8742049	0.8669056	0.8596140	0.8523285	0.8450506
160	0.9026500	0.8962421	0.8898418	0.8834476	0.8770603	0.8706790	0.8643024
180	0.9134159	0.9077154	0.9020210	0.8963327	0.8906505	0.8849729	0.8792998
200	0.9220375	0.9169040	0.9117758	0.9066530	0.9015348	0.8964218	0.8913127
250	0.9375771	0.9334658	0.9293582	0.9252552	0.9211560	0.9170599	0.9129676
300	0.9479519	0.9445232	0.9410975	0.9376757	0.9342569	0.9308412	0.9274277
350	0.9553693	0.9524290	0.9494910	0.9465568	0.9436241	0.9406945	0.9377671
400	0.9609369	0.9583629	0.9557912	0.9532218	0.9506554	0.9480905	0.9455280
450	0.9652691	0.9629805	0.9606934	0.9584094	0.9561269	0.9538467	0.9515680
500	0.9687365	0.9666754	0.9646175	0.9625610	0.9605068	0.9584534	0.9564023
550	0.9715741	0.9697003	0.9678287	0.9659594	0.9640908	0.9622246	0.9603591
600	0.9739395	0.9722217	0.9705062	0.9687915	0.9670790	0.9653674	0.9636572
650	0.9759419	0.9743561	0.9727719	0.9711892	0.9696080	0.9680276	0.9664487
700	0.9776581	0.9761853	0.9747140	0.9732442	0.9717760	0.9703085	0.9688425
750	0.9791455	0.9777710	0.9763980	0.9750258	0.9736551	0.9722859	0.9709167
800	0.9804480	0.9791591	0.9778717	0.9765851	0.9753000	0.9740157	0.9727321
850	0.9815972	0.9803841	0.9791718	0.9779610	0.9767517	0.9755424	0.9743347
900	0.9826183	0.9814729	0.9803282	0.9791843	0.9780419	0.9769002	0.9757594
950	0.9835324	0.9824469	0.9813623	0.9802791	0.9791960	0.9781143	0.9770335
1000	0.9843554	0.9833234	0.9822936	0.9812639	0.9802356	0.9792082	0.9781807

表 A.1 (续)

N/F	$\gamma=0.70$						
	0	1	2	3	4	5	6
1	0.3000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.5477226	0.1633400	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.6694330	0.3625311	0.1120960	0.0000000	0.0000000	0.0000000	0.0000000
4	0.7400828	0.4910948	0.2715987	0.0853088	0.0000000	0.0000000	0.0000000
5	0.7860031	0.5776734	0.3890819	0.2174021	0.0688501	0.0000000	0.0000000
6	0.8181888	0.6394341	0.4755147	0.3225376	0.1817475	0.0577134	0.0000000
7	0.8419824	0.6855756	0.5410544	0.4045868	0.2763493	0.1558577	0.0496772
8	0.8602807	0.7213114	0.5922563	0.4695665	0.3530078	0.2413280	0.1364345
9	0.8747871	0.7497858	0.6332872	0.5220488	0.4156450	0.3127614	0.2142136
10	0.8865682	0.7729991	0.6668721	0.5652262	0.4675426	0.3726032	0.2808048
11	0.8963251	0.7922816	0.6948540	0.6013286	0.5111435	0.4232053	0.3377115
12	0.9045379	0.8085514	0.7185190	0.6319420	0.5482420	0.4664452	0.3866397
13	0.9115459	0.8224620	0.7387902	0.6582184	0.5801666	0.5037669	0.4290434
14	0.9175960	0.8344909	0.7563461	0.6810118	0.6079163	0.5362794	0.4660876
15	0.9228719	0.8449953	0.7716965	0.7009679	0.6322510	0.5648405	0.4986980
16	0.9275132	0.8542473	0.7852313	0.7185829	0.6537586	0.5901187	0.5276065
17	0.9316278	0.8624580	0.7972540	0.7342443	0.6729025	0.6126431	0.5533999
18	0.9353005	0.8697938	0.8080040	0.7482591	0.6900500	0.6328373	0.5765478
19	0.9385990	0.8763873	0.8176731	0.7608731	0.7054950	0.6510415	0.5974334
20	0.9415775	0.8823457	0.8264162	0.7722859	0.7194797	0.6675342	0.6163692
21	0.9442805	0.8877564	0.8343600	0.7826609	0.7322002	0.6825449	0.6336136
22	0.9467445	0.8926916	0.8416094	0.7921332	0.7438200	0.6962643	0.6493830
23	0.9489998	0.8972113	0.8482513	0.8008155	0.7544755	0.7088503	0.6638576
24	0.9510720	0.9013658	0.8543591	0.8088025	0.7642823	0.7204382	0.6771884
25	0.9529823	0.9051977	0.8599946	0.8161745	0.7733370	0.7311412	0.6895070
26	0.9547491	0.9087431	0.8652105	0.8229997	0.7817232	0.7410575	0.7009221
27	0.9563880	0.9120330	0.8700520	0.8293368	0.7895116	0.7502697	0.7115313
28	0.9579123	0.9150940	0.8745579	0.8352362	0.7967649	0.7588503	0.7214148
29	0.9593337	0.9179492	0.8787621	0.8407417	0.8035351	0.7668619	0.7306457
30	0.9606622	0.9206186	0.8826936	0.8458914	0.8098692	0.7743592	0.7392856
31	0.9619067	0.9231199	0.8863784	0.8507188	0.8158083	0.7813896	0.7473897
32	0.9630748	0.9254684	0.8898387	0.8552532	0.8213886	0.7879962	0.7550060
33	0.9641735	0.9276778	0.8930947	0.8595204	0.8266405	0.7942158	0.7621778
34	0.9652087	0.9297599	0.8961637	0.8635434	0.8315926	0.8000820	0.7689422
35	0.9661857	0.9317255	0.8990616	0.8673425	0.8362700	0.8056226	0.7753331
36	0.9671094	0.9335842	0.9018021	0.8709358	0.8406943	0.8108650	0.7813805
37	0.9679839	0.9353443	0.9043978	0.8743398	0.8448868	0.8158318	0.7871111
38	0.9688132	0.9370136	0.9068599	0.8775689	0.8488639	0.8205454	0.7925496
39	0.9696005	0.9385988	0.9091983	0.8806362	0.8526425	0.8250240	0.7977174
40	0.9703492	0.9401063	0.9114223	0.8835537	0.8562374	0.8292842	0.8026340
41	0.9710618	0.9415415	0.9135399	0.8863321	0.8596606	0.8333424	0.8073179
42	0.9717410	0.9429095	0.9155587	0.8889810	0.8629249	0.8372119	0.8117849
43	0.9723890	0.9442150	0.9174854	0.8915093	0.8660408	0.8409066	0.8160494
44	0.9730080	0.9454621	0.9193261	0.8939250	0.8690178	0.8444366	0.8201255
45	0.9735998	0.9466547	0.9210865	0.8962355	0.8718658	0.8478141	0.8240248
46	0.9741662	0.9477962	0.9227718	0.8984476	0.8745932	0.8510482	0.8277588
47	0.9747089	0.9488900	0.9243865	0.9005673	0.8772063	0.8541474	0.8313373
48	0.9752292	0.9499388	0.9259352	0.9026004	0.8797133	0.8571206	0.8347705
49	0.9757285	0.9509455	0.9274217	0.9045520	0.8821197	0.8599750	0.8380669
50	0.9762081	0.9519125	0.9288497	0.9064270	0.8844313	0.8627178	0.8412346

表 A.1 (续)

N/F	$\gamma=0.70$						
	0	1	2	3	4	5	6
51	0.9766692	0.9528421	0.9302226	0.9082298	0.8866546	0.8653545	0.8442811
52	0.9771127	0.9537364	0.9315436	0.9099644	0.8887938	0.8678930	0.8472126
53	0.9775396	0.9545974	0.9328154	0.9116347	0.8908540	0.8703366	0.8500359
54	0.9779509	0.9554270	0.9340409	0.9132441	0.8928391	0.8726925	0.8527564
55	0.9783474	0.9562269	0.9352225	0.9147960	0.8947528	0.8749634	0.8553809
56	0.9787300	0.9569985	0.9363625	0.9162933	0.8966000	0.8771552	0.8579126
57	0.9790992	0.9577434	0.9374631	0.9177390	0.8983834	0.8792717	0.8603576
58	0.9794558	0.9584629	0.9385262	0.9191355	0.9001068	0.8813159	0.8627203
59	0.9798005	0.9591583	0.9395538	0.9204855	0.9017716	0.8832927	0.8650039
60	0.9801338	0.9598309	0.9405476	0.9217911	0.9033829	0.8852041	0.8672130
61	0.9804562	0.9604816	0.9415093	0.9230545	0.9049420	0.8870547	0.8693513
62	0.9807684	0.9611116	0.9424404	0.9242778	0.9064513	0.8888460	0.8714215
63	0.9810708	0.9617219	0.9433423	0.9254628	0.9079133	0.8905815	0.8734268
64	0.9813638	0.9623132	0.9442163	0.9266113	0.9093303	0.8922641	0.8753711
65	0.9816478	0.9628866	0.9450639	0.9277249	0.9107046	0.8938956	0.8772567
66	0.9819234	0.9634428	0.9458860	0.9288053	0.9120383	0.8954784	0.8790856
67	0.9821907	0.9639826	0.9466839	0.9298538	0.9133320	0.8970142	0.8808613
68	0.9824503	0.9645066	0.9474586	0.9308719	0.9145884	0.8985061	0.8825849
69	0.9827025	0.9650157	0.9482112	0.9318609	0.9158094	0.8999548	0.8842604
70	0.9829475	0.9655103	0.9489424	0.9328219	0.9169953	0.9013630	0.8858879
75	0.9840752	0.9677876	0.9523095	0.9372475	0.9224582	0.9078492	0.8933852
80	0.9850630	0.9697827	0.9552600	0.9411261	0.9272464	0.9135349	0.8999591
85	0.9859355	0.9715452	0.9578667	0.9445531	0.9314777	0.9185602	0.9057686
90	0.9867116	0.9731134	0.9601863	0.9476032	0.9352441	0.9230336	0.9109414
95	0.9874066	0.9745177	0.9622639	0.9503352	0.9386179	0.9270410	0.9155755
100	0.9880325	0.9757827	0.9641355	0.9527964	0.9416577	0.9306514	0.9197512
110	0.9891145	0.9779699	0.9673719	0.9570531	0.9469153	0.9368975	0.9269750
120	0.9900171	0.9797947	0.9700726	0.9606056	0.9513044	0.9421120	0.9330066
130	0.9907814	0.9813404	0.9723603	0.9636153	0.9550221	0.9465302	0.9381170
140	0.9914371	0.9826663	0.9743232	0.9661978	0.9582133	0.9503212	0.9425039
160	0.9925034	0.9848233	0.9775166	0.9703998	0.9634052	0.9564922	0.9496425
180	0.9933336	0.9865029	0.9800035	0.9736725	0.9674502	0.9612988	0.9552046
200	0.9939982	0.9878477	0.9819951	0.9762936	0.9706894	0.9651493	0.9596602
250	0.9951957	0.9902712	0.9855844	0.9810182	0.9765296	0.9720917	0.9676942
300	0.9959948	0.9918888	0.9879805	0.9841725	0.9804284	0.9767274	0.9730591
350	0.9965660	0.9930451	0.9896936	0.9864279	0.9832174	0.9800422	0.9768959
400	0.9969946	0.9939129	0.9909793	0.9881206	0.9853098	0.9825305	0.9797765
450	0.9973281	0.9945882	0.9919798	0.9894380	0.9869389	0.9844672	0.9820184
500	0.9975950	0.9951286	0.9927805	0.9904923	0.9882421	0.9860178	0.9838126
550	0.9978134	0.9955709	0.9934359	0.9913553	0.9893096	0.9872862	0.9852811
600	0.9979954	0.9959395	0.9939822	0.9920746	0.9901987	0.9883444	0.9865061
650	0.9981494	0.9962515	0.9944445	0.9926834	0.9909519	0.9892395	0.9875423
700	0.9982815	0.9965190	0.9948409	0.9932054	0.9915974	0.9900071	0.9884305
750	0.9983960	0.9967508	0.9951844	0.9936578	0.9921569	0.9906725	0.9892011
800	0.9984962	0.9969537	0.9954851	0.9940538	0.9926459	0.9912548	0.9898751
850	0.9985846	0.9971328	0.9957504	0.9944032	0.9930781	0.9917681	0.9904696
900	0.9986631	0.9972919	0.9959863	0.9947138	0.9934624	0.9922254	0.9909983
950	0.9987335	0.9974343	0.9961974	0.9949918	0.9938062	0.9926337	0.9914720
1000	0.9987968	0.9975625	0.9963874	0.9952420	0.9941159	0.9930014	0.9918977

表 A.1 (续)

N/F	$\gamma=0.70$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0436051	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.1213208	0.0388555	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.1925922	0.1092243	0.0350389	0.0000000	0.0000000	0.0000000	0.0000000
11	0.2548011	0.1749447	0.0993234	0.0319049	0.0000000	0.0000000	0.0000000
12	0.3088362	0.2332223	0.1602649	0.0910696	0.0292855	0.0000000	0.0000000
13	0.3559442	0.2845348	0.2150237	0.1478617	0.0840825	0.0270636	0.0000000
14	0.3972583	0.3297963	0.2637945	0.1994658	0.1372429	0.0780915	0.0251550
15	0.4337258	0.3698988	0.3072490	0.2458833	0.1860121	0.1280483	0.0728982
16	0.4661197	0.4056140	0.3460910	0.2876018	0.2302570	0.1742618	0.1200099
17	0.4950662	0.4375900	0.3809554	0.3251813	0.2703266	0.2165026	0.1639100
18	0.5210770	0.4663652	0.4123874	0.3591451	0.3066665	0.2550156	0.2043025
19	0.5445676	0.4923838	0.4408493	0.3899550	0.3397122	0.2901560	0.2413513
20	0.5658832	0.5160152	0.4667293	0.4180087	0.3698551	0.3222858	0.2753391
21	0.5853087	0.5375683	0.4903544	0.4436463	0.3974387	0.3517384	0.3065684
22	0.6030824	0.5573013	0.5120009	0.4671574	0.4227619	0.3788134	0.3353241
23	0.6194043	0.5754332	0.5319031	0.4887903	0.4460810	0.4037716	0.3618665
24	0.6344445	0.5921484	0.5502605	0.5087556	0.4676191	0.4268421	0.3864266
25	0.6483473	0.6076058	0.5672441	0.5272373	0.4875669	0.4482253	0.4092087
26	0.6612353	0.6219414	0.5830014	0.5443911	0.5060921	0.4680937	0.4303913
27	0.6732163	0.6352714	0.5976591	0.5603543	0.5233386	0.4865999	0.4501328
28	0.6843814	0.6476974	0.6113270	0.5752450	0.5394323	0.5038767	0.4685706
29	0.6948118	0.6593090	0.6241022	0.5891663	0.5544836	0.5200398	0.4858286
30	0.7045767	0.6701821	0.6360683	0.6022101	0.5685891	0.5351932	0.5020132
31	0.7137370	0.6803849	0.6472991	0.6144551	0.5818354	0.5494270	0.5172200
32	0.7223491	0.6899776	0.6578609	0.6259731	0.5942974	0.5628208	0.5315341
33	0.7304588	0.6990136	0.6678103	0.6368260	0.6060423	0.5754471	0.5450311
34	0.7381091	0.7075384	0.6771997	0.6470692	0.6171302	0.5873691	0.5577780
35	0.7453380	0.7155955	0.6860742	0.6567528	0.6276137	0.5986440	0.5698346
36	0.7521791	0.7232211	0.6944759	0.6659207	0.6375402	0.6093222	0.5812552
37	0.7586635	0.7304501	0.7024404	0.6746138	0.6469543	0.6194497	0.5920892
38	0.7648178	0.7373111	0.7100012	0.6828675	0.6558933	0.6290671	0.6023798
39	0.7706665	0.7438322	0.7171887	0.6907138	0.6643922	0.6382133	0.6121657
40	0.7762309	0.7500385	0.7240291	0.6981822	0.6724834	0.6469203	0.6214847
41	0.7815330	0.7559518	0.7305469	0.7052991	0.6801940	0.6552201	0.6303675
42	0.7865899	0.7615917	0.7367644	0.7120897	0.6875516	0.6631401	0.6388454
43	0.7914180	0.7669773	0.7427022	0.7185744	0.6945793	0.6707055	0.6469438
44	0.7960326	0.7721258	0.7483785	0.7247746	0.7012981	0.6779391	0.6546885
45	0.8004483	0.7770512	0.7538104	0.7307078	0.7077287	0.6848626	0.6621017
46	0.8046761	0.7817689	0.7590128	0.7363909	0.7138888	0.6914959	0.6692051
47	0.8087294	0.7862916	0.7640011	0.7418395	0.7197946	0.6978565	0.6760161
48	0.8126181	0.7906304	0.7687862	0.7470678	0.7254632	0.7039607	0.6825537
49	0.8163519	0.7947970	0.7733818	0.7520893	0.7309068	0.7098234	0.6888324
50	0.8199399	0.7988016	0.7777991	0.7569156	0.7361389	0.7154592	0.6948694

表 A.1 (续)

N/F	$\gamma=0.70$						
	7	8	9	10	11	12	13
51	0.8233900	0.8026522	0.7820472	0.7615574	0.7411721	0.7208807	0.7006762
52	0.8267108	0.8063593	0.7861359	0.7660255	0.7460173	0.7260998	0.7062671
53	0.8299091	0.8099288	0.7900744	0.7703299	0.7506845	0.7311277	0.7116532
54	0.8329920	0.8133695	0.7938706	0.7744785	0.7551833	0.7359744	0.7168455
55	0.8359646	0.8166887	0.7975319	0.7784803	0.7595233	0.7406503	0.7218551
56	0.8388339	0.8198911	0.8010658	0.7823435	0.7637120	0.7451636	0.7266908
57	0.8416045	0.8229841	0.8044782	0.7860737	0.7677578	0.7495227	0.7313608
58	0.8442811	0.8259723	0.8077765	0.7896784	0.7716672	0.7537353	0.7358752
59	0.8468691	0.8288618	0.8109650	0.7931637	0.7754478	0.7578082	0.7402395
60	0.8493722	0.8316572	0.8140491	0.7965348	0.7791044	0.7617496	0.7444627
61	0.8517951	0.8343626	0.8170345	0.7997988	0.7826447	0.7655646	0.7485509
62	0.8541412	0.8369822	0.8199262	0.8029594	0.7860736	0.7692594	0.7525109
63	0.8564147	0.8395201	0.8227270	0.8060224	0.7893956	0.7728397	0.7563480
64	0.8586177	0.8419803	0.8254428	0.8089916	0.7926166	0.7763111	0.7600681
65	0.8607544	0.8443673	0.8280763	0.8118707	0.7957407	0.7796778	0.7636768
66	0.8628278	0.8466822	0.8306319	0.8146648	0.7987717	0.7829450	0.7671786
67	0.8648403	0.8489293	0.8331128	0.8173771	0.8017148	0.7861165	0.7705777
68	0.8667949	0.8511117	0.8355215	0.8200115	0.8045724	0.7891967	0.7738797
69	0.8686934	0.8532325	0.8378624	0.8225708	0.8073494	0.7921899	0.7770875
70	0.8705387	0.8552932	0.8401369	0.8250585	0.8100480	0.7950993	0.7802063
75	0.8790379	0.8647868	0.8506172	0.8365194	0.8224842	0.8085047	0.7945770
80	0.8864910	0.8731121	0.8598087	0.8465717	0.8333927	0.8202655	0.8071849
85	0.8930784	0.8804715	0.8679354	0.8554597	0.8430388	0.8306653	0.8183360
90	0.8989437	0.8870247	0.8751712	0.8633750	0.8516299	0.8399290	0.8282686
95	0.9041993	0.8928963	0.8816558	0.8704688	0.8593291	0.8482314	0.8371717
100	0.9089348	0.8981879	0.8874997	0.8768627	0.8662691	0.8557160	0.8451979
110	0.9171287	0.9073443	0.8976133	0.8879265	0.8782802	0.8686690	0.8590898
120	0.9239698	0.9149903	0.9060581	0.8971670	0.8883126	0.8794895	0.8706946
130	0.9297678	0.9214704	0.9132166	0.9050002	0.8968165	0.8886626	0.8805339
140	0.9347438	0.9270326	0.9193610	0.9117245	0.9041178	0.8965378	0.8889822
160	0.9428440	0.9360866	0.9293643	0.9226718	0.9160053	0.9093616	0.9027385
180	0.9491554	0.9431421	0.9371593	0.9312032	0.9252700	0.9193574	0.9134623
200	0.9542108	0.9487943	0.9434052	0.9380397	0.9326949	0.9273676	0.9220563
250	0.9633280	0.9589877	0.9546696	0.9503697	0.9460859	0.9418166	0.9375595
300	0.9694175	0.9657966	0.9621939	0.9586073	0.9550329	0.9514707	0.9479184
350	0.9737725	0.9706666	0.9675768	0.9644991	0.9614329	0.9583767	0.9553295
400	0.9770415	0.9743225	0.9716173	0.9689228	0.9662381	0.9635626	0.9608940
450	0.9795864	0.9771688	0.9747627	0.9723666	0.9699788	0.9675994	0.9652261
500	0.9816233	0.9794463	0.9772800	0.9751229	0.9729733	0.9708306	0.9686941
550	0.9832905	0.9813106	0.9793413	0.9773790	0.9754250	0.9734763	0.9715330
600	0.9846799	0.9828653	0.9810590	0.9792603	0.9774685	0.9756813	0.9739002
650	0.9858566	0.9841808	0.9825134	0.9808529	0.9791977	0.9775486	0.9759033
700	0.9868654	0.9853094	0.9837603	0.9822181	0.9806812	0.9791488	0.9776211
750	0.9877396	0.9862872	0.9848410	0.9834016	0.9819668	0.9805366	0.9791102
800	0.9885053	0.9871431	0.9857871	0.9844371	0.9830918	0.9817510	0.9804140
850	0.9891802	0.9878984	0.9866220	0.9853517	0.9840852	0.9828233	0.9815644
900	0.9897804	0.9885693	0.9873644	0.9861640	0.9849682	0.9837755	0.9825865
950	0.9903179	0.9891707	0.9880288	0.9868915	0.9857580	0.9846283	0.9835024
1000	0.9908015	0.9897115	0.9886261	0.9875460	0.9864697	0.9853964	0.9843255

表 A.1 (续)

N/F	$\gamma=0.70$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0234979	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0683521	0.0220456	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.1129211	0.0643408	0.0207623	0.0000000	0.0000000	0.0000000	0.0000000
18	0.1547207	0.1066244	0.0607735	0.0196202	0.0000000	0.0000000	0.0000000
19	0.1934072	0.1465084	0.1009928	0.0575813	0.0185973	0.0000000	0.0000000
20	0.2290805	0.1836170	0.1391246	0.0959270	0.0547077	0.0176757	0.0000000
21	0.2619668	0.2180007	0.1747716	0.1324504	0.0913446	0.0521079	0.0168411
22	0.2923186	0.2498373	0.2079449	0.1667399	0.1263879	0.0871811	0.0497435
23	0.3203803	0.2793396	0.2387842	0.1987773	0.1594159	0.1208563	0.0833801
24	0.3463797	0.3067180	0.2674676	0.2286703	0.1903858	0.1527085	0.1157888
25	0.3705202	0.3321712	0.2941777	0.2565673	0.2193794	0.1826753	0.1465427
26	0.3929849	0.3558798	0.3190868	0.2826257	0.2465224	0.2108166	0.1755655
27	0.4139342	0.3780065	0.3423580	0.3069994	0.2719498	0.2372367	0.2028974
28	0.4335117	0.3986984	0.3641354	0.3298310	0.2957975	0.2620531	0.2286261
29	0.4518433	0.4180838	0.3845517	0.3512515	0.3181916	0.2853865	0.2528539
30	0.4690430	0.4362804	0.4037253	0.3713807	0.3392520	0.3073500	0.2756860
31	0.4852092	0.4533898	0.4217619	0.3903263	0.3590868	0.3280494	0.2972249
32	0.5004306	0.4695049	0.4387569	0.4081859	0.3777934	0.3475849	0.3175662
33	0.5147868	0.4847089	0.4547957	0.4250466	0.3954622	0.3660450	0.3368001
34	0.5283479	0.4990750	0.4699555	0.4409885	0.4121733	0.3835123	0.3550084
35	0.5411785	0.5126696	0.4843049	0.4560822	0.4280013	0.4000623	0.3722676
36	0.5533340	0.5255524	0.4979059	0.4703929	0.4430118	0.4157627	0.3886472
37	0.5648669	0.5377765	0.5108144	0.4839781	0.4572646	0.4306747	0.4042092
38	0.5758229	0.5493919	0.5230822	0.4968908	0.4708153	0.4448558	0.4190115
39	0.5862439	0.5604419	0.5347543	0.5091789	0.4837133	0.4583561	0.4331080
40	0.5961681	0.5709659	0.5458728	0.5208866	0.4960040	0.4712238	0.4465450
41	0.6056301	0.5810011	0.5564765	0.5320535	0.5077281	0.4835004	0.4593688
42	0.6146605	0.5905801	0.5665989	0.5427146	0.5189242	0.4952260	0.4716179
43	0.6232883	0.5997326	0.5762731	0.5529051	0.5296264	0.5064355	0.4833307
44	0.6315400	0.6084878	0.5855263	0.5626540	0.5398665	0.5171621	0.4945401
45	0.6394393	0.6168692	0.5943868	0.5719891	0.5496730	0.5274363	0.5052774
46	0.6470082	0.6249006	0.6028777	0.5809364	0.5590729	0.5372857	0.5155725
47	0.6542673	0.6326039	0.6110229	0.5895189	0.5680905	0.5467353	0.5254511
48	0.6612344	0.6399982	0.6188414	0.5977593	0.5767490	0.5558096	0.5349374
49	0.6679276	0.6471021	0.6263529	0.6056761	0.5850696	0.5645295	0.5440549
50	0.6743613	0.6539318	0.6335755	0.6132894	0.5930704	0.5729155	0.5528240

表 A.1 (续)

N/F	$\gamma=0.70$						
	14	15	16	17	18	19	20
51	0.6805519	0.6605030	0.6405252	0.6206152	0.6007693	0.5809868	0.5612637
52	0.6865122	0.6668305	0.6472175	0.6276694	0.6081846	0.5887600	0.5693934
53	0.6922543	0.6729263	0.6536654	0.6344672	0.6153299	0.5962514	0.5772279
54	0.6977906	0.6788044	0.6598823	0.6410228	0.6222212	0.6034753	0.5847844
55	0.7031316	0.6844752	0.6658814	0.6473471	0.6288700	0.6104471	0.5920769
56	0.7082874	0.6899496	0.6716728	0.6534540	0.6352902	0.6171790	0.5991189
57	0.7132677	0.6952378	0.6772675	0.6593528	0.6414923	0.6236830	0.6059224
58	0.7180807	0.7003488	0.6826748	0.6650558	0.6474878	0.6299703	0.6125000
59	0.7227357	0.7052922	0.6879051	0.6705706	0.6532873	0.6360521	0.6188626
60	0.7272398	0.7100749	0.6929650	0.6759077	0.6588993	0.6419373	0.6250204
61	0.7315998	0.7147051	0.6978645	0.6810744	0.6643323	0.6476359	0.6309831
62	0.7358226	0.7191908	0.7026109	0.6860805	0.6695960	0.6531572	0.6367596
63	0.7399157	0.7235369	0.7072099	0.6909310	0.6746979	0.6585075	0.6423590
64	0.7438831	0.7277515	0.7116695	0.6956348	0.6796443	0.6636966	0.6477886
65	0.7477322	0.7318394	0.7159955	0.7001982	0.6844435	0.6687301	0.6530563
66	0.7514671	0.7358067	0.7201943	0.7046263	0.6891017	0.6736160	0.6581692
67	0.7550931	0.7396581	0.7242704	0.7089261	0.6936239	0.6783605	0.6631346
68	0.7586153	0.7433998	0.7282301	0.7131030	0.6980172	0.6829688	0.6679577
69	0.7620371	0.7470347	0.7320773	0.7171619	0.7022861	0.6874478	0.6726453
70	0.7653644	0.7505690	0.7358179	0.7211080	0.7064363	0.6918019	0.6772018
75	0.7806951	0.7668559	0.7530572	0.7392951	0.7255681	0.7118747	0.6982117
80	0.7941471	0.7811489	0.7681865	0.7552585	0.7423625	0.7294963	0.7166591
85	0.8060456	0.7937919	0.7815709	0.7693820	0.7572221	0.7450897	0.7329839
90	0.8166448	0.8050545	0.7934956	0.7819656	0.7704631	0.7589858	0.7475321
95	0.8261457	0.8151517	0.8041867	0.7932484	0.7823353	0.7714466	0.7605792
100	0.8347119	0.8242549	0.8138262	0.8034218	0.7930411	0.7826825	0.7723453
110	0.8495388	0.8400146	0.8305140	0.8210355	0.8115783	0.8021403	0.7927206
120	0.8619263	0.8531810	0.8444578	0.8357545	0.8270694	0.8184019	0.8097512
130	0.8724296	0.8643459	0.8562821	0.8482365	0.8402078	0.8321950	0.8241968
140	0.8814479	0.8739335	0.8664367	0.8589566	0.8514918	0.8440414	0.8366049
160	0.8961345	0.8895465	0.8829746	0.8764163	0.8698711	0.8633388	0.8568172
180	0.9075833	0.9017195	0.8958687	0.8900301	0.8842037	0.8783872	0.8725806
200	0.9167595	0.9114757	0.9062041	0.9009432	0.8956922	0.8904511	0.8852184
250	0.9333131	0.9290773	0.9248515	0.9206333	0.9164235	0.9122213	0.9080252
300	0.9443752	0.9408412	0.9373141	0.9337946	0.9302812	0.9267740	0.9232720
350	0.9522901	0.9492574	0.9462317	0.9432113	0.9401970	0.9371872	0.9341828
400	0.9582331	0.9555774	0.9529279	0.9502837	0.9476433	0.9450083	0.9423770
450	0.9628589	0.9604979	0.9581414	0.9557887	0.9534413	0.9510970	0.9487566
500	0.9665629	0.9644370	0.9623149	0.9601966	0.9580830	0.9559723	0.9538648
550	0.9695951	0.9676609	0.9657313	0.9638056	0.9618829	0.9599632	0.9580466
600	0.9721229	0.9703494	0.9685805	0.9668139	0.9650511	0.9632906	0.9615332
650	0.9742626	0.9726258	0.9709919	0.9693612	0.9677334	0.9661080	0.9644848
700	0.9760971	0.9745770	0.9730592	0.9715451	0.9700327	0.9685232	0.9670161
750	0.9776876	0.9762688	0.9748523	0.9734381	0.9720269	0.9706173	0.9692107
800	0.9790801	0.9777492	0.9764206	0.9750951	0.9737718	0.9724501	0.9711307
850	0.9803086	0.9790558	0.9778054	0.9765572	0.9753120	0.9740677	0.9728264
900	0.9814006	0.9802178	0.9790365	0.9778575	0.9766808	0.9755063	0.9743327
950	0.9823780	0.9812568	0.9801385	0.9790210	0.9779066	0.9767929	0.9756816
1000	0.9832583	0.9821927	0.9811301	0.9800684	0.9790096	0.9779516	0.9768959

表 A.1 (续)

N/F	$\gamma=0.80$						
	0	1	2	3	4	5	6
1	0.2000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.4472136	0.1055728	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.5848035	0.2870274	0.0716822	0.0000000	0.0000000	0.0000000	0.0000000
4	0.6687403	0.4172687	0.2125937	0.0542584	0.0000000	0.0000000	0.0000000
5	0.7247797	0.5095778	0.3263659	0.1692823	0.0436475	0.0000000	0.0000000
6	0.7647245	0.5773818	0.4143152	0.2685828	0.1398392	0.0365075	0.0000000
7	0.7945974	0.6290130	0.4829872	0.3498478	0.2283330	0.1194915	0.0313749
8	0.8177654	0.6695415	0.5376941	0.4160809	0.3032604	0.1985843	0.1043247
9	0.8362510	0.7021593	0.5821526	0.4706263	0.3661356	0.2675573	0.1757302
10	0.8513399	0.7289570	0.6189300	0.5161399	0.4191905	0.3268520	0.2394405
11	0.8638877	0.7513548	0.6498266	0.5546082	0.4643684	0.3778958	0.2952753
12	0.8744853	0.7703485	0.6761313	0.5875067	0.5032103	0.4220963	0.3441295
13	0.8835540	0.7866563	0.6987875	0.6159400	0.5369156	0.4606417	0.3870237
14	0.8914019	0.8008082	0.7184994	0.6407460	0.5664116	0.4945001	0.4248804
15	0.8982599	0.8132041	0.7358022	0.6625688	0.5924248	0.5244459	0.4584796
16	0.9043038	0.8241508	0.7511100	0.6819110	0.6155284	0.5511020	0.4884693
17	0.9096705	0.8338880	0.7647474	0.6991694	0.6361776	0.5749699	0.5153789
18	0.9144676	0.8426051	0.7769726	0.7146612	0.6547404	0.5964581	0.5396467
19	0.9187812	0.8504544	0.7879936	0.7286429	0.6715137	0.6159005	0.5616362
20	0.9226808	0.8575589	0.7979793	0.7413239	0.6867427	0.6335720	0.5816463
21	0.9262233	0.8640199	0.8070687	0.7528769	0.7006298	0.6497019	0.5999290
22	0.9294556	0.8699208	0.8153770	0.7634453	0.7133436	0.6644810	0.6166964
23	0.9324166	0.8753313	0.8230006	0.7731495	0.7250268	0.6780718	0.6321262
24	0.9351392	0.8803102	0.8300206	0.7820910	0.7357987	0.6906096	0.6463710
25	0.9376510	0.8849069	0.8365058	0.7903560	0.7457613	0.7022129	0.6595616
26	0.9399755	0.8891639	0.8425151	0.7980183	0.7550020	0.7129813	0.6718091
27	0.9421330	0.8931175	0.8480988	0.8051414	0.7635969	0.7230007	0.6832110
28	0.9441408	0.8967989	0.8533006	0.8117801	0.7716100	0.7323473	0.6938514
29	0.9460140	0.9002353	0.8581583	0.8179821	0.7791004	0.7410860	0.7038032
30	0.9477657	0.9034503	0.8627049	0.8237891	0.7861151	0.7492738	0.7131313
31	0.9494073	0.9064647	0.8669693	0.8292375	0.7926996	0.7569610	0.7218923
32	0.9509489	0.9092967	0.8709771	0.8343596	0.7988910	0.7641925	0.7301357
33	0.9523994	0.9119623	0.8747507	0.8391838	0.8047243	0.7710064	0.7379064
34	0.9537665	0.9144757	0.8783099	0.8437352	0.8102295	0.7774395	0.7452430
35	0.9550573	0.9168497	0.8816726	0.8480364	0.8154334	0.7835211	0.7521817
36	0.9562780	0.9190955	0.8848546	0.8521074	0.8203597	0.7892799	0.7587533
37	0.9574342	0.9212233	0.8878700	0.8559662	0.8250305	0.7947411	0.7649866
38	0.9585308	0.9232420	0.8907317	0.8596289	0.8294646	0.7999270	0.7709066
39	0.9595723	0.9251598	0.8934509	0.8631101	0.8336797	0.8048579	0.7765359
40	0.9605628	0.9269842	0.8960382	0.8664230	0.8376918	0.8095516	0.7818958
41	0.9615059	0.9287218	0.8985029	0.8695794	0.8415156	0.8140255	0.7870054
42	0.9624050	0.9303787	0.9008535	0.8725902	0.8451629	0.8182940	0.7918813
43	0.9632630	0.9319603	0.9030977	0.8754653	0.8486465	0.8223710	0.7965395
44	0.9640828	0.9334716	0.9052427	0.8782135	0.8519776	0.8262695	0.8009940
45	0.9648667	0.9349173	0.9072947	0.8808432	0.8551643	0.8300010	0.8052580
46	0.9656172	0.9363015	0.9092598	0.8833617	0.8582174	0.8335761	0.8093437
47	0.9663363	0.9376280	0.9111434	0.8857761	0.8611449	0.8370034	0.8132618
48	0.9670259	0.9389005	0.9129504	0.8880926	0.8639532	0.8402932	0.8170215
49	0.9676879	0.9401220	0.9146853	0.8903170	0.8666511	0.8434521	0.8206338
50	0.9683238	0.9412957	0.9163525	0.8924548	0.8692435	0.8464893	0.8241059

表 A.1 (续)

N/F	$\gamma=0.80$						
	0	1	2	3	4	5	6
51	0.9689351	0.9424243	0.9179558	0.8945109	0.8717376	0.8494106	0.8274468
52	0.9695233	0.9435103	0.9194989	0.8964899	0.8741377	0.8522229	0.8306629
53	0.9700897	0.9445561	0.9209849	0.8983960	0.8764505	0.8549329	0.8337616
54	0.9706354	0.9455639	0.9224171	0.9002332	0.8786789	0.8575443	0.8367492
55	0.9711615	0.9465357	0.9237984	0.9020052	0.8808291	0.8600640	0.8396307
56	0.9716691	0.9474735	0.9251313	0.9037154	0.8829045	0.8624961	0.8424134
57	0.9721592	0.9483789	0.9264184	0.9053669	0.8849091	0.8648453	0.8451003
58	0.9726325	0.9492536	0.9276620	0.9069627	0.8868454	0.8671160	0.8476980
59	0.9730901	0.9500992	0.9288643	0.9085057	0.8887185	0.8693111	0.8502097
60	0.9735326	0.9509170	0.9300272	0.9099983	0.8905297	0.8714346	0.8526401
61	0.9739608	0.9517085	0.9311528	0.9114430	0.8922841	0.8734912	0.8549928
62	0.9743754	0.9524749	0.9322428	0.9128421	0.8939826	0.8754823	0.8572712
63	0.9747769	0.9532173	0.9332987	0.9141976	0.8956284	0.8774124	0.8594802
64	0.9751661	0.9539369	0.9343223	0.9155117	0.8972232	0.8792827	0.8616214
65	0.9755434	0.9546347	0.9353149	0.9167861	0.8987707	0.8810973	0.8636977
66	0.9759095	0.9553117	0.9362780	0.9180227	0.9002723	0.8828586	0.8657136
67	0.9762648	0.9559688	0.9372129	0.9192231	0.9017300	0.8845676	0.8676708
68	0.9766097	0.9566068	0.9381207	0.9203888	0.9031458	0.8862281	0.8695714
69	0.9769447	0.9572266	0.9390026	0.9215214	0.9045217	0.8878416	0.8714178
70	0.9772703	0.9578290	0.9398598	0.9226222	0.9058587	0.8894101	0.8732132
75	0.9787694	0.9606030	0.9438080	0.9276934	0.9120189	0.8966363	0.8814879
80	0.9800830	0.9630346	0.9472697	0.9321410	0.9174228	0.9029766	0.8887485
85	0.9812436	0.9651836	0.9503298	0.9360732	0.9222011	0.9085848	0.8951714
90	0.9822763	0.9670964	0.9530542	0.9395747	0.9264574	0.9135801	0.9008935
95	0.9832012	0.9688100	0.9554953	0.9427126	0.9302713	0.9180574	0.9060227
100	0.9840344	0.9703539	0.9576951	0.9455407	0.9337102	0.9220943	0.9106478
110	0.9854753	0.9730246	0.9615011	0.9504346	0.9396606	0.9290812	0.9186551
120	0.9866776	0.9752539	0.9646788	0.9545215	0.9446318	0.9349183	0.9253452
130	0.9876960	0.9771428	0.9673719	0.9579859	0.9488454	0.9398683	0.9310193
140	0.9885698	0.9787638	0.9696835	0.9609598	0.9524637	0.9441185	0.9358915
160	0.9899914	0.9814018	0.9734460	0.9658013	0.9583545	0.9510395	0.9438266
180	0.9910985	0.9834568	0.9763777	0.9695745	0.9629466	0.9564350	0.9500143
200	0.9919851	0.9851029	0.9787265	0.9725979	0.9666268	0.9607593	0.9549742
250	0.9935829	0.9880704	0.9829618	0.9780506	0.9732650	0.9685616	0.9639230
300	0.9946496	0.9900521	0.9857907	0.9816935	0.9777002	0.9737756	0.9699051
350	0.9954122	0.9914692	0.9878140	0.9842993	0.9808737	0.9775061	0.9741851
400	0.9959845	0.9925329	0.9893329	0.9862557	0.9832559	0.9803081	0.9773999
450	0.9964299	0.9933607	0.9905151	0.9877786	0.9851109	0.9824889	0.9799020
500	0.9967863	0.9940233	0.9914614	0.9889976	0.9865957	0.9842348	0.9819060
550	0.9970780	0.9945657	0.9922361	0.9899956	0.9878112	0.9856642	0.9835462
600	0.9973212	0.9950178	0.9928818	0.9908275	0.9888247	0.9868559	0.9849138
650	0.9975270	0.9954004	0.9934284	0.9915317	0.9896824	0.9878647	0.9860714
700	0.9977034	0.9957285	0.9938970	0.9921355	0.9904179	0.9887300	0.9870641
750	0.9978564	0.9960129	0.9943033	0.9926589	0.9910560	0.9894793	0.9879247
800	0.9979902	0.9962618	0.9946588	0.9931170	0.9916136	0.9901363	0.9886780
850	0.9981083	0.9964814	0.9949726	0.9935213	0.9921061	0.9907153	0.9893427
900	0.9982133	0.9966767	0.9952515	0.9938807	0.9925438	0.9912305	0.9899340
950	0.9983073	0.9968514	0.9955012	0.9942024	0.9929356	0.9916908	0.9904626
1000	0.9983919	0.9970087	0.9957258	0.9944919	0.9932888	0.9921057	0.9909387

表 A.1 (续)

N/F	$\gamma=0.80$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0275075	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0925808	0.0244889	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.1576122	0.0832172	0.0220672	0.0000000	0.0000000	0.0000000	0.0000000
11	0.2167060	0.1428913	0.0755754	0.0200814	0.0000000	0.0000000	0.0000000
12	0.2693152	0.1979356	0.1306926	0.0692206	0.0184235	0.0000000	0.0000000
13	0.3159722	0.2475826	0.1821700	0.1204165	0.0638517	0.0170184	0.0000000
14	0.3574184	0.2921177	0.2291166	0.1687392	0.1116417	0.0592569	0.0158125
15	0.3943719	0.3320753	0.2716396	0.2132264	0.1571585	0.1040607	0.0552789
16	0.4274646	0.3680149	0.3101230	0.2538639	0.1994068	0.1470691	0.0974445
17	0.4572351	0.4004524	0.3450034	0.2909177	0.2382847	0.1872759	0.1381997
18	0.4841377	0.4298364	0.3767002	0.3247297	0.2739692	0.2245156	0.1765414
19	0.5085538	0.4565569	0.4055916	0.3556425	0.3067265	0.2588984	0.2122582
20	0.5308032	0.4809429	0.4320095	0.3839740	0.3368389	0.2906307	0.2454082
21	0.5511541	0.5032788	0.4562418	0.4100098	0.3645728	0.3199414	0.2761500
22	0.5698365	0.5238045	0.4785384	0.4340001	0.3901737	0.3470575	0.3046715
23	0.5870427	0.5427268	0.4991142	0.4561662	0.4138606	0.3721912	0.3311643
24	0.6029397	0.5602217	0.5181552	0.4766999	0.4358297	0.3955333	0.3558106
25	0.6176688	0.5764435	0.5358240	0.4957691	0.4562512	0.4172568	0.3787796
26	0.6313524	0.5915230	0.5522596	0.5135211	0.4752785	0.4375151	0.4002231
27	0.6440980	0.6055763	0.5675848	0.5300839	0.4930438	0.4564462	0.4202805
28	0.6559978	0.6187019	0.5819073	0.5455712	0.5096662	0.4741709	0.4390738
29	0.6671316	0.6309895	0.5953203	0.5600830	0.5252485	0.4907978	0.4567156
30	0.6775717	0.6425156	0.6079067	0.5737066	0.5398850	0.5064228	0.4733038
31	0.6873805	0.6533479	0.6197402	0.5865200	0.5536569	0.5211311	0.4889272
32	0.6966121	0.6635463	0.6308856	0.5985919	0.5666370	0.5350002	0.5036655
33	0.7053169	0.6731652	0.6414012	0.6099858	0.5788916	0.5480987	0.5175905
34	0.7135378	0.6822523	0.6513368	0.6207547	0.5904785	0.5604869	0.5307654
35	0.7213146	0.6908503	0.6607408	0.6309494	0.6014502	0.5722219	0.5432492
36	0.7286813	0.6989970	0.6696529	0.6406140	0.6118535	0.5833517	0.5550933
37	0.7356700	0.7067272	0.6781118	0.6497885	0.6217323	0.5939225	0.5663446
38	0.7423088	0.7140719	0.6861502	0.6585084	0.6311237	0.6039748	0.5770464
39	0.7486237	0.7210593	0.6937977	0.6668078	0.6400628	0.6135444	0.5872374
40	0.7546367	0.7277141	0.7010837	0.6747142	0.6485813	0.6226657	0.5969531
41	0.7603698	0.7340593	0.7080325	0.6822568	0.6567084	0.6313698	0.6062258
42	0.7658417	0.7401172	0.7146659	0.6894586	0.6644704	0.6396835	0.6150836
43	0.7710696	0.7459057	0.7210065	0.6963423	0.6718902	0.6476327	0.6235552
44	0.7760702	0.7514425	0.7270718	0.7029286	0.6789905	0.6552401	0.6316637
45	0.7808568	0.7567440	0.7328800	0.7092364	0.6857919	0.6625283	0.6394325
46	0.7854439	0.7618241	0.7384470	0.7152834	0.6923122	0.6695164	0.6468823
47	0.7898436	0.7666978	0.7437870	0.7210845	0.6985681	0.6762219	0.6540320
48	0.7940664	0.7713753	0.7489139	0.7266539	0.7045754	0.6826618	0.6608992
49	0.7981230	0.7758700	0.7538406	0.7320065	0.7103494	0.6888510	0.6675006
50	0.8020230	0.7801920	0.7585776	0.7371547	0.7159020	0.6948049	0.6738512

表 A.1 (续)

N/F	$\gamma=0.80$						
	7	8	9	10	11	12	13
51	0.8057760	0.7843500	0.7631362	0.7421085	0.7212472	0.7005361	0.6799639
52	0.8093890	0.7883546	0.7675263	0.7468795	0.7263952	0.7060567	0.6858539
53	0.8128704	0.7922134	0.7717570	0.7514784	0.7313570	0.7113782	0.6915307
54	0.8162272	0.7959333	0.7758370	0.7559124	0.7361419	0.7165110	0.6970075
55	0.8194653	0.7995234	0.7797737	0.7601919	0.7407604	0.7214647	0.7022933
56	0.8225917	0.8029897	0.7835747	0.7643244	0.7452206	0.7262496	0.7073991
57	0.8256117	0.8063375	0.7872472	0.7683171	0.7495304	0.7308726	0.7123331
58	0.8285309	0.8095745	0.7907973	0.7721773	0.7536969	0.7353424	0.7171039
59	0.8313547	0.8127050	0.7942307	0.7759106	0.7577270	0.7396670	0.7217191
60	0.8340867	0.8157339	0.7975536	0.7795243	0.7616278	0.7438526	0.7261872
61	0.8367309	0.8186666	0.8007710	0.7830233	0.7654053	0.7479064	0.7305135
62	0.8392928	0.8215082	0.8038884	0.7864128	0.7690653	0.7518339	0.7347061
63	0.8417762	0.8242613	0.8069090	0.7896985	0.7726132	0.7556408	0.7387714
64	0.8441829	0.8269313	0.8098392	0.7928852	0.7760539	0.7593334	0.7427135
65	0.8465187	0.8295220	0.8126809	0.7959764	0.7793917	0.7629161	0.7465389
66	0.8487851	0.8320353	0.8154395	0.7989765	0.7826325	0.7663938	0.7502520
67	0.8509854	0.8344761	0.8181180	0.8018902	0.7857785	0.7697713	0.7538587
68	0.8531230	0.8368477	0.8207197	0.8047207	0.7888361	0.7730529	0.7573621
69	0.8551993	0.8391517	0.8232483	0.8074715	0.7918068	0.7762422	0.7607683
70	0.8572185	0.8413916	0.8257067	0.8101461	0.7946954	0.7793431	0.7640800
75	0.8665249	0.8517176	0.8370407	0.8224783	0.8080166	0.7936449	0.7793541
80	0.8746936	0.8607814	0.8469913	0.8333072	0.8197154	0.8062068	0.7927738
85	0.8819197	0.8688023	0.8557978	0.8428918	0.8300719	0.8173299	0.8046565
90	0.8883587	0.8759498	0.8636463	0.8514350	0.8393046	0.8272459	0.8152521
95	0.8941315	0.8823586	0.8706848	0.8590980	0.8475859	0.8361418	0.8247580
100	0.8993378	0.8881384	0.8770329	0.8660091	0.8550562	0.8441673	0.8333342
110	0.9083511	0.8981471	0.8880270	0.8779809	0.8679973	0.8580710	0.8481950
120	0.9158835	0.9065118	0.8972179	0.8879889	0.8788178	0.8696979	0.8606245
130	0.9222719	0.9136076	0.9050142	0.8964804	0.8879992	0.8795653	0.8711727
140	0.9277584	0.9197023	0.9117110	0.9037755	0.8958880	0.8880440	0.8802374
160	0.9366962	0.9296314	0.9226230	0.9156627	0.9087444	0.9018627	0.8950138
180	0.9436653	0.9373758	0.9311351	0.9249364	0.9187743	0.9126450	0.9065447
200	0.9492533	0.9435842	0.9379601	0.9323726	0.9268187	0.9212938	0.9157941
250	0.9593363	0.9547900	0.9502789	0.9457982	0.9413420	0.9369094	0.9324966
300	0.9660766	0.9622825	0.9585166	0.9547758	0.9510564	0.9473554	0.9436703
350	0.9709000	0.9676438	0.9644128	0.9612025	0.9580096	0.9548336	0.9516705
400	0.9745230	0.9716713	0.9688417	0.9660297	0.9632337	0.9604514	0.9576813
450	0.9773433	0.9748075	0.9722901	0.9697886	0.9673016	0.9648261	0.9623620
500	0.9796016	0.9773177	0.9750507	0.9727989	0.9705586	0.9683297	0.9661108
550	0.9814503	0.9793735	0.9773119	0.9752633	0.9732262	0.9711990	0.9691801
600	0.9829923	0.9810876	0.9791966	0.9773186	0.9754498	0.9735909	0.9717396
650	0.9842972	0.9825382	0.9807930	0.9790585	0.9773331	0.9756161	0.9739067
700	0.9854166	0.9837828	0.9821612	0.9805503	0.9789478	0.9773529	0.9757656
750	0.9863861	0.9848613	0.9833479	0.9818437	0.9803479	0.9788589	0.9773768
800	0.9872357	0.9858057	0.9843863	0.9829761	0.9815727	0.9801770	0.9787866
850	0.9879847	0.9866388	0.9853029	0.9839754	0.9826547	0.9813401	0.9800317
900	0.9886512	0.9873799	0.9861177	0.9848632	0.9836163	0.9823747	0.9811385
950	0.9892475	0.9880430	0.9868470	0.9856586	0.9844770	0.9833008	0.9821291
1000	0.9897845	0.9886396	0.9875038	0.9863741	0.9852512	0.9841337	0.9830208

表 A.1 (续)

N/F	$\gamma=0.80$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0147661	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0518016	0.0138497	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.0916206	0.0487364	0.0130403	0.0000000	0.0000000	0.0000000	0.0000000
18	0.1303410	0.0864546	0.0460136	0.0123203	0.0000000	0.0000000	0.0000000
19	0.1669737	0.1233303	0.0818399	0.0435791	0.0116757	0.0000000	0.0000000
20	0.2012744	0.1583919	0.1170352	0.0776935	0.0413890	0.0110952	0.0000000
21	0.2332612	0.1913749	0.1506513	0.1113536	0.0739480	0.0394088	0.0105696
22	0.2630530	0.2222646	0.1824070	0.1436335	0.1061982	0.0705467	0.0376097
23	0.2908034	0.2511483	0.2122621	0.1742435	0.1372418	0.1014990	0.0674445
24	0.3166746	0.2781505	0.2402787	0.2031248	0.1667819	0.1313958	0.0971991
25	0.3408251	0.3034091	0.2665586	0.2303160	0.1947434	0.1599336	0.1260272
26	0.3634026	0.3270613	0.2912174	0.2558992	0.2211486	0.1870282	0.1536272
27	0.3845427	0.3492368	0.3143741	0.2799731	0.2460633	0.2126861	0.1799024
28	0.4043697	0.3700577	0.3361439	0.3026414	0.2695692	0.2369585	0.2048491
29	0.4229950	0.3896338	0.3566342	0.3240046	0.2917580	0.2599150	0.2285062
30	0.4405198	0.4080676	0.3759459	0.3441590	0.3127170	0.2816343	0.2509316
31	0.4570353	0.4254502	0.3941695	0.3631954	0.3325342	0.3021949	0.2721936
32	0.4726230	0.4418659	0.4113910	0.3811977	0.3512897	0.3216747	0.2923618
33	0.4873569	0.4573895	0.4276846	0.3982415	0.3690599	0.3401462	0.3115064
34	0.5013025	0.4720898	0.4431221	0.4143970	0.3859153	0.3576784	0.3296926
35	0.5145206	0.4860277	0.4577653	0.4297295	0.4019194	0.3743368	0.3469838
36	0.5270660	0.4992615	0.4716730	0.4442980	0.4171337	0.3901799	0.3634389
37	0.5389865	0.5118396	0.4848973	0.4581556	0.4316108	0.4052628	0.3791123
38	0.5503285	0.5238105	0.4974864	0.4713506	0.4454018	0.4196369	0.3940558
39	0.5611309	0.5352153	0.5094834	0.4839302	0.4585523	0.4333477	0.4083163
40	0.5714321	0.5460926	0.5209278	0.4959339	0.4711048	0.4464405	0.4219379
41	0.5812649	0.5564779	0.5318580	0.5073991	0.4830989	0.4589528	0.4349593
42	0.5906599	0.5664033	0.5423054	0.5183624	0.4945681	0.4709218	0.4474198
43	0.5996463	0.5758976	0.5523023	0.5288535	0.5055473	0.4823815	0.4593523
44	0.6082498	0.5849892	0.5618751	0.5389022	0.5160658	0.4933622	0.4707890
45	0.6164931	0.5937018	0.5710515	0.5485363	0.5261516	0.5038935	0.4817606
46	0.6243993	0.6020597	0.5798544	0.5577796	0.5358299	0.5140015	0.4922922
47	0.6319887	0.6100827	0.5883071	0.5666559	0.5451252	0.5237113	0.5024103
48	0.6392785	0.6177905	0.5964277	0.5751855	0.5540592	0.5330443	0.5121384
49	0.6462868	0.6252011	0.6042375	0.5833892	0.5626522	0.5420228	0.5214979
50	0.6530295	0.6323322	0.6117523	0.5912839	0.5709238	0.5506659	0.5305095

表 A.1 (续)

N/F	$\gamma=0.80$						
	14	15	16	17	18	19	20
51	0.6595207	0.6391980	0.6189889	0.5988875	0.5788897	0.5589920	0.5391911
52	0.6657747	0.6458138	0.6259620	0.6062155	0.5865682	0.5670177	0.5475604
53	0.6718044	0.6521919	0.6326861	0.6132819	0.5939737	0.5747594	0.5556344
54	0.6776216	0.6583462	0.6391746	0.6201007	0.6011206	0.5822305	0.5634274
55	0.6832372	0.6642871	0.6454384	0.6266844	0.6080219	0.5894464	0.5709549
56	0.6886608	0.6700262	0.6514893	0.6330454	0.6146901	0.5964187	0.5782289
57	0.6939020	0.6755731	0.6573387	0.6391944	0.6211364	0.6031599	0.5852620
58	0.6989706	0.6809365	0.6629955	0.6451423	0.6273714	0.6096807	0.5920662
59	0.7038751	0.6861271	0.6684700	0.6508975	0.6334059	0.6159921	0.5986516
60	0.7086226	0.6911518	0.6737695	0.6564703	0.6392490	0.6221032	0.6050299
61	0.7132206	0.6960192	0.6789032	0.6618681	0.6449101	0.6280246	0.6112100
62	0.7176761	0.7007353	0.6838784	0.6671001	0.6503966	0.6337648	0.6172001
63	0.7219958	0.7053080	0.6887025	0.6721734	0.6557175	0.6393302	0.6230101
64	0.7261859	0.7097438	0.6933818	0.6770946	0.6608791	0.6447307	0.6286472
65	0.7302517	0.7140484	0.6979230	0.6818708	0.6658888	0.6499724	0.6341186
66	0.7341987	0.7182270	0.7023324	0.6865087	0.6707530	0.6550621	0.6394329
67	0.7380323	0.7222860	0.7066144	0.6910139	0.6754789	0.6600073	0.6445951
68	0.7417567	0.7262299	0.7107763	0.6953914	0.6800714	0.6648123	0.6496121
69	0.7453775	0.7300639	0.7148219	0.6996471	0.6845356	0.6694844	0.6544903
70	0.7488979	0.7337920	0.7187563	0.7037863	0.6888780	0.6740285	0.6592354
75	0.7651380	0.7509914	0.7369081	0.7228843	0.7089162	0.6950015	0.6811364
80	0.7794086	0.7661067	0.7528629	0.7396739	0.7265354	0.7134456	0.7004009
85	0.7920465	0.7794952	0.7669965	0.7545490	0.7421480	0.7297912	0.7174764
90	0.8033162	0.7914353	0.7796040	0.7678192	0.7560771	0.7443770	0.7327143
95	0.8134290	0.8021505	0.7909184	0.7797299	0.7685818	0.7574710	0.7463962
100	0.8225537	0.8118198	0.8011293	0.7904801	0.7798674	0.7692914	0.7587474
110	0.8383656	0.8285773	0.8188280	0.8091153	0.7994346	0.7897860	0.7801664
120	0.8515916	0.8425968	0.8336363	0.8247086	0.8158107	0.8069403	0.7980958
130	0.8628181	0.8544979	0.8462082	0.8379491	0.8297158	0.8215085	0.8133241
140	0.8724658	0.8647264	0.8570151	0.8493306	0.8416704	0.8340332	0.8264181
160	0.8881946	0.8814022	0.8746349	0.8678897	0.8611659	0.8544620	0.8477764
180	0.9004704	0.8944189	0.8883896	0.8823801	0.8763889	0.8704145	0.8644561
200	0.9103173	0.9048618	0.8994254	0.8940065	0.8886037	0.8832169	0.8778430
250	0.9281022	0.9237238	0.9193606	0.9150112	0.9106740	0.9063490	0.9020346
300	0.9400013	0.9363452	0.9327006	0.9290682	0.9254457	0.9218332	0.9182290
350	0.9485211	0.9453825	0.9422545	0.9391357	0.9360260	0.9329239	0.9298295
400	0.9549219	0.9521732	0.9494329	0.9467009	0.9439766	0.9412592	0.9385478
450	0.9599071	0.9574614	0.9550240	0.9525935	0.9501691	0.9477516	0.9453394
500	0.9639002	0.9616973	0.9595019	0.9573127	0.9551296	0.9529519	0.9507794
550	0.9671697	0.9651661	0.9631686	0.9611773	0.9591912	0.9572105	0.9552344
600	0.9698959	0.9680583	0.9662269	0.9644008	0.9625792	0.9607622	0.9589499
650	0.9722042	0.9705071	0.9688160	0.9671296	0.9654477	0.9637696	0.9620961
700	0.9741837	0.9726071	0.9710366	0.9694700	0.9679071	0.9663488	0.9647944
750	0.9759000	0.9744286	0.9729617	0.9714987	0.9700402	0.9685856	0.9671340
800	0.9774024	0.9760227	0.9746468	0.9732755	0.9719072	0.9705428	0.9691814
850	0.9787286	0.9774293	0.9761345	0.9748436	0.9735558	0.9722710	0.9709892
900	0.9799076	0.9786805	0.9774572	0.9762378	0.9750206	0.9738073	0.9725970
950	0.9809628	0.9797995	0.9786409	0.9774852	0.9763327	0.9751824	0.9740359
1000	0.9819125	0.9808080	0.9797065	0.9786081	0.9775135	0.9764204	0.9753304

表 A.1 (续)

N/F	$\gamma=0.90$						
	0	1	2	3	4	5	6
1	0.1000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.3162278	0.0513167	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.4641589	0.1968259	0.0345106	0.0000000	0.0000000	0.0000000	0.0000000
4	0.5623413	0.3207696	0.1444887	0.0259963	0.0000000	0.0000000	0.0000000
5	0.6309573	0.4161914	0.2474422	0.1147513	0.0208516	0.0000000	0.0000000
6	0.6812921	0.4896909	0.3335349	0.2021571	0.0925019	0.0174068	0.0000000
7	0.7196857	0.5474180	0.4039633	0.2792314	0.1695836	0.0787429	0.0149388
8	0.7498942	0.5937309	0.4618396	0.3449507	0.2396572	0.1467922	0.0685573
9	0.7742637	0.6316138	0.5099332	0.4007533	0.3009996	0.2103688	0.1294358
10	0.7943282	0.6631304	0.5503916	0.4483613	0.3542682	0.2673183	0.1875263
11	0.8111308	0.6897372	0.5848300	0.4892861	0.4005928	0.3177388	0.2405184
12	0.8254042	0.7124853	0.6144619	0.5247554	0.4410696	0.3623086	0.2881740
13	0.8376776	0.7321497	0.6402065	0.5557438	0.4766479	0.4018023	0.3308729
14	0.8483429	0.7493127	0.6627689	0.5830218	0.5081123	0.4369386	0.3691530
15	0.8576959	0.7644202	0.6826969	0.6072007	0.5361062	0.4683409	0.4035602
16	0.8659643	0.7778184	0.7004212	0.6287694	0.5611537	0.4965404	0.4345905
17	0.8733262	0.7897807	0.7162846	0.6481216	0.5836836	0.5219792	0.4626780
18	0.8799225	0.8005251	0.7305634	0.6655773	0.6040492	0.5450291	0.4881987
19	0.8858668	0.8102280	0.7434819	0.6813990	0.6225421	0.5660020	0.5114707
20	0.8912509	0.8190334	0.7552245	0.6958033	0.6394046	0.5851581	0.5327686
21	0.8961505	0.8270598	0.7659440	0.7089708	0.6548405	0.6027201	0.5523247
22	0.9006280	0.8344061	0.7757677	0.7210530	0.6690206	0.6188748	0.5703379
23	0.9047357	0.8411550	0.7848031	0.7321778	0.6820916	0.6337817	0.5869808
24	0.9085176	0.8473763	0.7931409	0.7424539	0.6941773	0.6475782	0.6024005
25	0.9120108	0.8531296	0.8008586	0.7519743	0.7053835	0.6603832	0.6167241
26	0.9152473	0.8584655	0.8080228	0.7608189	0.7158029	0.6722976	0.6300641
27	0.9182543	0.8634278	0.8146907	0.7690570	0.7255140	0.6834110	0.6425150
28	0.9210553	0.8680544	0.8209121	0.7767486	0.7345876	0.6938006	0.6541633
29	0.9236709	0.8723782	0.8267302	0.7839460	0.7430828	0.7035341	0.6650833
30	0.9261187	0.8764280	0.8321830	0.7906953	0.7510534	0.7126717	0.6753398
31	0.9284145	0.8802289	0.8373037	0.7970369	0.7585459	0.7212662	0.6849912
32	0.9305720	0.8838032	0.8421216	0.8030066	0.7656024	0.7293636	0.6940891
33	0.9326033	0.8871706	0.8466629	0.8086360	0.7722605	0.7370060	0.7026792
34	0.9345192	0.8903484	0.8509507	0.8139534	0.7785508	0.7442310	0.7108031
35	0.9363292	0.8933524	0.8550055	0.8189839	0.7845046	0.7510709	0.7184971
36	0.9380419	0.8961962	0.8588459	0.8237501	0.7901481	0.7575561	0.7257951
37	0.9396648	0.8988924	0.8624883	0.8282722	0.7955032	0.7637124	0.7327249
38	0.9412050	0.9014522	0.8659477	0.8325685	0.8005935	0.7695658	0.7393155
39	0.9426685	0.9038857	0.8692375	0.8366555	0.8054364	0.7751369	0.7455896
40	0.9440609	0.9062020	0.8723699	0.8405480	0.8100505	0.7804454	0.7515705
41	0.9453873	0.9084093	0.8753559	0.8442596	0.8144511	0.7855099	0.7572767
42	0.9466523	0.9105152	0.8782055	0.8478026	0.8186531	0.7903468	0.7627287
43	0.9478600	0.9125265	0.8809278	0.8511882	0.8226692	0.7949708	0.7679415
44	0.9490142	0.9144493	0.8835312	0.8544267	0.8265122	0.7993957	0.7729308
45	0.9501185	0.9162896	0.8860233	0.8575273	0.8301918	0.8036345	0.7777105
46	0.9511760	0.9180523	0.8884111	0.8604989	0.8337187	0.8076980	0.7822937
47	0.9521895	0.9197424	0.8907010	0.8633491	0.8371023	0.8115968	0.7866924
48	0.9531619	0.9213642	0.8928988	0.8660853	0.8403517	0.8153413	0.7909175
49	0.9540955	0.9229218	0.8950100	0.8687143	0.8434737	0.8189403	0.7949784
50	0.9549926	0.9244189	0.8970397	0.8712420	0.8464766	0.8224018	0.7988854

表 A.1 (续)

N/F	$\gamma=0.90$						
	0	1	2	3	4	5	6
51	0.9558553	0.9258590	0.8989925	0.8736744	0.8493659	0.8257335	0.8026467
52	0.9566856	0.9272452	0.9008726	0.8760167	0.8521491	0.8289426	0.8062695
53	0.9574852	0.9285806	0.9026840	0.8782738	0.8548311	0.8320363	0.8097626
54	0.9582559	0.9298679	0.9044304	0.8804502	0.8574181	0.8350193	0.8131324
55	0.9589991	0.9311096	0.9061153	0.8825502	0.8599147	0.8378991	0.8163847
56	0.9597163	0.9323081	0.9077419	0.8845778	0.8623250	0.8406806	0.8195260
57	0.9604088	0.9334656	0.9093131	0.8865367	0.8646538	0.8433678	0.8225616
58	0.9610780	0.9345842	0.9108317	0.8884302	0.8669052	0.8459658	0.8254972
59	0.9617249	0.9356659	0.9123003	0.8902615	0.8690834	0.8484798	0.8283370
60	0.9623506	0.9367123	0.9137213	0.8920338	0.8711915	0.8509123	0.8310863
61	0.9629563	0.9377253	0.9150970	0.8937498	0.8732327	0.8532687	0.8337495
62	0.9635427	0.9387064	0.9164296	0.8954121	0.8752097	0.8555513	0.8363293
63	0.9641109	0.9396570	0.9177210	0.8970232	0.8771272	0.8577645	0.8388305
64	0.9646616	0.9405786	0.9189731	0.8985855	0.8789859	0.8599109	0.8412570
65	0.9651957	0.9414725	0.9201877	0.9001011	0.8807890	0.8619940	0.8436118
66	0.9657139	0.9423399	0.9213664	0.9015721	0.8825400	0.8640155	0.8458970
67	0.9662169	0.9431820	0.9225109	0.9030005	0.8842396	0.8659786	0.8481167
68	0.9667053	0.9439998	0.9236225	0.9043880	0.8858912	0.8678863	0.8502736
69	0.9671799	0.9447944	0.9247027	0.9057364	0.8874967	0.8697401	0.8523703
70	0.9676411	0.9455668	0.9257527	0.9070473	0.8890573	0.8715428	0.8544090
75	0.9697654	0.9491259	0.9305925	0.9130906	0.8962527	0.8798569	0.8638137
80	0.9716280	0.9522481	0.9348400	0.9183964	0.9025730	0.8871616	0.8720775
85	0.9732744	0.9550093	0.9385978	0.9230919	0.9081669	0.8936289	0.8793968
90	0.9747402	0.9574687	0.9419459	0.9272765	0.9131540	0.8993955	0.8859246
95	0.9760536	0.9596732	0.9449478	0.9310294	0.9176279	0.9045686	0.8917817
100	0.9772372	0.9616604	0.9476546	0.9344140	0.9216626	0.9092365	0.8970674
110	0.9792850	0.9651001	0.9523412	0.9402760	0.9286536	0.9173250	0.9062283
120	0.9809947	0.9679734	0.9562576	0.9451762	0.9344992	0.9240899	0.9138927
130	0.9824437	0.9704096	0.9595793	0.9493334	0.9394597	0.9298318	0.9203985
140	0.9836875	0.9725014	0.9624322	0.9529046	0.9437212	0.9347664	0.9259908
160	0.9857119	0.9759077	0.9670793	0.9587235	0.9506679	0.9428104	0.9351093
180	0.9872893	0.9785632	0.9707033	0.9632627	0.9560877	0.9490887	0.9422278
200	0.9885531	0.9806914	0.9736086	0.9669025	0.9604347	0.9541248	0.9479384
250	0.9908319	0.9845308	0.9788518	0.9734731	0.9682838	0.9632210	0.9582551
300	0.9923541	0.9870965	0.9823569	0.9778670	0.9735345	0.9693063	0.9651596
350	0.9934428	0.9889323	0.9848654	0.9810122	0.9772940	0.9736647	0.9701042
400	0.9942601	0.9903108	0.9867494	0.9833748	0.9801180	0.9769390	0.9738202
450	0.9948962	0.9913839	0.9882162	0.9852144	0.9823170	0.9794890	0.9767144
500	0.9954054	0.9922430	0.9893907	0.9866876	0.9840780	0.9815309	0.9790320
550	0.9958222	0.9929463	0.9903522	0.9878937	0.9855201	0.9832037	0.9809308
600	0.9961697	0.9935327	0.9911540	0.9888995	0.9867228	0.9845984	0.9825136
650	0.9964638	0.9940291	0.9918327	0.9897509	0.9877414	0.9857788	0.9838543
700	0.9967160	0.9944547	0.9924147	0.9904811	0.9886143	0.9867921	0.9850034
750	0.9969346	0.9948237	0.9929193	0.9911141	0.9893714	0.9876696	0.9860006
800	0.9971259	0.9951466	0.9933609	0.9916682	0.9900336	0.9884380	0.9868729
850	0.9972947	0.9954316	0.9937507	0.9921573	0.9906192	0.9891169	0.9876429
900	0.9974448	0.9956850	0.9940972	0.9925921	0.9911385	0.9897199	0.9883280
950	0.9975792	0.9959118	0.9944073	0.9929812	0.9916043	0.9902595	0.9889406
1000	0.9977001	0.9961159	0.9946865	0.9933315	0.9920231	0.9907454	0.9894921

表 A.1 (续)

N/F	$\gamma=0.90$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0130837	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0607113	0.0116385	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.1157680	0.0544788	0.0104807	0.0000000	0.0000000	0.0000000	0.0000000
11	0.1691947	0.1047207	0.0494088	0.0095325	0.0000000	0.0000000	0.0000000
12	0.2186598	0.1541489	0.0956050	0.0452032	0.0087416	0.0000000	0.0000000
13	0.2637223	0.2004772	0.1415740	0.0879526	0.0416585	0.0080719	0.0000000
14	0.3045558	0.2431431	0.1851080	0.1309043	0.0814377	0.0386293	0.0074975
15	0.3415307	0.2821798	0.2255748	0.1719422	0.1217352	0.0758228	0.0360108
16	0.3750568	0.3178295	0.2629070	0.2103946	0.1605342	0.1137714	0.0709331
17	0.4055299	0.3504021	0.2972548	0.2461255	0.1971432	0.1505526	0.1067880
18	0.4333061	0.3802110	0.3288478	0.2792157	0.2313773	0.1854714	0.1417445
19	0.4587025	0.4075511	0.3579354	0.3098326	0.2632649	0.2183100	0.1751122
20	0.4819942	0.4326869	0.3847598	0.3381725	0.2929248	0.2490558	0.2066501
21	0.5034193	0.4558567	0.4095454	0.3644364	0.3205115	0.2777882	0.2363139
22	0.5231851	0.4772690	0.4324965	0.3888127	0.3461902	0.3046289	0.2641549
23	0.5414701	0.4971060	0.4537949	0.4114770	0.3701197	0.3297123	0.2902647
24	0.5584306	0.5155287	0.4736026	0.4325898	0.3924528	0.3531742	0.3147530
25	0.5742018	0.5326790	0.4920633	0.4522930	0.4133275	0.3751448	0.3377358
26	0.5889017	0.5486793	0.5093045	0.4707164	0.4328728	0.3957464	0.3593250
27	0.6026338	0.5636383	0.5254393	0.4879750	0.4512019	0.4150911	0.3796280
28	0.6154895	0.5776535	0.5405675	0.5041703	0.4684194	0.4332835	0.3987449
29	0.6275488	0.5908085	0.5547777	0.5193955	0.4846182	0.4504162	0.4167673
30	0.6388815	0.6031792	0.5681491	0.5337317	0.4998833	0.4665737	0.4337805
31	0.6495517	0.6148324	0.5807517	0.5472523	0.5142900	0.4818336	0.4498616
32	0.6596141	0.6258281	0.5926502	0.5600238	0.5279064	0.4962665	0.4650814
33	0.6691200	0.6362191	0.6038996	0.5721051	0.5407934	0.5099342	0.4795038
34	0.6781130	0.6460546	0.6145516	0.5835499	0.5530075	0.5228946	0.4931875
35	0.6866336	0.6553759	0.6246514	0.5944061	0.5645987	0.5352003	0.5061856
36	0.6947176	0.6642230	0.6342403	0.6047169	0.5756124	0.5468969	0.5185477
37	0.7023967	0.6726307	0.6433561	0.6145224	0.5860900	0.5580284	0.5303163
38	0.7097015	0.6806300	0.6520323	0.6238580	0.5960682	0.5686347	0.5415338
39	0.7166588	0.6882506	0.6603001	0.6327563	0.6055826	0.5787498	0.5522360
40	0.7232913	0.6955181	0.6681865	0.6412471	0.6146633	0.5884075	0.5624576
41	0.7296219	0.7024568	0.6757175	0.6493573	0.6233397	0.5976380	0.5722292
42	0.7356705	0.7090877	0.6829162	0.6571115	0.6316373	0.6064674	0.5815791
43	0.7414561	0.7154309	0.6898046	0.6645331	0.6395805	0.6149216	0.5905336
44	0.7469939	0.7215040	0.6964017	0.6716419	0.6471912	0.6230235	0.5991175
45	0.7523008	0.7273251	0.7027255	0.6784579	0.6544893	0.6307946	0.6073524
46	0.7573906	0.7329087	0.7087923	0.6849985	0.6614940	0.6382549	0.6152592
47	0.7622756	0.7382699	0.7146183	0.6912795	0.6682222	0.6454212	0.6228568
48	0.7669691	0.7434197	0.7202159	0.6973166	0.6746903	0.6523120	0.6301626
49	0.7714811	0.7483721	0.7255996	0.7031231	0.6809121	0.6589415	0.6371936
50	0.7758222	0.7531381	0.7307814	0.7087123	0.6869019	0.6653249	0.6439638

表 A.1 (续)

N/F	$\gamma=0.90$						
	7	8	9	10	11	12	13
51	0.7800016	0.7577273	0.7357711	0.7140957	0.6926721	0.6714758	0.6504885
52	0.7840289	0.7621492	0.7405800	0.7192855	0.6982344	0.6774052	0.6567790
53	0.7879116	0.7664130	0.7452182	0.7242903	0.7036005	0.6831258	0.6628495
54	0.7916567	0.7705274	0.7496934	0.7291203	0.7087791	0.6886485	0.6687101
55	0.7952732	0.7744989	0.7540153	0.7337850	0.7137813	0.6939828	0.6743712
56	0.7987659	0.7783369	0.7581902	0.7382921	0.7186154	0.6991385	0.6798440
57	0.8021422	0.7820463	0.7622266	0.7426503	0.7232891	0.7041240	0.6851366
58	0.8054070	0.7856334	0.7661307	0.7468653	0.7278111	0.7089477	0.6902575
59	0.8085665	0.7891050	0.7699090	0.7509457	0.7321883	0.7136172	0.6952162
60	0.8116242	0.7924666	0.7735675	0.7548966	0.7364271	0.7181400	0.7000184
61	0.8145866	0.7957220	0.7771122	0.7587244	0.7405350	0.7225226	0.7046727
62	0.8174575	0.7988779	0.7805478	0.7624351	0.7445162	0.7267721	0.7091858
63	0.8202414	0.8019377	0.7838789	0.7660337	0.7483784	0.7308933	0.7135638
64	0.8229412	0.8049060	0.7871112	0.7695255	0.7521259	0.7348926	0.7178119
65	0.8255615	0.8077866	0.7902482	0.7729143	0.7557635	0.7387751	0.7219364
66	0.8281049	0.8105837	0.7932937	0.7762058	0.7592957	0.7425466	0.7259432
67	0.8305760	0.8133007	0.7962529	0.7794026	0.7627278	0.7462102	0.7298360
68	0.8329768	0.8159409	0.7991286	0.7825107	0.7660638	0.7497718	0.7336202
69	0.8353110	0.8185081	0.8019241	0.7855316	0.7693078	0.7532350	0.7373002
70	0.8375804	0.8210044	0.8046435	0.7884702	0.7724633	0.7566044	0.7408806
75	0.8480519	0.8325236	0.8171936	0.8020353	0.7870304	0.7721605	0.7574142
80	0.8572559	0.8426509	0.8282298	0.8139690	0.7998477	0.7858524	0.7719707
85	0.8654097	0.8516255	0.8380122	0.8245469	0.8112128	0.7979947	0.7848819
90	0.8726834	0.8596313	0.8467403	0.8339881	0.8213580	0.8088362	0.7964121
95	0.8792115	0.8668190	0.8545776	0.8424667	0.8304694	0.8185744	0.8067717
100	0.8851029	0.8733069	0.8616529	0.8501216	0.8386980	0.8273705	0.8161292
110	0.8953163	0.8845553	0.8739225	0.8633996	0.8529727	0.8426321	0.8323678
120	0.9038626	0.8939706	0.8841937	0.8745176	0.8649285	0.8554164	0.8459745
130	0.9111192	0.9019659	0.8929186	0.8839629	0.8750872	0.8662818	0.8575389
140	0.9173580	0.9088411	0.9004218	0.8920865	0.8838251	0.8756287	0.8674894
160	0.9275310	0.9200542	0.9126621	0.9053417	0.8980853	0.8908847	0.8837337
180	0.9354751	0.9288118	0.9222232	0.9156980	0.9092284	0.9028085	0.8964320
200	0.9418497	0.9358403	0.9298973	0.9240116	0.9181761	0.9123834	0.9066304
250	0.9533671	0.9485423	0.9437695	0.9390424	0.9343534	0.9296996	0.9250763
300	0.9610771	0.9570464	0.9530593	0.9491087	0.9451910	0.9413014	0.9374371
350	0.9665993	0.9631379	0.9597147	0.9563219	0.9529574	0.9496166	0.9462971
400	0.9707495	0.9677170	0.9647173	0.9617450	0.9587964	0.9558684	0.9529595
450	0.9739818	0.9712843	0.9686142	0.9659693	0.9633458	0.9607398	0.9581514
500	0.9765712	0.9741409	0.9717365	0.9693543	0.9669903	0.9646432	0.9623106
550	0.9786915	0.9764812	0.9742937	0.9721261	0.9699761	0.9678405	0.9657187
600	0.9804601	0.9784326	0.9764264	0.9744385	0.9724659	0.9705071	0.9685612
650	0.9819580	0.9800854	0.9782326	0.9763966	0.9745751	0.9727657	0.9709679
700	0.9832422	0.9815031	0.9797816	0.9780761	0.9763835	0.9747032	0.9730327
750	0.9843560	0.9827319	0.9811247	0.9795320	0.9779522	0.9763831	0.9748232
800	0.9853307	0.9838075	0.9823004	0.9808071	0.9793251	0.9778539	0.9763910
850	0.9861910	0.9847574	0.9833383	0.9819322	0.9805375	0.9791520	0.9777749
900	0.9869560	0.9856015	0.9842615	0.9829329	0.9816151	0.9803064	0.9790053
950	0.9876408	0.9863577	0.9850876	0.9838290	0.9825803	0.9813400	0.9801073
1000	0.9882579	0.9870382	0.9858314	0.9846353	0.9834484	0.9822699	0.9810990

表 A.1 (续)

N/F	$\gamma=0.90$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0069994	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0337253	0.0065634	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.0666374	0.0317129	0.0061785	0.0000000	0.0000000	0.0000000	0.0000000
18	0.1006138	0.0628325	0.0299271	0.0058363	0.0000000	0.0000000	0.0000000
19	0.1339132	0.0951168	0.0594393	0.0283317	0.0055299	0.0000000	0.0000000
20	0.1658542	0.1269046	0.0901897	0.0563938	0.0268985	0.0052542	0.0000000
21	0.1961800	0.1575301	0.1205953	0.0857493	0.0536462	0.0256027	0.0050046
22	0.2248229	0.1867254	0.1500050	0.1148852	0.0817262	0.0511535	0.0244261
23	0.2518088	0.2144043	0.1781442	0.1431682	0.1096928	0.0780638	0.0488822
24	0.2772078	0.2405750	0.2049149	0.1703207	0.1369296	0.1049500	0.0747168
25	0.3011087	0.2652858	0.2303075	0.1962340	0.1631577	0.1312128	0.1006015
26	0.3236093	0.2886115	0.2543560	0.2208856	0.1882629	0.1565755	0.1259562
27	0.3448065	0.3106321	0.2771198	0.2442980	0.2122094	0.1809158	0.1505056
28	0.3647945	0.3314323	0.2986668	0.2665162	0.2350103	0.2041926	0.1741240
29	0.3836617	0.3510939	0.3190693	0.2875979	0.2567010	0.2264075	0.1967625
30	0.4014909	0.3696971	0.3384001	0.3076058	0.2773281	0.2475882	0.2184167
31	0.4183581	0.3873147	0.3567290	0.3266033	0.2969461	0.2677718	0.2391042
32	0.4343349	0.4040172	0.3741229	0.3446529	0.3156108	0.2870075	0.2588581
33	0.4494853	0.4198682	0.3906447	0.3618141	0.3333773	0.3053417	0.2777181
34	0.4638689	0.4349263	0.4063522	0.3781436	0.3502997	0.3228251	0.2957258
35	0.4775386	0.4492472	0.4213006	0.3936958	0.3664306	0.3395055	0.3129269
36	0.4905464	0.4628800	0.4355402	0.4085200	0.3818164	0.3554310	0.3293645
37	0.5029353	0.4758716	0.4491162	0.4226621	0.3965049	0.3706444	0.3450807
38	0.5147488	0.4882652	0.4620730	0.4361661	0.4105385	0.3851886	0.3601164
39	0.5260235	0.5000979	0.4744500	0.4490715	0.4239569	0.3991032	0.3745097
40	0.5367954	0.5114070	0.4862834	0.4614147	0.4367969	0.4124248	0.3882976
41	0.5470959	0.5222257	0.4976066	0.4732316	0.4490939	0.4251889	0.4015151
42	0.5569555	0.5325837	0.5084522	0.4845527	0.4608790	0.4374273	0.4141930
43	0.5664004	0.5425091	0.5188474	0.4954077	0.4721832	0.4491700	0.4263628
44	0.5754564	0.5520279	0.5288193	0.5058242	0.4830344	0.4604452	0.4380528
45	0.5841468	0.5611639	0.5383939	0.5158268	0.4934574	0.4712794	0.4492891
46	0.5924924	0.5699400	0.5475920	0.5254401	0.5034767	0.4816964	0.4600961
47	0.6005128	0.5783764	0.5564361	0.5346849	0.5131146	0.4917205	0.4704973
48	0.6082268	0.5864917	0.5649458	0.5435822	0.5223918	0.5013708	0.4805138
49	0.6156517	0.5943036	0.5731394	0.5521498	0.5313282	0.5106690	0.4901670
50	0.6228026	0.6018299	0.5810333	0.5604062	0.5399408	0.5196326	0.4994747

表 A.1 (续)

N/F	$\gamma=0.90$						
	14	15	16	17	18	19	20
51	0.6296951	0.6090840	0.5886445	0.5683676	0.5482478	0.5282783	0.5084553
52	0.6363421	0.6160806	0.5959861	0.5760497	0.5562642	0.5366238	0.5171244
53	0.6427563	0.6228340	0.6030734	0.5834654	0.5640039	0.5446828	0.5254975
54	0.6489495	0.6293560	0.6099182	0.5906291	0.5714819	0.5524705	0.5335904
55	0.6549336	0.6356578	0.6165337	0.5975539	0.5787106	0.5600000	0.5414154
56	0.6607181	0.6417501	0.6229301	0.6042498	0.5857022	0.5672828	0.5489862
57	0.6663133	0.6476441	0.6291184	0.6107284	0.5924681	0.5743315	0.5563138
58	0.6717283	0.6533479	0.6351086	0.6170005	0.5990183	0.5811566	0.5634102
59	0.6769708	0.6588719	0.6409095	0.6230754	0.6053640	0.5877687	0.5702862
60	0.6820501	0.6642238	0.6465302	0.6289618	0.6115131	0.5941773	0.5769507
61	0.6869724	0.6694109	0.6519791	0.6346693	0.6174756	0.6003917	0.5834146
62	0.6917461	0.6744413	0.6572633	0.6402042	0.6232588	0.6064203	0.5896855
63	0.6963763	0.6793215	0.6623910	0.6455758	0.6288712	0.6122718	0.5957724
64	0.7008708	0.6840587	0.6673679	0.6507900	0.6343204	0.6179531	0.6016826
65	0.7052341	0.6886586	0.6722013	0.6558546	0.6396132	0.6234710	0.6074241
66	0.7094734	0.6931271	0.6768968	0.6607748	0.6447558	0.6288337	0.6130047
67	0.7135923	0.6974698	0.6814603	0.6655576	0.6497541	0.6340467	0.6184294
68	0.7175967	0.7016922	0.6858976	0.6702075	0.6546151	0.6391158	0.6237049
69	0.7214907	0.7057978	0.6902140	0.6747310	0.6593441	0.6440480	0.6288388
70	0.7252796	0.7097937	0.6944132	0.6791326	0.6639459	0.6488476	0.6338348
75	0.7427808	0.7282512	0.7138185	0.6994750	0.6852170	0.6710383	0.6569367
80	0.7581920	0.7445087	0.7309139	0.7174014	0.7039668	0.6906040	0.6773113
85	0.7718652	0.7589362	0.7460882	0.7333164	0.7206163	0.7079826	0.6954115
90	0.7840772	0.7718247	0.7596470	0.7475395	0.7354984	0.7235183	0.7115969
95	0.7950513	0.7834073	0.7718342	0.7603260	0.7488796	0.7374904	0.7261546
100	0.8049657	0.7938739	0.7828478	0.7718827	0.7609755	0.7501210	0.7393176
110	0.8221737	0.8120429	0.8019701	0.7919522	0.7819847	0.7720645	0.7621885
120	0.8365945	0.8272716	0.8180021	0.8087807	0.7996051	0.7904707	0.7813767
130	0.8488540	0.8402203	0.8316353	0.8230939	0.8145936	0.8061315	0.7977045
140	0.8594036	0.8513650	0.8433699	0.8354153	0.8274981	0.8196159	0.8117666
160	0.8766284	0.8695629	0.8625354	0.8555423	0.8485813	0.8416500	0.8347461
180	0.8900952	0.8837935	0.8775246	0.8712862	0.8650752	0.8588903	0.8527289
200	0.9009117	0.8952251	0.8895674	0.8839365	0.8783307	0.8727471	0.8671848
250	0.9204804	0.9159090	0.9113604	0.9068325	0.9023236	0.8978330	0.8933585
300	0.9335949	0.9297740	0.9259707	0.9221850	0.9184145	0.9146592	0.9109170
350	0.9429967	0.9397138	0.9364470	0.9331939	0.9299545	0.9267273	0.9235116
400	0.9500673	0.9471897	0.9443257	0.9414748	0.9386345	0.9358057	0.9329860
450	0.9555767	0.9530157	0.9504670	0.9479289	0.9454007	0.9428825	0.9403726
500	0.9599917	0.9576842	0.9553874	0.9531013	0.9508236	0.9485543	0.9462926
550	0.9636083	0.9615086	0.9594195	0.9573389	0.9552659	0.9532013	0.9511436
600	0.9666252	0.9646991	0.9627830	0.9608737	0.9589728	0.9570788	0.9551909
650	0.9691799	0.9674019	0.9656308	0.9638680	0.9621121	0.9603623	0.9586186
700	0.9713722	0.9697194	0.9680749	0.9664365	0.9648050	0.9631796	0.9615595
750	0.9732724	0.9717293	0.9701938	0.9686644	0.9671411	0.9656231	0.9641097
800	0.9749365	0.9734897	0.9720490	0.9706143	0.9691858	0.9677619	0.9663433
850	0.9764054	0.9750435	0.9736870	0.9723366	0.9709908	0.9696503	0.9683143
900	0.9777118	0.9764245	0.9751432	0.9738673	0.9725968	0.9713300	0.9700678
950	0.9788815	0.9776618	0.9764474	0.9752384	0.9740339	0.9728333	0.9716372
1000	0.9799342	0.9787747	0.9776214	0.9764719	0.9753277	0.9741873	0.9730507

表 A.1 (续)

N/F	$\gamma=0.95$						
	0	1	2	3	4	5	6
1	0.0500000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.2236068	0.0253206	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.3684031	0.1372774	0.0169524	0.0000000	0.0000000	0.0000000	0.0000000
4	0.4728708	0.2495491	0.1006883	0.0127415	0.0000000	0.0000000	0.0000000
5	0.5492803	0.3430738	0.1910424	0.0801401	0.0102062	0.0000000	0.0000000
6	0.6069622	0.4184610	0.2723769	0.1555593	0.0626801	0.0085124	0.0000000
7	0.6518363	0.4794517	0.3418931	0.2268406	0.1286122	0.0532424	0.0073008
8	0.6876560	0.5294162	0.4007136	0.2902309	0.1928287	0.1109798	0.0462833
9	0.7168712	0.5709262	0.4506266	0.3456101	0.2513513	0.1686635	0.0976289
10	0.7411344	0.6058781	0.4932839	0.3938430	0.3035625	0.2223979	0.1499416
11	0.7615958	0.6356691	0.5300451	0.4359621	0.3498642	0.2712399	0.1995218
12	0.7790778	0.6613396	0.5619911	0.4729218	0.3909323	0.3152520	0.2452724
13	0.7941833	0.6836750	0.5899736	0.5055380	0.4274620	0.3548304	0.2870420
14	0.8073638	0.7032767	0.6146655	0.5344881	0.4600822	0.3904580	0.3250368
15	0.8189637	0.7206120	0.6366011	0.5603291	0.4893388	0.4226072	0.3595851
16	0.8292503	0.7360489	0.6562087	0.5835181	0.5156955	0.4517085	0.3910396
17	0.8384339	0.7498804	0.6738340	0.6044316	0.5395422	0.4781414	0.4197391
18	0.8466824	0.7623429	0.6897592	0.6233805	0.5612078	0.5022328	0.4459921
19	0.8541315	0.7736288	0.7042160	0.6406237	0.5809680	0.5242646	0.4700736
20	0.8608917	0.7838962	0.7173964	0.6563773	0.5990581	0.5444789	0.4922237
21	0.8670541	0.7932766	0.7294607	0.6708236	0.6156757	0.5630846	0.5126549
22	0.8726946	0.8018794	0.7405439	0.6841166	0.6309896	0.5802601	0.5315493
23	0.8778766	0.8097973	0.7507601	0.6963875	0.6451463	0.5961596	0.5490688
24	0.8826538	0.8171086	0.7602066	0.7077485	0.6582688	0.6109182	0.5653522
25	0.8870719	0.8238802	0.7689666	0.7182963	0.6704653	0.6246509	0.5805234
26	0.8911696	0.8301695	0.7771120	0.7281144	0.6818292	0.6374595	0.5946889
27	0.8949808	0.8360262	0.7847049	0.7372755	0.6924421	0.6494335	0.6079432
28	0.8985343	0.8414933	0.7917996	0.7458429	0.7023756	0.6606493	0.6203711
29	0.9018554	0.8466084	0.7984432	0.7538721	0.7116916	0.6711771	0.6320442
30	0.9049661	0.8514043	0.8046773	0.7614121	0.7204466	0.6810776	0.6430301
31	0.9078859	0.8559099	0.8105385	0.7685059	0.7286883	0.6904037	0.6533857
32	0.9106318	0.8601508	0.8160592	0.7751918	0.7364604	0.6992038	0.6631633
33	0.9132188	0.8641496	0.8212680	0.7815039	0.7438029	0.7075214	0.6724096
34	0.9156604	0.8679264	0.8261908	0.7874725	0.7507485	0.7153942	0.6811660
35	0.9179684	0.8714991	0.8308503	0.7931249	0.7573302	0.7228574	0.6894703
36	0.9201535	0.8748839	0.8352670	0.7984853	0.7635741	0.7299408	0.6973557
37	0.9222253	0.8780952	0.8394594	0.8035759	0.7695060	0.7366731	0.7048540
38	0.9241923	0.8811460	0.8434441	0.8084163	0.7751487	0.7430796	0.7119924
39	0.9260624	0.8840479	0.8472362	0.8130245	0.7805226	0.7491840	0.7187952
40	0.9278425	0.8868117	0.8508492	0.8174168	0.7856471	0.7550052	0.7252857
41	0.9295389	0.8894469	0.8542955	0.8216080	0.7905383	0.7605641	0.7314848
42	0.9311574	0.8919623	0.8575865	0.8256116	0.7952118	0.7658778	0.7374128
43	0.9327032	0.8943660	0.8607323	0.8294397	0.7996818	0.7709611	0.7430849
44	0.9341812	0.8966650	0.8637423	0.8331038	0.8039612	0.7758293	0.7485190
45	0.9355957	0.8988662	0.8666251	0.8366141	0.8080620	0.7804954	0.7537283
46	0.9369507	0.9009757	0.8693886	0.8399800	0.8119955	0.7849723	0.7587272
47	0.9382499	0.9029990	0.8720401	0.8432103	0.8157713	0.7892701	0.7635288
48	0.9394966	0.9049414	0.8745862	0.8463130	0.8193991	0.7934006	0.7681430
49	0.9406940	0.9068075	0.8770330	0.8492954	0.8228866	0.7973720	0.7725816
50	0.9418449	0.9086019	0.8793864	0.8521645	0.8262427	0.8011951	0.7768532

表 A.1 (续)

N/F	$\gamma=0.95$						
	0	1	2	3	4	5	6
51	0.9429520	0.9103285	0.8816514	0.8549266	0.8294738	0.8048763	0.7809685
52	0.9440178	0.9119911	0.8838330	0.8575875	0.8325873	0.8084241	0.7849354
53	0.9450445	0.9135932	0.8859357	0.8601527	0.8355897	0.8118451	0.7887612
54	0.9460342	0.9151380	0.8879637	0.8626272	0.8384865	0.8151470	0.7924537
55	0.9469889	0.9166287	0.8899209	0.8650158	0.8412829	0.8183350	0.7960196
56	0.9479105	0.9180678	0.8918110	0.8673228	0.8439839	0.8214148	0.7994661
57	0.9488005	0.9194582	0.8936373	0.8695525	0.8465951	0.8243927	0.8027976
58	0.9496607	0.9208022	0.8954030	0.8717085	0.8491203	0.8272723	0.8060209
59	0.9504924	0.9221021	0.8971111	0.8737944	0.8515641	0.8300602	0.8091399
60	0.9512971	0.9233600	0.8987644	0.8758137	0.8539300	0.8327589	0.8121616
61	0.9520760	0.9245780	0.9003654	0.8777695	0.8562214	0.8353740	0.8150888
62	0.9528305	0.9257578	0.9019166	0.8796646	0.8584426	0.8379083	0.8179265
63	0.9535615	0.9269014	0.9034202	0.8815020	0.8605958	0.8403656	0.8206779
64	0.9542703	0.9280102	0.9048785	0.8832842	0.8626850	0.8427501	0.8233486
65	0.9549577	0.9290860	0.9062934	0.8850135	0.8647123	0.8450644	0.8259406
66	0.9556248	0.9301300	0.9076669	0.8866925	0.8666808	0.8473118	0.8284577
67	0.9562724	0.9311438	0.9090007	0.8883231	0.8685932	0.8494946	0.8309033
68	0.9569014	0.9321286	0.9102966	0.8899075	0.8704508	0.8516166	0.8332799
69	0.9575126	0.9330857	0.9115560	0.8914477	0.8722577	0.8536794	0.8355910
70	0.9581067	0.9340161	0.9127807	0.8929453	0.8740144	0.8556859	0.8378387
75	0.9608441	0.9383054	0.9184282	0.8998542	0.8821200	0.8649453	0.8482169
80	0.9632458	0.9420713	0.9233892	0.9059258	0.8892467	0.8730899	0.8573482
85	0.9653699	0.9454040	0.9277816	0.9113037	0.8955616	0.8803087	0.8654442
90	0.9672620	0.9483741	0.9316977	0.9161002	0.9011957	0.8867505	0.8726716
95	0.9689580	0.9510378	0.9352111	0.9204048	0.9062525	0.8925355	0.8791627
100	0.9704870	0.9534402	0.9383808	0.9242894	0.9108182	0.8977580	0.8850244
110	0.9731336	0.9576009	0.9438729	0.9310225	0.9187331	0.9068163	0.8951939
120	0.9753446	0.9610791	0.9484663	0.9366562	0.9253584	0.9144006	0.9037113
130	0.9772194	0.9640300	0.9523648	0.9414392	0.9309849	0.9208436	0.9109487
140	0.9788292	0.9665650	0.9557150	0.9455508	0.9358233	0.9263855	0.9171751
160	0.9814509	0.9706955	0.9611762	0.9522554	0.9437152	0.9354267	0.9273364
180	0.9834948	0.9739177	0.9654384	0.9574900	0.9498789	0.9424908	0.9352774
200	0.9851330	0.9765015	0.9688574	0.9616903	0.9548256	0.9481616	0.9416541
250	0.9880886	0.9811660	0.9750323	0.9692791	0.9637664	0.9584137	0.9531854
300	0.9900639	0.9842854	0.9791638	0.9743586	0.9697534	0.9652810	0.9609116
350	0.9914773	0.9865184	0.9821221	0.9779967	0.9740423	0.9702010	0.9664482
400	0.9925386	0.9881957	0.9843448	0.9807308	0.9772661	0.9739002	0.9706113
450	0.9933649	0.9895018	0.9860760	0.9828605	0.9797772	0.9767821	0.9738557
500	0.9940264	0.9905477	0.9874624	0.9845663	0.9817891	0.9790911	0.9764548
550	0.9945680	0.9914041	0.9885977	0.9859633	0.9834373	0.9809828	0.9785840
600	0.9950196	0.9921182	0.9895445	0.9871284	0.9848116	0.9825598	0.9803598
650	0.9954018	0.9927227	0.9903461	0.9881149	0.9859752	0.9838958	0.9818638
700	0.9957295	0.9932411	0.9910336	0.9889609	0.9869733	0.9850419	0.9831541
750	0.9960137	0.9936906	0.9916296	0.9896945	0.9878387	0.9860354	0.9842725
800	0.9962623	0.9940840	0.9921514	0.9903367	0.9885962	0.9869052	0.9852516
850	0.9964818	0.9944313	0.9926119	0.9909035	0.9892649	0.9876727	0.9861162
900	0.9966769	0.9947400	0.9930213	0.9914076	0.9898598	0.9883558	0.9868853
950	0.9968516	0.9950163	0.9933878	0.9918587	0.9903920	0.9889663	0.9875734
1000	0.9970088	0.9952650	0.9937177	0.9922648	0.9908711	0.9895163	0.9881928

表 A.1 (续)

N/F	$\gamma=0.95$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0063912	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0409375	0.0056830	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0871608	0.0367008	0.0051162	0.0000000	0.0000000	0.0000000	0.0000000
11	0.1349929	0.0787289	0.0332605	0.0046522	0.0000000	0.0000000	0.0000000
12	0.1809668	0.1227738	0.0717899	0.0304106	0.0042653	0.0000000	0.0000000
13	0.2239198	0.1656017	0.1125953	0.0659784	0.0280111	0.0039379	0.0000000
14	0.2635662	0.2060338	0.1526624	0.1039832	0.0610397	0.0259629	0.0036571
15	0.2999818	0.2437025	0.1908237	0.1416116	0.0965994	0.0567897	0.0241944
16	0.3333797	0.2785900	0.2266635	0.1777255	0.1320613	0.0901981	0.0530948
17	0.3640222	0.3108261	0.2600966	0.2118779	0.1663235	0.1237246	0.0845956
18	0.3921745	0.3406015	0.2911914	0.2439405	0.1989226	0.1563058	0.1163821
19	0.4180882	0.3681246	0.3200837	0.2739325	0.2296986	0.1874743	0.1474335
20	0.4419925	0.3935985	0.3469341	0.3019466	0.2586345	0.2170452	0.1772815
21	0.4640936	0.4172161	0.3719080	0.3281067	0.2857901	0.2449772	0.2057269
22	0.4845742	0.4391517	0.3951659	0.3525464	0.3112583	0.2713007	0.2327057
23	0.5035970	0.4595656	0.4168579	0.3753991	0.3351458	0.2960858	0.2582296
24	0.5213044	0.4786001	0.4371219	0.3967927	0.3575651	0.3194164	0.2823457
25	0.5378241	0.4963829	0.4560831	0.4168483	0.3786260	0.3413870	0.3051232
26	0.5532664	0.5130264	0.4738553	0.4356746	0.3984307	0.3620901	0.3266384
27	0.5677308	0.5286338	0.4905409	0.4533726	0.4170764	0.3816156	0.3469704
28	0.5813045	0.5432945	0.5062306	0.4700349	0.4346540	0.4000490	0.3661979
29	0.5940664	0.5570896	0.5210077	0.4857445	0.4512449	0.4174702	0.3843951
30	0.6060843	0.5700913	0.5349475	0.5005766	0.4669251	0.4339527	0.4016325
31	0.6174213	0.5823647	0.5481162	0.5146000	0.4817629	0.4495651	0.4179784
32	0.6281314	0.5939685	0.5605739	0.5278758	0.4958209	0.4643695	0.4334934
33	0.6382659	0.6049531	0.5723750	0.5404606	0.5091574	0.4784248	0.4482347
34	0.6478678	0.6153677	0.5835702	0.5524060	0.5218231	0.4917827	0.4622557
35	0.6569789	0.6252543	0.5942025	0.5637566	0.5338660	0.5044920	0.4756039
36	0.6656351	0.6346507	0.6043132	0.5745564	0.5453298	0.5165960	0.4883254
37	0.6738681	0.6435925	0.6139386	0.5848425	0.5562545	0.5281365	0.5004602
38	0.6817093	0.6521113	0.6231131	0.5946504	0.5666752	0.5391501	0.5120477
39	0.6891846	0.6602370	0.6318661	0.6040118	0.5766260	0.5496719	0.5231215
40	0.6963199	0.6679943	0.6402263	0.6129558	0.5861362	0.5597316	0.5337146
41	0.7031372	0.6754083	0.6482188	0.6215100	0.5952353	0.5693604	0.5438563
42	0.7096566	0.6825016	0.6558677	0.6296984	0.6039479	0.5785828	0.5535747
43	0.7158977	0.6892933	0.6631940	0.6375434	0.6122986	0.5874245	0.5628946
44	0.7218779	0.6958021	0.6702168	0.6450665	0.6203082	0.5959078	0.5718393
45	0.7276128	0.7020466	0.6769557	0.6522868	0.6279970	0.6040536	0.5804314
46	0.7331169	0.7080407	0.6834268	0.6592211	0.6353839	0.6118816	0.5886898
47	0.7384047	0.7138002	0.6896450	0.6658873	0.6424859	0.6194103	0.5966338
48	0.7434874	0.7193375	0.6956256	0.6722990	0.6493195	0.6266551	0.6042807
49	0.7483770	0.7246668	0.7013816	0.6784718	0.6558992	0.6336317	0.6116466
50	0.7530850	0.7297975	0.7069251	0.6844181	0.6622376	0.6403554	0.6187464

表 A.1 (续)

N/F	$\gamma=0.95$						
	7	8	9	10	11	12	13
51	0.7576208	0.7347422	0.7122680	0.6901493	0.6683495	0.6468389	0.6255938
52	0.7619937	0.7395097	0.7174202	0.6956779	0.6742460	0.6530949	0.6322017
53	0.7662113	0.7441093	0.7223925	0.7010138	0.6799371	0.6591352	0.6385834
54	0.7702837	0.7485502	0.7271935	0.7061665	0.6854347	0.6649699	0.6447492
55	0.7742161	0.7528406	0.7318321	0.7111463	0.6907482	0.6706102	0.6507102
56	0.7780170	0.7569875	0.7363167	0.7159609	0.6958859	0.6760642	0.6564759
57	0.7816924	0.7609976	0.7406537	0.7206181	0.7008564	0.6813426	0.6620554
58	0.7852486	0.7648784	0.7448516	0.7251261	0.7056683	0.6864532	0.6674586
59	0.7886912	0.7686353	0.7489159	0.7294910	0.7103286	0.6914027	0.6726927
60	0.7920251	0.7722746	0.7528537	0.7337212	0.7148451	0.6962001	0.6777657
61	0.7952560	0.7758016	0.7566700	0.7378207	0.7192231	0.7008506	0.6826856
62	0.7983886	0.7792215	0.7603703	0.7417969	0.7234690	0.7053625	0.6874573
63	0.8014266	0.7825384	0.7639608	0.7456548	0.7275891	0.7097408	0.6920894
64	0.8043750	0.7857578	0.7674457	0.7493991	0.7315891	0.7139911	0.6965869
65	0.8072365	0.7888832	0.7708291	0.7530359	0.7354738	0.7181201	0.7009555
66	0.8100164	0.7919192	0.7741157	0.7565685	0.7392480	0.7221319	0.7052004
67	0.8127171	0.7948696	0.7773098	0.7600026	0.7429166	0.7260312	0.7093282
68	0.8153430	0.7977372	0.7804159	0.7633404	0.7464838	0.7298240	0.7133412
69	0.8178954	0.8005263	0.7834358	0.7665878	0.7499543	0.7335130	0.7172464
70	0.8203792	0.8032401	0.7863741	0.7697469	0.7533303	0.7371029	0.7210472
75	0.8318463	0.8157717	0.7999489	0.7843458	0.7689358	0.7536997	0.7386192
80	0.8419391	0.8268054	0.8119052	0.7972080	0.7826900	0.7683307	0.7541164
85	0.8508909	0.8365940	0.8225153	0.8086259	0.7949019	0.7813260	0.7678843
90	0.8588841	0.8453369	0.8319942	0.8188285	0.8058169	0.7929442	0.7801965
95	0.8660645	0.8531929	0.8405136	0.8279998	0.8156318	0.8033926	0.7912710
100	0.8725502	0.8602904	0.8482114	0.8362888	0.8245027	0.8128395	0.8012854
110	0.8838051	0.8726092	0.8615759	0.8506829	0.8399127	0.8292517	0.8186883
120	0.8932348	0.8829339	0.8727802	0.8627531	0.8528374	0.8430210	0.8332922
130	0.9012498	0.8917105	0.8823069	0.8730193	0.8638331	0.8547370	0.8457218
140	0.9081455	0.8992639	0.8905074	0.8818577	0.8733012	0.8648277	0.8564283
160	0.9194026	0.9115978	0.9039005	0.8962955	0.8887714	0.8813182	0.8739292
180	0.9282028	0.9212419	0.9143756	0.9075902	0.9008765	0.8942246	0.8876292
200	0.9352709	0.9289891	0.9227921	0.9166675	0.9106063	0.9046007	0.8986447
250	0.9480555	0.9430057	0.9380230	0.9330976	0.9282217	0.9233901	0.9185974
300	0.9566230	0.9524017	0.9482352	0.9441168	0.9400388	0.9359975	0.9319874
350	0.9627648	0.9591386	0.9555589	0.9520197	0.9485156	0.9450420	0.9415959
400	0.9673834	0.9642044	0.9610666	0.9579647	0.9548924	0.9518469	0.9488251
450	0.9709827	0.9681532	0.9653603	0.9625987	0.9598637	0.9571525	0.9544626
500	0.9738665	0.9713172	0.9688007	0.9663124	0.9638478	0.9614053	0.9589803
550	0.9762287	0.9739093	0.9716196	0.9693551	0.9671128	0.9648895	0.9626837
600	0.9781995	0.9760721	0.9739713	0.9718941	0.9698369	0.9677971	0.9657734
650	0.9798683	0.9779034	0.9759636	0.9740444	0.9721443	0.9702603	0.9683907
700	0.9813005	0.9794745	0.9776721	0.9758895	0.9741237	0.9723732	0.9706356
750	0.9825417	0.9808368	0.9791540	0.9774888	0.9758404	0.9742057	0.9725832
800	0.9836286	0.9820299	0.9804511	0.9788898	0.9773438	0.9758100	0.9742884
850	0.9845881	0.9830828	0.9815965	0.9801263	0.9786706	0.9772271	0.9757943
900	0.9854415	0.9840192	0.9826151	0.9812263	0.9798504	0.9784868	0.9771330
950	0.9862049	0.9848578	0.9835267	0.9822108	0.9809072	0.9796150	0.9783320
1000	0.9868930	0.9856122	0.9843475	0.9830973	0.9818585	0.9806303	0.9794106

表 A.1 (续)

N/F	$\gamma=0.95$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0034137	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0226512	0.0032007	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.0498520	0.0212935	0.0030127	0.0000000	0.0000000	0.0000000	0.0000000
18	0.0796498	0.0469824	0.0200890	0.0028456	0.0000000	0.0000000	0.0000000
19	0.1098661	0.0752514	0.0444261	0.0190139	0.0026960	0.0000000	0.0000000
20	0.1395191	0.1040431	0.0713146	0.0421337	0.0180477	0.0025614	0.0000000
21	0.1681474	0.1324157	0.0988080	0.0677699	0.0400666	0.0171752	0.0024396
22	0.1955398	0.1599134	0.1260029	0.0940760	0.0645616	0.0381927	0.0163832
23	0.2216170	0.1863213	0.1524530	0.1201854	0.0897778	0.0616438	0.0364867
24	0.2463743	0.2115467	0.1779382	0.1456610	0.1148830	0.0858559	0.0589789
25	0.2698419	0.2355716	0.2023595	0.1702819	0.1394509	0.1100298	0.0822635
26	0.2920747	0.2584128	0.2256847	0.1939424	0.1632613	0.1337506	0.1055712
27	0.3131339	0.2801114	0.2479221	0.2166018	0.1862023	0.1567985	0.1285002
28	0.3330875	0.3007187	0.2691014	0.2382592	0.2082273	0.1790598	0.1508307
29	0.3520035	0.3202918	0.2892643	0.2589343	0.2293275	0.2004805	0.1724477
30	0.3699479	0.3388904	0.3084585	0.2786614	0.2495159	0.2210470	0.1932938
31	0.3869845	0.3565713	0.3267363	0.2974811	0.2688181	0.2407646	0.2133481
32	0.4031719	0.3733913	0.3441463	0.3154369	0.2872684	0.2596530	0.2326114
33	0.4185664	0.3894040	0.3607413	0.3325737	0.3049044	0.2777416	0.2510991
34	0.4332201	0.4046605	0.3765671	0.3489360	0.3217658	0.2950624	0.2688337
35	0.4471813	0.4192072	0.3916719	0.3645676	0.3378928	0.3116491	0.2858426
36	0.4604950	0.4330895	0.4060968	0.3795092	0.3533229	0.3275379	0.3021573
37	0.4732035	0.4463489	0.4198842	0.3938017	0.3680945	0.3427635	0.3178079
38	0.4853444	0.4590233	0.4330721	0.4074811	0.3822449	0.3573596	0.3328261
39	0.4969525	0.4711482	0.4456956	0.4205841	0.3958067	0.3713604	0.3472437
40	0.5080623	0.4827579	0.4577876	0.4331424	0.4088138	0.3847972	0.3610904
41	0.5187031	0.4938825	0.4693801	0.4451875	0.4212955	0.3976994	0.3743956
42	0.5289030	0.5045495	0.4805020	0.4567489	0.4332819	0.4100957	0.3871857
43	0.5386882	0.5147877	0.4911794	0.4678526	0.4447990	0.4220124	0.3994890
44	0.5480829	0.5246201	0.5014374	0.4785247	0.4558729	0.4334760	0.4113286
45	0.5571090	0.5340696	0.5112996	0.4887883	0.4665271	0.4445094	0.4227296
46	0.5657872	0.5431578	0.5207877	0.4986663	0.4767845	0.4551354	0.4337137
47	0.5741380	0.5519047	0.5299216	0.5081781	0.4866642	0.4653739	0.4443018
48	0.5821778	0.5603290	0.5387213	0.5173440	0.4961879	0.4752462	0.4545136
49	0.5899239	0.5684468	0.5472025	0.5261809	0.5053722	0.4847702	0.4643682
50	0.5973914	0.5762744	0.5553833	0.5347066	0.5142359	0.4939635	0.4738834

表 A.1 (续)

N/F	$\gamma=0.95$						
	14	15	16	17	18	19	20
51	0.6045951	0.5838276	0.5632783	0.5429365	0.5227931	0.5028412	0.4830746
52	0.6115488	0.5911195	0.5709022	0.5508849	0.5310598	0.5114201	0.4919589
53	0.6182652	0.5981636	0.5782680	0.5585661	0.5390505	0.5197141	0.5005494
54	0.6247552	0.6049725	0.5853889	0.5659938	0.5467787	0.5277360	0.5088605
55	0.6310307	0.6115565	0.5922761	0.5731787	0.5542553	0.5354998	0.5169060
56	0.6371013	0.6179273	0.5989410	0.5801325	0.5614934	0.5430167	0.5246965
57	0.6429773	0.6240945	0.6053940	0.5868667	0.5685042	0.5502989	0.5322446
58	0.6486685	0.6300675	0.6116451	0.5933913	0.5752970	0.5573552	0.5395607
59	0.6541818	0.6358563	0.6177040	0.5997149	0.5818823	0.5641977	0.5466557
60	0.6595266	0.6414676	0.6235780	0.6058477	0.5882686	0.5708344	0.5535383
61	0.6647098	0.6469103	0.6292755	0.6117972	0.5944653	0.5772746	0.5602189
62	0.6697383	0.6521910	0.6348054	0.6175708	0.6004805	0.5835275	0.5667050
63	0.6746195	0.6573183	0.6401742	0.6231782	0.6063225	0.5895995	0.5730048
64	0.6793598	0.6622974	0.6453884	0.6286243	0.6119968	0.5954997	0.5791271
65	0.6839642	0.6671345	0.6504552	0.6339170	0.6175124	0.6012344	0.5850784
66	0.6884399	0.6718365	0.6553803	0.6390623	0.6228747	0.6068107	0.5908657
67	0.6927907	0.6764081	0.6601697	0.6440656	0.6280897	0.6122350	0.5964956
68	0.6970225	0.6808556	0.6648283	0.6489337	0.6331643	0.6175124	0.6019741
69	0.7011400	0.6851824	0.6693621	0.6536716	0.6381031	0.6226506	0.6073079
70	0.7051479	0.6893943	0.6737757	0.6582845	0.6429116	0.6276532	0.6125015
75	0.7236829	0.7088787	0.6941972	0.6796302	0.6651715	0.6508151	0.6365556
80	0.7400340	0.7260730	0.7122241	0.6984806	0.6848354	0.6712833	0.6578189
85	0.7545648	0.7413573	0.7282537	0.7152461	0.7023294	0.6894981	0.6767469
90	0.7675618	0.7550317	0.7425976	0.7302528	0.7179920	0.7058097	0.6937022
95	0.7792554	0.7673368	0.7555073	0.7437619	0.7320949	0.7205005	0.7089747
100	0.7898305	0.7784679	0.7671884	0.7559876	0.7448592	0.7337987	0.7228031
110	0.8082134	0.7978194	0.7875010	0.7772511	0.7670654	0.7569400	0.7468717
120	0.8236436	0.8140682	0.8045591	0.7951127	0.7857242	0.7763891	0.7671052
130	0.8367790	0.8279026	0.8190873	0.8103276	0.8016206	0.7929624	0.7843500
140	0.8480953	0.8398233	0.8316069	0.8234425	0.8153254	0.8072525	0.7992223
160	0.8665973	0.8593173	0.8520847	0.8448970	0.8377490	0.8306400	0.8235652
180	0.8810841	0.8745840	0.8681259	0.8617059	0.8553218	0.8489705	0.8426505
200	0.8927338	0.8868633	0.8810301	0.8752306	0.8694623	0.8637238	0.8580120
250	0.9138398	0.9091135	0.9044169	0.8997463	0.8951001	0.8904768	0.8858740
300	0.9280071	0.9240520	0.9201213	0.9162119	0.9123232	0.9084520	0.9045991
350	0.9381742	0.9347738	0.9313948	0.9280334	0.9246887	0.9213601	0.9180467
400	0.9458246	0.9428432	0.9398793	0.9369315	0.9339989	0.9310793	0.9281719
450	0.9517910	0.9491362	0.9464974	0.9438723	0.9412602	0.9386596	0.9360712
500	0.9565729	0.9541808	0.9518024	0.9494361	0.9470821	0.9447388	0.9424047
550	0.9604924	0.9583149	0.9561504	0.9539972	0.9518540	0.9497208	0.9475967
600	0.9637626	0.9617656	0.9597793	0.9578029	0.9558371	0.9538798	0.9519301
650	0.9665333	0.9646881	0.9628529	0.9610276	0.9592114	0.9574029	0.9556019
700	0.9689102	0.9671956	0.9654908	0.9637944	0.9621065	0.9604261	0.9587527
750	0.9709722	0.9693703	0.9677783	0.9661940	0.9646180	0.9630482	0.9614852
800	0.9727774	0.9712749	0.9697815	0.9682958	0.9668169	0.9653449	0.9638790
850	0.9743707	0.9729569	0.9715500	0.9701515	0.9687592	0.9673729	0.9659919
900	0.9757885	0.9744523	0.9731237	0.9718021	0.9704865	0.9691762	0.9678721
950	0.9770573	0.9757911	0.9745317	0.9732791	0.9720320	0.9707909	0.9695544
1000	0.9782000	0.9769963	0.9758003	0.9746095	0.9734249	0.9722448	0.9710701

表 A.1 (续)

N/F	$\gamma=0.99$						
	0	1	2	3	4	5	6
1	0.0100000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.1000000	0.0050126	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.2154435	0.0616714	0.0033445	0.0000000	0.0000000	0.0000000	0.0000000
4	0.3162278	0.1429335	0.0459344	0.0025094	0.0000000	0.0000000	0.0000000
5	0.3981072	0.2234577	0.1089534	0.0371265	0.0020080	0.0000000	0.0000000
6	0.4641589	0.2952479	0.1755031	0.0887793	0.0264764	0.0016737	0.0000000
7	0.5179475	0.3572833	0.2380806	0.1454380	0.0704671	0.0224502	0.0014347
8	0.5623413	0.4105214	0.2945152	0.2006066	0.1206905	0.0605617	0.0194911
9	0.5994843	0.4563073	0.3446435	0.2518560	0.1707979	0.1050267	0.0531167
10	0.6309573	0.4959046	0.3889864	0.2985257	0.2182498	0.1502712	0.0930049
11	0.6579332	0.5303824	0.4282431	0.3406851	0.2621742	0.1938641	0.1342234
12	0.6812921	0.5606125	0.4631040	0.3786814	0.3024214	0.2348290	0.1744907
13	0.7017038	0.5872983	0.4941878	0.4129450	0.3391502	0.2728617	0.2127973
14	0.7196857	0.6110058	0.5220273	0.4439054	0.3726387	0.3079736	0.2487520
15	0.7356423	0.6321927	0.5470738	0.4719592	0.4031960	0.3403158	0.2822681
16	0.7498942	0.6512309	0.5697068	0.4974586	0.4311290	0.3700977	0.3134075
17	0.7626986	0.6684249	0.5902449	0.5207116	0.4567198	0.3975415	0.3423045
18	0.7742637	0.6840255	0.6089563	0.5419847	0.4802218	0.4228680	0.3691206
19	0.7847600	0.6982412	0.6260675	0.5615080	0.5018619	0.4462808	0.3940245
20	0.7943282	0.7112460	0.6417702	0.5794799	0.5218395	0.4679676	0.4171787
21	0.8030857	0.7231866	0.6562278	0.5960717	0.5403271	0.4880968	0.4387390
22	0.8111308	0.7341871	0.6695800	0.6114315	0.5574796	0.5068180	0.4588465
23	0.8185467	0.7443533	0.6819468	0.6256881	0.5734307	0.5242663	0.4776302
24	0.8254042	0.7537759	0.6934319	0.6389534	0.5882981	0.5405600	0.4952075
25	0.8317638	0.7625329	0.7041249	0.6513251	0.6021845	0.5558055	0.5116834
26	0.8376776	0.7706920	0.7141042	0.6628889	0.6151826	0.5700965	0.5271535
27	0.8431909	0.7783121	0.7234382	0.6737202	0.6273717	0.5835178	0.5417025
28	0.8483429	0.7854444	0.7321868	0.6838852	0.6388250	0.5961436	0.5554068
29	0.8531679	0.7921343	0.7404029	0.6934429	0.6496044	0.6080408	0.5683357
30	0.8576959	0.7984213	0.7481334	0.7024453	0.6597677	0.6192689	0.5805499
31	0.8619536	0.8043407	0.7554197	0.7109389	0.6693651	0.6298820	0.5921064
32	0.8659643	0.8099236	0.7622988	0.7189653	0.6784419	0.6399280	0.6030544
33	0.8697490	0.8151979	0.7688036	0.7265616	0.6870386	0.6494503	0.6134412
34	0.8733262	0.8201883	0.7749638	0.7337610	0.6951920	0.6584888	0.6233062
35	0.8767124	0.8249171	0.7808058	0.7405937	0.7029357	0.6670774	0.6326886
36	0.8799225	0.8294042	0.7863535	0.7470868	0.7102981	0.6752502	0.6416207
37	0.8829700	0.8336677	0.7916285	0.7532648	0.7173080	0.6830354	0.6501345
38	0.8858668	0.8377239	0.7966504	0.7591499	0.7239890	0.6904596	0.6582585
39	0.8886238	0.8415874	0.8014368	0.7647624	0.7303639	0.6975467	0.6660173
40	0.8912509	0.8452717	0.8060040	0.7701205	0.7364520	0.7043193	0.6734363
41	0.8937571	0.8487888	0.8103664	0.7752412	0.7422739	0.7107983	0.6805357
42	0.8961505	0.8521499	0.8145376	0.7801398	0.7478454	0.7170008	0.6873357
43	0.8984385	0.8553652	0.8185299	0.7848304	0.7531822	0.7229455	0.6938555
44	0.9006280	0.8584438	0.8223544	0.7893259	0.7582993	0.7286468	0.7001105
45	0.9027252	0.8613944	0.8260214	0.7936381	0.7632095	0.7341201	0.7061178
46	0.9047357	0.8642246	0.8295406	0.7977779	0.7679251	0.7393783	0.7118904
47	0.9066649	0.8669418	0.8329205	0.8017556	0.7724577	0.7444329	0.7174427
48	0.9085176	0.8695526	0.8361694	0.8055803	0.7768171	0.7492974	0.7227863
49	0.9102982	0.8720630	0.8392946	0.8092607	0.7810134	0.7539806	0.7279328
50	0.9120108	0.8744787	0.8423030	0.8128048	0.7850554	0.7584926	0.7328934

表 A.1 (续)

N/F	$\gamma=0.99$						
	0	1	2	3	4	5	6
51	0.9136594	0.8768051	0.8452012	0.8162200	0.7889510	0.7628437	0.7376771
52	0.9152473	0.8790468	0.8479950	0.8195132	0.7927092	0.7670408	0.7422932
53	0.9167779	0.8812086	0.8506899	0.8226908	0.7963357	0.7710934	0.7467507
54	0.9182543	0.8832945	0.8532911	0.8257587	0.7998383	0.7750072	0.7510581
55	0.9196792	0.8853086	0.8558034	0.8287225	0.8032229	0.7787903	0.7552214
56	0.9210553	0.8872544	0.8582312	0.8315875	0.8064946	0.7824485	0.7592485
57	0.9223851	0.8891353	0.8605788	0.8343584	0.8096604	0.7859886	0.7631461
58	0.9236709	0.8909545	0.8628500	0.8370398	0.8127243	0.7894160	0.7669195
59	0.9249147	0.8927151	0.8650485	0.8396359	0.8156914	0.7927349	0.7705756
60	0.9261187	0.8944198	0.8671778	0.8421508	0.8185662	0.7959517	0.7741191
61	0.9272847	0.8960712	0.8692410	0.8445883	0.8213528	0.7990703	0.7775552
62	0.9284145	0.8976718	0.8712412	0.8469517	0.8240561	0.8020952	0.7808888
63	0.9295098	0.8992239	0.8731812	0.8492445	0.8266778	0.8050308	0.7841247
64	0.9305720	0.9007297	0.8750637	0.8514697	0.8292232	0.8078807	0.7872661
65	0.9316028	0.9021911	0.8768912	0.8536303	0.8316954	0.8106490	0.7903182
66	0.9326033	0.9036102	0.8786660	0.8557290	0.8340972	0.8133388	0.7932839
67	0.9335751	0.9049887	0.8803905	0.8577685	0.8364315	0.8159534	0.7961674
68	0.9345192	0.9063284	0.8820667	0.8597513	0.8387003	0.8184959	0.7989721
69	0.9354369	0.9076309	0.8836966	0.8616796	0.8409078	0.8209692	0.8017004
70	0.9363292	0.9088976	0.8852822	0.8635557	0.8430558	0.8233761	0.8043564
75	0.9404449	0.9147440	0.8926034	0.8722222	0.8529826	0.8345040	0.8166365
80	0.9440609	0.9198856	0.8990469	0.8798548	0.8617298	0.8443149	0.8274700
85	0.9472630	0.9244427	0.9047615	0.8866278	0.8694958	0.8530298	0.8370972
90	0.9501185	0.9285094	0.9098643	0.8926786	0.8764368	0.8608220	0.8457078
95	0.9526807	0.9321609	0.9144484	0.8981167	0.8826776	0.8678300	0.8534562
100	0.9549926	0.9354576	0.9185889	0.9030307	0.8883184	0.8741672	0.8604639
110	0.9589991	0.9411752	0.9257743	0.9115626	0.8981170	0.8851793	0.8726468
120	0.9623506	0.9459624	0.9317947	0.9187152	0.9063360	0.8944207	0.8828746
130	0.9651957	0.9500293	0.9369119	0.9247978	0.9133282	0.9022866	0.8915830
140	0.9676411	0.9535269	0.9413151	0.9300338	0.9193499	0.9090622	0.8990873
160	0.9716280	0.9592339	0.9485038	0.9385864	0.9291896	0.9201381	0.9113590
180	0.9747402	0.9636926	0.9541237	0.9452762	0.9368907	0.9288098	0.9209715
200	0.9772372	0.9672723	0.9586378	0.9506519	0.9430802	0.9357837	0.9287032
250	0.9817479	0.9737441	0.9668042	0.9603820	0.9542899	0.9484169	0.9427156
300	0.9847667	0.9780790	0.9722777	0.9669073	0.9618112	0.9568970	0.9521256
350	0.9869286	0.9811853	0.9762018	0.9715872	0.9672072	0.9629828	0.9588806
400	0.9885531	0.9835206	0.9791527	0.9751075	0.9712671	0.9675632	0.9639653
450	0.9898185	0.9853402	0.9814526	0.9778516	0.9744328	0.9711341	0.9679307
500	0.9908319	0.9867979	0.9832954	0.9800508	0.9769697	0.9739977	0.9711096
550	0.9916619	0.9879920	0.9848052	0.9818528	0.9790489	0.9763441	0.9737157
600	0.9923541	0.9889879	0.9860647	0.9833562	0.9807841	0.9783018	0.9758905
650	0.9929402	0.9898313	0.9871313	0.9846295	0.9822536	0.9799606	0.9777324
700	0.9934428	0.9905547	0.9880463	0.9857219	0.9835141	0.9813836	0.9793134
750	0.9938786	0.9911821	0.9888398	0.9866693	0.9846076	0.9826182	0.9806844
800	0.9942601	0.9917312	0.9895345	0.9874988	0.9855651	0.9836986	0.9818855
850	0.9945968	0.9922160	0.9901478	0.9882311	0.9864100	0.9846529	0.9829455
900	0.9948962	0.9926471	0.9906932	0.9888824	0.9871620	0.9855016	0.9838885
950	0.9951642	0.9930330	0.9911814	0.9894653	0.9878346	0.9862617	0.9847330
1000	0.9954054	0.9933803	0.9916209	0.9899902	0.9884411	0.9869460	0.9854928

表 A.1 (续)

N/F	$\gamma=0.99$						
	7	8	9	10	11	12	13
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0012555	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0172238	0.0011161	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0473117	0.0154300	0.0010045	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0834762	0.0426563	0.0139756	0.0009132	0.0000000	0.0000000	0.0000000
12	0.1213141	0.0757317	0.0388378	0.0127722	0.0008372	0.0000000	0.0000000
13	0.1587007	0.1106953	0.0693107	0.0356493	0.0117597	0.0007728	0.0000000
14	0.1946323	0.1455700	0.1018011	0.0638984	0.0329460	0.0108964	0.0007176
15	0.2286668	0.1793776	0.1344699	0.0942399	0.0592741	0.0306247	0.0101517
16	0.2606555	0.2116527	0.1663745	0.1249591	0.0877307	0.0552761	0.0286098
17	0.2905984	0.2422028	0.1970397	0.1551526	0.1167164	0.0820681	0.0517854
18	0.3185715	0.2709840	0.2262470	0.1843457	0.1453650	0.1095010	0.0770953
19	0.3446889	0.2980311	0.2539185	0.2123015	0.1732098	0.1367510	0.1031319
20	0.3690776	0.3234188	0.2800571	0.2389209	0.2000034	0.1633579	0.1291088
21	0.3918692	0.3472439	0.3047106	0.2641831	0.2256300	0.1890713	0.1545779
22	0.4131906	0.3696090	0.3279471	0.2881101	0.2500507	0.2137644	0.1792878
23	0.4331594	0.3906180	0.3498497	0.3107522	0.2732707	0.2373829	0.2031026
24	0.4518883	0.4103720	0.3705005	0.3321717	0.2953215	0.2599185	0.2259583
25	0.4694785	0.4289635	0.3899851	0.3524372	0.3162519	0.2813895	0.2478356
26	0.4860232	0.4464836	0.4083837	0.3716174	0.3361145	0.3018293	0.2687412
27	0.5016061	0.4630127	0.4257727	0.3897824	0.3549693	0.3212839	0.2886986
28	0.5163043	0.4786255	0.4422253	0.4069993	0.3728735	0.3397990	0.3077414
29	0.5301870	0.4933918	0.4578066	0.4233299	0.3898863	0.3574253	0.3259112
30	0.5433180	0.5073746	0.4725803	0.4388342	0.4060624	0.3742130	0.3432478
31	0.5557536	0.5206307	0.4866019	0.4535671	0.4214548	0.3902099	0.3597951
32	0.5675456	0.5332134	0.4999248	0.4675815	0.4361118	0.4054637	0.3755962
33	0.5787422	0.5451707	0.5125956	0.4809231	0.4500815	0.4200172	0.3906915
34	0.5893847	0.5565450	0.5246606	0.4936375	0.4634063	0.4339145	0.4051214
35	0.5995128	0.5673777	0.5361589	0.5057655	0.4761275	0.4471937	0.4189236
36	0.6091623	0.5777056	0.5471286	0.5173436	0.4882817	0.4598913	0.4321334
37	0.6183650	0.5875607	0.5576047	0.5284079	0.4999038	0.4720436	0.4447854
38	0.6271507	0.5969752	0.5676168	0.5389893	0.5110271	0.4836806	0.4569116
39	0.6355469	0.6059775	0.5771954	0.5491183	0.5216806	0.4948341	0.4685408
40	0.6435779	0.6145923	0.5863674	0.5588222	0.5318927	0.5055316	0.4797008
41	0.6512680	0.6228441	0.5951572	0.5681256	0.5416892	0.5157990	0.4904185
42	0.6586365	0.6307552	0.6035872	0.5770533	0.5510938	0.5256607	0.5007182
43	0.6657031	0.6383458	0.6116788	0.5856261	0.5601288	0.5351395	0.5106226
44	0.6724867	0.6456342	0.6194524	0.5938648	0.5688144	0.5442553	0.5201525
45	0.6790029	0.6526387	0.6269245	0.6017878	0.5771707	0.5530297	0.5293289
46	0.6852677	0.6593743	0.6341135	0.6094118	0.5852153	0.5614795	0.5381688
47	0.6912947	0.6658563	0.6410335	0.6167548	0.5929651	0.5696224	0.5466910
48	0.6970968	0.6720994	0.6477001	0.6238302	0.6004357	0.5774738	0.5549116
49	0.7026869	0.6781153	0.6541267	0.6306523	0.6076410	0.5850493	0.5628459
50	0.7080754	0.6839159	0.6603247	0.6372348	0.6145943	0.5923627	0.5705072

表 A.1 (续)

N/F	$\gamma=0.99$						
	7	8	9	10	11	12	13
51	0.7132741	0.6895136	0.6663078	0.6435902	0.6213097	0.5994268	0.5779101
52	0.7182925	0.6949182	0.6720856	0.6497283	0.6277983	0.6062543	0.5850665
53	0.7231388	0.7001395	0.6776683	0.6556616	0.6340707	0.6128568	0.5919884
54	0.7278223	0.7051862	0.6830659	0.6613995	0.6401382	0.6192439	0.5986868
55	0.7323513	0.7100672	0.6882873	0.6669508	0.6460094	0.6254266	0.6051726
56	0.7367328	0.7147901	0.6933411	0.6723246	0.6516949	0.6314147	0.6114549
57	0.7409735	0.7193632	0.6982350	0.6775296	0.6572025	0.6372165	0.6175425
58	0.7450814	0.7237926	0.7029761	0.6825739	0.6625401	0.6428406	0.6234454
59	0.7490610	0.7280851	0.7075715	0.6874638	0.6677161	0.6482951	0.6291715
60	0.7529191	0.7322470	0.7120281	0.6922059	0.6727369	0.6535869	0.6347283
61	0.7566612	0.7362844	0.7163517	0.6968081	0.6776101	0.6587241	0.6401220
62	0.7602912	0.7402025	0.7205480	0.7012756	0.6823413	0.6637121	0.6453614
63	0.7638160	0.7440062	0.7246229	0.7056142	0.6869366	0.6685581	0.6504520
64	0.7672389	0.7477008	0.7285816	0.7098293	0.6914020	0.6732677	0.6553996
65	0.7705642	0.7512909	0.7324288	0.7139261	0.6957422	0.6778460	0.6602107
66	0.7737966	0.7547808	0.7361686	0.7179096	0.6999642	0.6822996	0.6648905
67	0.7769390	0.7581746	0.7398068	0.7217847	0.7040707	0.6866322	0.6694447
68	0.7799960	0.7614754	0.7433455	0.7255558	0.7080667	0.6908492	0.6738774
69	0.7829703	0.7646888	0.7467903	0.7292259	0.7119576	0.6949548	0.6781946
70	0.7858653	0.7678168	0.7501444	0.7328001	0.7157464	0.6989544	0.6823990
75	0.7992596	0.7822909	0.7656685	0.7493490	0.7332966	0.7174837	0.7018875
80	0.8110813	0.7950717	0.7793842	0.7639759	0.7488155	0.7338756	0.7191364
85	0.8215910	0.8064389	0.7915873	0.7769959	0.7626349	0.7484792	0.7345080
90	0.8309956	0.8166145	0.8025150	0.7886596	0.7750186	0.7615690	0.7482927
95	0.8394600	0.8257759	0.8123565	0.7991668	0.7861777	0.7733687	0.7607222
100	0.8471176	0.8340666	0.8212659	0.8086811	0.7962854	0.7840599	0.7719855
110	0.8604362	0.8484912	0.8367704	0.8252442	0.8138881	0.8026831	0.7916139
120	0.8716223	0.8606110	0.8498043	0.8391729	0.8286965	0.8183559	0.8081389
130	0.8811494	0.8709371	0.8609125	0.8510481	0.8413249	0.8317268	0.8222400
140	0.8893619	0.8798410	0.8704924	0.8612919	0.8522219	0.8432663	0.8344137
160	0.9027965	0.8944110	0.8861751	0.8780681	0.8700725	0.8621753	0.8543674
180	0.9133232	0.9058321	0.8984729	0.8912259	0.8840781	0.8770165	0.8700335
200	0.9217935	0.9150243	0.9083733	0.9018222	0.8953597	0.8889749	0.8826589
250	0.9371500	0.9316959	0.9263340	0.9210530	0.9158400	0.9106887	0.9055916
300	0.9474662	0.9428992	0.9384093	0.9339850	0.9296178	0.9253011	0.9210293
350	0.9548744	0.9509461	0.9470834	0.9432772	0.9395198	0.9358051	0.9321286
400	0.9604506	0.9570053	0.9536165	0.9502773	0.9469800	0.9437194	0.9404923
450	0.9648014	0.9617323	0.9587136	0.9557391	0.9528012	0.9498969	0.9470216
500	0.9682895	0.9655227	0.9628017	0.9601197	0.9574712	0.9548524	0.9522596
550	0.9711483	0.9686298	0.9661532	0.9637117	0.9613007	0.9589164	0.9565558
600	0.9735342	0.9712236	0.9689504	0.9667100	0.9644970	0.9623085	0.9601421
650	0.9755562	0.9734211	0.9713211	0.9692509	0.9672058	0.9651837	0.9631813
700	0.9772905	0.9753065	0.9733553	0.9714309	0.9695308	0.9676513	0.9657902
750	0.9787957	0.9769428	0.9751196	0.9733224	0.9715473	0.9697921	0.9680537
800	0.9801137	0.9783754	0.9766654	0.9749790	0.9733139	0.9716672	0.9700365
850	0.9812769	0.9796404	0.9780298	0.9764421	0.9748742	0.9733232	0.9717874
900	0.9823120	0.9807652	0.9792436	0.9777434	0.9762615	0.9747964	0.9733450
950	0.9832386	0.9817725	0.9803307	0.9789088	0.9775045	0.9761154	0.9747400
1000	0.9840724	0.9826795	0.9813095	0.9799578	0.9786228	0.9773032	0.9759957

表 A.1 (续)

N/F	$\gamma=0.99$						
	14	15	16	17	18	19	20
1	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
2	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
3	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
4	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
5	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
6	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
7	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
8	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
9	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
10	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
11	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
12	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
13	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
14	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
15	0.0006698	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
16	0.0095020	0.0006279	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000
17	0.0268440	0.0089300	0.0005910	0.0000000	0.0000000	0.0000000	0.0000000
18	0.0487103	0.0252836	0.0084236	0.0005582	0.0000000	0.0000000	0.0000000
19	0.0726927	0.0459812	0.0238954	0.0079712	0.0005288	0.0000000	0.0000000
20	0.0974674	0.0687686	0.0435420	0.0226512	0.0075656	0.0005024	0.0000000
21	0.1222817	0.0923949	0.0652471	0.0413493	0.0215310	0.0071990	0.0004785
22	0.1467025	0.1161451	0.0878277	0.0620700	0.0393678	0.0205162	0.0068663
23	0.1704780	0.1395966	0.1105982	0.0836917	0.0591885	0.0375669	0.0195930
24	0.1934675	0.1625019	0.1331528	0.1055601	0.0799297	0.0565636	0.0359244
25	0.2156000	0.1847164	0.1552449	0.1272811	0.1009636	0.0764925	0.0541622
26	0.2368455	0.2061621	0.1767300	0.1486139	0.1219085	0.0967519	0.0733393
27	0.2572028	0.2268041	0.1975269	0.1694132	0.1425301	0.1169738	0.0928793
28	0.2766872	0.2466324	0.2175915	0.1895929	0.1626829	0.1369288	0.1124251
29	0.2953232	0.2656546	0.2369099	0.2091075	0.1822785	0.1564711	0.1317532
30	0.3131440	0.2838901	0.2554846	0.2279366	0.2012683	0.1755133	0.1507195
31	0.3301852	0.3013642	0.2733267	0.2460766	0.2196268	0.1940017	0.1692357
32	0.3464832	0.3181064	0.2904575	0.2635356	0.2373470	0.2119090	0.1872463
33	0.3620753	0.3341496	0.3069022	0.2803284	0.2544329	0.2292247	0.2047216
34	0.3769983	0.3495244	0.3226860	0.2964763	0.2708953	0.2459475	0.2216459
35	0.3912883	0.3642649	0.3378389	0.3120018	0.2867507	0.2620863	0.2380177
36	0.4049782	0.3784020	0.3523897	0.3269297	0.3020183	0.2776539	0.2538411
37	0.4181010	0.3919666	0.3663655	0.3412855	0.3167197	0.2926659	0.2691262
38	0.4306882	0.4049888	0.3797946	0.3550931	0.3308770	0.3071414	0.2838866
39	0.4427685	0.4174960	0.3927041	0.3683792	0.3445127	0.3210995	0.2981372
40	0.4543697	0.4295154	0.4051197	0.3811679	0.3576503	0.3345607	0.3118961
41	0.4655178	0.4410726	0.4170654	0.3934816	0.3703105	0.3475453	0.3251807
42	0.4762365	0.4521920	0.4285664	0.4053443	0.3825159	0.3600728	0.3380097
43	0.4865489	0.4628956	0.4396428	0.4167776	0.3942871	0.3721635	0.3504007
44	0.4964769	0.4732049	0.4503181	0.4278014	0.4056441	0.3838369	0.3623723
45	0.5060393	0.4831396	0.4606099	0.4384358	0.4166057	0.3951106	0.3739420
46	0.5152562	0.4927191	0.4705382	0.4486999	0.4271912	0.4060021	0.3851266
47	0.5241449	0.5019612	0.4801215	0.4586099	0.4374165	0.4165298	0.3959429
48	0.5327218	0.5108821	0.4893751	0.4681847	0.4472988	0.4267088	0.4064065
49	0.5410027	0.5194981	0.4983148	0.4774375	0.4568546	0.4365555	0.4165319
50	0.5490012	0.5278240	0.5069565	0.4863858	0.4660974	0.4460837	0.4263355

表 A.1 (续)

N/F	$\gamma=0.99$						
	14	15	16	17	18	19	20
51	0.5567321	0.5358731	0.5153146	0.4950415	0.4750430	0.4553085	0.4358296
52	0.5642084	0.5436585	0.5234009	0.5034194	0.4837034	0.4642422	0.4450274
53	0.5714405	0.5511932	0.5312282	0.5115317	0.4920916	0.4728979	0.4539430
54	0.5784417	0.5584881	0.5388092	0.5193904	0.5002203	0.4812883	0.4625874
55	0.5852216	0.5655543	0.5461541	0.5270064	0.5080998	0.4894243	0.4709717
56	0.5917903	0.5724020	0.5532730	0.5343905	0.5157414	0.4973166	0.4791070
57	0.5981577	0.5790414	0.5601776	0.5415527	0.5231551	0.5049749	0.4870046
58	0.6043325	0.5854806	0.5668758	0.5485030	0.5303507	0.5124105	0.4946732
59	0.6103227	0.5917301	0.5733764	0.5552493	0.5373381	0.5196313	0.5021222
60	0.6161375	0.5977962	0.5796883	0.5618017	0.5441242	0.5266465	0.5093611
61	0.6217830	0.6036875	0.5858200	0.5681670	0.5507191	0.5334651	0.5163972
62	0.6272670	0.6094107	0.5917771	0.5743541	0.5571288	0.5400941	0.5232403
63	0.6325968	0.6149736	0.5975686	0.5803689	0.5633621	0.5465408	0.5298964
64	0.6377773	0.6203823	0.6032009	0.5862186	0.5694263	0.5528133	0.5363735
65	0.6428158	0.6256436	0.6086789	0.5919104	0.5753264	0.5589187	0.5426788
66	0.6477172	0.6307621	0.6140107	0.5974501	0.5810702	0.5648620	0.5488179
67	0.6524877	0.6357442	0.6192006	0.6028432	0.5866628	0.5706503	0.5547979
68	0.6571322	0.6405960	0.6242544	0.6080959	0.5921105	0.5762892	0.5606243
69	0.6616548	0.6453211	0.6291774	0.6132137	0.5974186	0.5817845	0.5663029
70	0.6660610	0.6499244	0.6339747	0.6182005	0.6025926	0.5871411	0.5718391
75	0.6864896	0.6712749	0.6562303	0.6413451	0.6266103	0.6120166	0.5975571
80	0.7045787	0.6901897	0.6759571	0.6618701	0.6479190	0.6340976	0.6203983
85	0.7207070	0.7070609	0.6935582	0.6801906	0.6669481	0.6538246	0.6408132
90	0.7351735	0.7221986	0.7093579	0.6966415	0.6840412	0.6715507	0.6591631
95	0.7482221	0.7358571	0.7236165	0.7114926	0.6994762	0.6875621	0.6757442
100	0.7600499	0.7482410	0.7365488	0.7249649	0.7134825	0.7020947	0.6907970
110	0.7806683	0.7698348	0.7591059	0.7484731	0.7379295	0.7274699	0.7170889
120	0.7980325	0.7880283	0.7781164	0.7682915	0.7585475	0.7488775	0.7392792
130	0.8128547	0.8035618	0.7943535	0.7852238	0.7761674	0.7671788	0.7582536
140	0.8256542	0.8169793	0.8083815	0.7998562	0.7913973	0.7830002	0.7746618
160	0.8466388	0.8389835	0.8313945	0.8238666	0.8163951	0.8089771	0.8016086
180	0.8631199	0.8562704	0.8494781	0.8427400	0.8360515	0.8294088	0.8228095
200	0.8764054	0.8702083	0.8640624	0.8579645	0.8519109	0.8458977	0.8399227
250	0.9005441	0.8955400	0.8905764	0.8856495	0.8807576	0.8758970	0.8710662
300	0.9167973	0.9126018	0.9084391	0.9043070	0.9002031	0.8961252	0.8920708
350	0.9284856	0.9248739	0.9212904	0.9177322	0.9141975	0.9106858	0.9071939
400	0.9372949	0.9341243	0.9309781	0.9278548	0.9247513	0.9216669	0.9186000
450	0.9441723	0.9413473	0.9385437	0.9357592	0.9329930	0.9302444	0.9275102
500	0.9496905	0.9471427	0.9446139	0.9421028	0.9396084	0.9371285	0.9346631
550	0.9542165	0.9518963	0.9495937	0.9473070	0.9450349	0.9427765	0.9405303
600	0.9579948	0.9558650	0.9537513	0.9516520	0.9495665	0.9474932	0.9454313
650	0.9611973	0.9592293	0.9572758	0.9553360	0.9534085	0.9514916	0.9495862
700	0.9639458	0.9621167	0.9603013	0.9584981	0.9567056	0.9549246	0.9531534
750	0.9663312	0.9646225	0.9629260	0.9612417	0.9595681	0.9579037	0.9562484
800	0.9684203	0.9668170	0.9652260	0.9636457	0.9620745	0.9605132	0.9589603
850	0.9702653	0.9687554	0.9672570	0.9657685	0.9642891	0.9628181	0.9613556
900	0.9719066	0.9704797	0.9690634	0.9676570	0.9662590	0.9648694	0.9634867
950	0.9733761	0.9720237	0.9706811	0.9693477	0.9680227	0.9667053	0.9653956
1000	0.9746997	0.9734143	0.9721381	0.9708711	0.9696117	0.9683592	0.9671143

附录 B

(资料性附录)

二项分布可靠度单侧置信下限的近似公式

对本标准未给出的 γ 、 n 、 F 诸参数,二项分布可靠度单侧置信下限 R_L ,当 $F=0$ 、 $F=n-1$ 及 $F=n$ 时,分别按式(B.1)、式(B.3)及式(B.4)计算;当 $F=1,2,3$ 时,按对数伽玛近似式计算;其他场合用贝泽-普拉特(Peizer-Pratt)近似计算。后面给出了用这些公式近似计算的 S-PLUS 程序。

B.1 计算公式

B.1.1 对数伽玛近似

R_L 的对数伽玛近似按以下步骤计算:

a) 计算

$$c = \frac{\ln\left(\frac{n+1}{n-F}\right)}{\ln\left(\frac{n+2}{n-F+1}\right)} \dots\dots\dots (B.1)$$

b) 计算

$$\eta = \frac{3-c}{2(c-1)-0.355(c-1)^3} \dots\dots\dots (B.2)$$

c) 计算

$$Z = \frac{\ln\left(\frac{n+1}{n-F}\right)}{\ln\left(\frac{\eta+1}{\eta}\right)} \dots\dots\dots (B.3)$$

d) 查 χ^2 分布表确定参数为 $2Z$ 的 χ^2 分布的 γ 分位数 $\chi^2_{\gamma}(2Z)$ 。

e) 计算 R_L 的对数伽玛近似。

$$R_L = \exp\left\{-\frac{\chi^2_{\gamma}(2Z)}{2\eta}\right\} \dots\dots\dots (B.4)$$

示例: $n=50, F=1, \gamma=0.90$

按式(B.1)、式(B.2)、式(B.3)算得:

$c=1.020\ 005$

$\eta=49.490\ 31$

$Z=1.999\ 812$

利用 χ^2 分布表插值得:

$$\chi^2_{0.90}(2 * 1.999\ 812) = 7.778\ 882$$

按式(B.4)算得:

$$R_L = 0.924\ 418\ 9$$

B.1.2 贝泽-普拉特近似

R_L 的贝泽-普拉特近似由式(B.5)迭代求解:

$$h(R_L) + U_{\gamma} = 0 \dots\dots\dots (B.5)$$

式中:

$$h(R) = \frac{d}{|F+0.5-n(1-R)|} \left\{ \frac{2}{1+\frac{1}{6n}} \left[(F+0.5) \ln \frac{F+0.5}{n(1-R)} + (n-F-0.5) \ln \frac{n-F-0.5}{nR} \right] \right\}^{\frac{1}{2}}$$

$$d = F + 0.5 + \frac{1}{6} - \left(n + \frac{1}{3}\right)(1 - R) + 0.02 \left(\frac{R}{F+1} - \frac{1-R}{n-F} + \frac{R-0.5}{n+1}\right)$$

U_γ ——标准正态分布的 γ 分位数。

